What are the proven biological recovery methods and quantified performance metrics for microalgae-based extraction of Neodymium (Nd), Dysprosium (Dy), Praseodymium (Pr), Terbium (Tb), and Samarium (Sm) from NdFeB and SmCo magnet manufacturing waste streams, including grinding coolant, sintering residues, and electroplating bath waste? Microalgae focus: • Genera: Chlorella, Spirulina, Nannochloropsis, Galdieria, Scenedesmus, Dunaliella • Extremophiles capable of tolerating acidic pH (2-4) and elevated temperatures (40-55℃) • Comparison of dead/immobilized biomass vs living cell systems Waste stream characterization needed: 1. NdFeB magnet manufacturing waste types: • Grinding/machining coolant: suspended Nd-Fe-B powder (concentration in ppm) • Sintering furnace effluent: oxidized REE residues • Ni-Cu-Ni electroplating bath waste: Nd, Pr, Dy, Ni, Cu co-contamination • Acid leaching solutions from scrap magnet recycling 2. SmCo magnet production waste: • Samarium and Cobalt concentrations (ppm) • pH and chemical composition of waste streams Required quantitative data: • Biosorption capacity (mg Nd/g, mg Dy/g biomass) - report q_max values • Recovery efficiency (%): Nd, Dy, Pr, Tb individually • Selectivity ratios: Nd/Ca, Nd/Fe, Dy/Fe (critical for separating REEs from iron matrix) • Kinetics: adsorption equilibrium time (minutes/hours) • Desorption efficiency (%) for metal recovery from loaded biomass • Eluent type and concentration for regeneration (HCl, H2SO4, EDTA concentrations) • Final purity of recovered REE concentrate (% REO, grade suitable for magnet remanufacturing) • Economic metrics: cost per kg Nd recovered, OPEX per m^3 wastewater treated • Comparison to hydrometallurgical alternatives: solvent extraction, ion exchange resins Critical questions to address: • Can microalgae selectively separate heavy REEs (Dy, Tb) from light REEs (Nd, Pr)? Report separation factors. • How does co-presence of iron (Fe) from NdFeB matrix affect REE biosorption? • What is the minimum economically viable REE concentration (ppm) in waste streams? • Can bio-recovered REE concentrate be directly reused in magnet manufacturing, or does it require further refining? Scale and validation: • Prioritize pilot-scale studies (>100L) or industrial demonstrations • Include techno-economic analyses (TEA) and life cycle assessments (LCA) • Studies from magnet manufacturing facilities or in collaboration with magnet producers Timeframe: 2019-2026 strongly preferred. Include 2015-2018 if reporting industrial case studies or comparative economic data.

While microalgae including Chlorella vulgaris and Spirulina species demonstrate biosorption capacities up to 188.68 mg Nd/g and recovery efficiencies exceeding 80% for REEs from synthetic solutions at laboratory scale, no studies

have examined actual NdFeB or SmCo magnet manufacturing waste streams (grinding coolant, sintering residues, electroplating bath waste), and critical data gaps remain regarding iron matrix interference, heavy/light REE selectivity, economic viability, and scalability beyond bench-scale experiments.

Abstract

Microalgae-based biosorption demonstrates technical feasibility for REE recovery from aqueous solutions at laboratory scale, with maximum biosorption capacities of 188.68 mg Nd/g for Chlorella vulgaris, 72.5 mg Nd/g for Spirulina platensis, and 51.10-55.31 mg Pr/g for Spirulina dry powder. Recovery efficiencies exceed 80% for Nd from phosphogypsum leachate and 79% for Pr under optimized conditions, with rapid equilibrium times of 5-80 minutes. The Langmuir isotherm model consistently provides excellent fit (R^2 >0.98), indicating monolayer chemisorption with ion exchange as the dominant mechanism. Desorption efficiencies of 86-95% are achievable using 0.1 M HNO3 or Na2EDTA, with biofilm systems demonstrating superior reusability over suspended biomass. Galdieria sulphuraria offers potential for acidic waste applications due to pH 2-4 tolerance, though biosorption performance is compromised at its growth optimum and improves up to 24-fold at pH 5.0.

However, critical gaps preclude direct application to magnet manufacturing: no studies examined actual NdFeB grinding coolant, sintering residues, or electroplating bath waste, and no selectivity ratios for Nd/Fe or Dy/Fe separation from iron matrices were reported. All experiments operated at lab bench scale (<500 mL), with no pilot-scale (>100L) or industrial demonstrations identified. Economic metrics including cost per kg REE recovered and minimum viable waste stream concentrations remain unquantified, and no studies assessed whether bio-recovered REE concentrate meets purity requirements for magnet remanufacturing. The evidence supports proof-of-concept for microalgae-based REE biosorption but cannot yet address the specific requirements for magnet manufacturing waste treatment at industrial scale.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the proven biological recovery methods and quantified performance metrics for microalgae-based extraction of Neodymium (Nd), Dysprosium (Dy), Praseodymium (Pr), Terbium (Tb), and Samarium (Sm) from NdFeB and SmCo magnet manufacturing waste streams, including grinding coolant, sintering residues, and electroplating bath waste?

Microalgae focus: • Genera: Chlorella, Spirulina, Nannochloropsis, Galdieria, Scenedesmus, Dunaliella • Extremophiles capable of tolerating acidic pH (2-4) and elevated temperatures (40-55℃) • Comparison of dead/immobilized biomass vs living cell systems

Waste stream characterization needed:

1. NdFeB magnet manufacturing waste types: • Grinding/machining coolant: suspended Nd-Fe-B powder (concentration in ppm) • Sintering furnace effluent: oxidized REE residues • Ni-Cu-Ni electroplating bath waste: Nd, Pr, Dy, Ni, Cu co-contamination • Acid leaching solutions from scrap magnet recycling
2. SmCo magnet production waste: • Samarium and Cobalt concentrations (ppm) • pH and chemical composition of waste streams

Required quantitative data: • Biosorption capacity (mg Nd/g, mg Dy/g biomass) - report q_max values • Recovery efficiency (%): Nd, Dy, Pr, Tb individually • Selectivity ratios: Nd/Ca, Nd/Fe, Dy/Fe (critical for separating REEs from

iron matrix) • Kinetics: adsorption equilibrium time (minutes/hours) • Desorption efficiency (%) for metal recovery from loaded biomass • Eluent type and concentration for regeneration (HCl, H2SO4, EDTA concentrations) • Final purity of recovered REE concentrate (% REO, grade suitable for magnet remanufacturing) • Economic metrics: cost per kg Nd recovered, OPEX per m^3 wastewater treated • Comparison to hydrometallurgical alternatives: solvent extraction, ion exchange resins

Critical questions to address: • Can microalgae selectively separate heavy REEs (Dy, Tb) from light REEs (Nd, Pr)? Report separation factors. • How does co-presence of iron (Fe) from NdFeB matrix affect REE biosorption? • What is the minimum economically viable REE concentration (ppm) in waste streams? • Can bio-recovered REE concentrate be directly reused in magnet manufacturing, or does it require further refining?

Scale and validation: • Prioritize pilot-scale studies (>100L) or industrial demonstrations • Include techno-economic analyses (TEA) and life cycle assessments (LCA) • Studies from magnet manufacturing facilities or in collaboration with magnet producers

Timeframe: 2019-2026 strongly preferred. Include 2015-2018 if reporting industrial case studies or comparative economic data."

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Microalgae Focus**: Does this study use microalgae as the primary biological system for rare earth element extraction (not bacteria, fungi, plants, or other non-algae organisms)?
- **Target REE Elements**: Does this study investigate extraction of at least one of the following rare earth elements: Neodymium (Nd), Dysprosium (Dy), Praseodymium (Pr), Terbium (Tb), or Samarium (Sm)?
- **Magnet Waste Source**: Does this study use waste streams from NdFeB or SmCo magnet manufacturing/recycling processes (not mining ores, non-magnet electronic waste, or other non-magnet industrial waste streams)?
- **Quantitative Data**: Does this study report at least one quantitative performance metric such as biosorption capacity, recovery efficiency, selectivity ratios, kinetics data, or desorption efficiency?
- **Empirical Research**: Does this study provide empirical data from laboratory experiments, pilot-scale studies, or industrial demonstrations (not purely theoretical or modeling studies without experimental validation)?
- **Full Research Paper**: Is this a full research paper with sufficient methodological detail and comprehensive results (not a conference abstract, poster, or brief communication)?
- **Recovery Application**: Does this study focus on actual REE extraction and recovery using microalgae (not solely REE detection, sensing, monitoring, or microalgae cultivation optimization)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

30 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Microalgae Focus**: Does this study use microalgae as the primary biological system for rare earth element extraction (not bacteria, fungi, plants, or other non-algae organisms)?
- **Target REE Elements**: Does this study investigate extraction of at least one of the following rare earth elements: Neodymium (Nd), Dysprosium (Dy), Praseodymium (Pr), Terbium (Tb), or Samarium (Sm)?
- **Magnet Waste Source**: Does this study use waste streams from NdFeB or SmCo magnet manufacturing/recycling processes (not mining ores, non-magnet electronic waste, or other non-magnet industrial waste streams)?
- **Quantitative Data**: Does this study report at least one quantitative performance metric such as biosorption capacity, recovery efficiency, selectivity ratios, kinetics data, or desorption efficiency?
- **Empirical Research**: Does this study provide empirical data from laboratory experiments, pilot-scale studies, or industrial demonstrations (not purely theoretical or modeling studies without experimental validation)?
- **Full Research Paper**: Is this a full research paper with sufficient methodological detail and comprehensive results (not a conference abstract, poster, or brief communication)?
- **Recovery Application**: Does this study focus on actual REE extraction and recovery using microalgae (not solely REE detection, sensing, monitoring, or microalgae cultivation optimization)?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

15 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Microalgae Species**:

  Extract details about the microalgae species used for REE recovery from magnet manufacturing waste, including:

- Genus and species name (prioritize Chlorella, Spirulina, Nannochloropsis, Galdieria, Scenedesmus, Dunaliella)
- Biomass state (living cells, dead biomass, immobilized biomass)
- Biomass pretreatment methods (drying, chemical treatment, functionalization)
- Extremophile characteristics (acid tolerance pH 2-4, temperature tolerance 40-55℃)
- Cultivation conditions and biomass preparation methods

- **Waste Stream Characterization**:

  Extract characterization data for magnet manufacturing waste streams used in the study, including:

- Waste type (NdFeB grinding coolant, sintering residues, electroplating bath waste, SmCo production waste, acid leaching solutions)
- REE concentrations in ppm for Nd, Dy, Pr, Tb, Sm individually
- Co-contaminant concentrations (Fe, Ni, Cu, Co, Ca) in ppm
- pH and chemical composition of waste streams
- Whether real industrial waste or synthetic solutions were used
- Source facility or manufacturing process details

- **REE Recovery Performance**:

Extract quantified performance metrics for recovery of Nd, Dy, Pr, Tb, and Sm from magnet waste, including:

- Maximum biosorption capacity (q_max) in mg REE/g biomass for each element individually
- Recovery efficiency percentages for Nd, Dy, Pr, Tb, Sm individually
- Selectivity ratios (Nd/Ca, Nd/Fe, Dy/Fe, Heavy REE/Light REE separation factors)
- Comparison data showing performance differences between target REEs
- Performance in mixed-metal vs single-metal solutions

- **Process Kinetics**:

  Extract kinetic and operational parameters for the biosorption process, including:

- Adsorption equilibrium time (minutes/hours)
- Optimal pH range for maximum REE uptake
- Temperature effects and optimal operating temperature
- Contact time and mixing conditions
- Adsorption isotherm model fit (Langmuir, Freundlich parameters)
- Mechanism of biosorption (ion exchange, complexation, physical adsorption)

- **Metal Recovery**:

  Extract data on recovering REEs from loaded biomass for reuse, including:

- Desorption/elution efficiency percentages for each REE
- Eluent type and concentration (HCl, H2SO4, EDTA, etc.)
- Number of adsorption-desorption cycles possible
- Final purity of recovered REE concentrate (% REO)
- Grade suitability for magnet remanufacturing
- Biomass regeneration and reusability potential

- **Comparative Performance**:

  Extract comparative data showing microalgae performance vs alternative methods and species, including:

- Comparison with hydrometallurgical methods (solvent extraction, ion exchange resins)
- Performance differences between microalgae genera/species tested
- Dead vs living biomass performance comparison
- Effectiveness compared to other biosorbents (bacteria, fungi, other algae)
- Advantages and limitations identified by authors

- **Economic Analysis**:

  Extract economic and feasibility data for microalgae-based REE recovery, including:

- Cost per kg of REE recovered (Nd, Dy, Pr, Tb, Sm)
- Operating expenses (OPEX) per m^3 wastewater treated
- Minimum economically viable REE concentration in waste streams (ppm)
- Techno-economic analysis (TEA) results if provided
- Life cycle assessment (LCA) findings
- Capital investment requirements and payback period estimates

- **Scale and Validation**:

  Extract information about study scale and industrial validation, including:

- Scale of experiments (lab bench, pilot >100L, industrial demonstration)
- Collaboration with magnet manufacturing facilities
- Real vs simulated waste stream testing
- Industrial case studies or field demonstrations
- Challenges identified for scale-up
- Regulatory or environmental approval status

Characteristics of Included Studies

The systematic review identified 15 studies investigating microalgae and related phototrophic organisms for rare earth element (REE) recovery from aqueous solutions. However, a critical finding is that none of the studies specifically addressed NdFeB or SmCo magnet manufacturing waste streams as defined in the research question (grinding coolant, sintering residues, electroplating bath waste). Most studies utilized synthetic solutions or waste streams unrelated to magnet production.

Study	Full Text Retrieved?	Organism Type	Target REEs	Waste Stream Type	Scale
M. A. Kucuker et al., 2017	Yes	Chlorella vulgaris	Nd	Acid leaching from HDD magnets	Lab (500 mL)
Z. Birungi et al., 2018	Yes	C. vulgaris, C. reinhardtii	La, Tl	Industrial wastewater (mining)	Lab bench
C. García-Balboa et al., 2022	Yes	Euglena sp., Chlamy-domonas sp.	Gd, Pr, Ce, La	Electronic scraps	Lab (100 mL)
Pro Danièle et al., 2016	Yes	Coccomyxa genus	Mixed REEs	Not specified	Not specified
Marcus Heilmann et al., 2021	Yes	C. kessleri, C. brevissima, P. patens	Nd, Eu	Synthetic solutions	Lab (0.1-15 L)
Mic Paper et al., 2023	Yes	Cyanobacteria (12 strains)	Ce, Nd, Tb, La	Not specified	Not specified
M. Palmieri et al., 2022	Yes	Galdieria sulphuraria	Y, Ce, Eu, Tb	Synthetic quaternary solutions	Lab bench
C. Fischer et al., 2019	Yes	Anabaena spec., A. cylindrica	Eu, Sm, Nd	Synthetic solutions	Lab (250 mL)
Jens Kastenhofer et al., 2024	Yes	Galdieria sulphuraria	La (model REE)	Synthetic solutions	Lab scale
Zhenxiang Hu et al., 2025	Yes	Spirulina	Pr	Synthetic solutions	Not specified
João Pinto et al., 2023	Yes	Multiple algae (review)	Various REEs	E-waste (review)	Lab scale (mostly)

Study	Full Text Retrieved?	Organism Type	Target REEs	Waste Stream Type	Scale
M. Vítová et al., 2024	Yes	C. reinhardtii, G. sulphuraria, others	Various REEs	Various (review)	Not specified
E. Lima et al., 2022	Yes	Spirulina platensis (LEB-18, LEB-52)	Nd	Phosphogypsum leachate	Lab bench
Paloma Martínez-Alesón García et al., 2024	Yes	Chlamydomonas reinhardtii	Sm	Synthetic solutions	Lab scale
M. T. Zak et al., 2024	Yes	Euglena mutabilis	Mixed REEs	Simulated solutions	Lab bench

The predominance of lab-scale studies using synthetic solutions represents a significant gap between current research and industrial applicability for magnet manufacturing waste treatment. Only Kucuker et al. (2017) utilized solutions derived from actual permanent magnets (hard disk drives), and Lima et al. (2022) tested real phosphogypsum leachate containing Nd at 95.7 mg/L.

Microalgae Species and Biomass Characteristics

Species Distribution and Extremophile Properties

Among the prioritized genera specified in the research question, the following were represented in the literature:

Species	Biomass State	Pretreatment	pH Tolerance	Temperature Tolerance
Chlorella vulgaris	Dead (oven-dried 90℃)	Oven drying	Optimal pH 5	Optimal 35℃
Chlorella kessleri	Dried (lyophilized)	Lyophilization	~pH 5	Room temperature
Spirulina platensis	Dead (dried, ground)	Drying, grinding <125 µm	pH 6.0	298 K optimal
Spirulina (dry powder)	Dead (dry powder)	Two-stage drying, cryogenic grinding	pH 5	298-318 K
Galdieria sulphuraria	Freeze-dried	Chemical treatment with H2SO4	pH 2-4 tolerant	37℃ cultivation
Galdieria sulphuraria	Living cells	None specified	pH 2-4 (growth optimum pH 1.5-2.5)	42℃
Chlamydomonas reinhardtii	Dead biomass	Drying at 50℃	pH 6	23℃
Chlamydomonas reinhardtii	Living cells	Settling selection	pH 4.5 tolerant	22℃

Species	Biomass State	Pretreatment	pH Tolerance	Temperature Tolerance
Euglena mutabilis	Living cells and biofilms	Centrifugation, washing	Desorption tested pH 1.5-4	25℃

Galdieria sulphuraria emerges as the most suitable extremophile for acidic waste stream applications, demonstrating acid tolerance at pH 2-4 and temperature tolerance at 37-42℃. This polyextremophilic red alga can be cultivated mixotrophically in Allen medium at pH 2.0. However, biosorption performance at the organism's growth optimum (pH 1.5-2.5) was poor, improving up to 24-fold at pH 5.0 in phosphate-free conditions.

Dead versus Living Biomass Performance

A critical distinction exists between dead and living biomass systems. Dried sorbents reach equilibrium faster (100-150 minutes) compared to living sorbents, which require several days. Metabolic activity was found to have a negative impact on REE sorption in G. sulphuraria, challenging the feasibility of REE biosorption under ideal growth conditions for acidophiles. For Euglena mutabilis biofilms versus suspensions, the desorption efficiency was higher in biofilms (86-95%) compared to suspensions (72-74%), with biofilm biosorption capacity increasing by 280% after repeated adsorption-desorption cycles using HCl.

REE Recovery Performance

Maximum Biosorption Capacities

Organism	Target REE	q_max (mg/g)	Conditions	Reference
Chlorella vulgaris	Nd	188.68	pH 5, 35℃	Kucuker et al., 2017
Chlamydomonas reinhardtii	La	143	pH 6	Birungi et al., 2018
Chlorella vulgaris	La	75	pH 6	Birungi et al., 2018
Nostoc sp. 20.02	Ce, Nd, Tb, La	84.2-91.5	pH 5-6	Paper et al., 2023
Synechococcus elongatus	Ce, Nd, Tb, La	69.5-83.4	pH 5-6	Paper et al., 2023
Calothrix brevissima	Ce, Nd, Tb, La	68.6-83.5	pH 5-6	Paper et al., 2023
Spirulina platensis LEB-18	Nd	72.5	pH 6, 298 K	Lima et al., 2022
Spirulina platensis LEB-52	Nd	48.2	pH 6, 298 K	Lima et al., 2022
Spirulina (dry powder)	Pr	51.10-55.31	pH 5, 298-318 K	Hu et al., 2025
Coccomyxa sp.	Mixed REEs	Up to 15	Not specified	Danièle et al., 2016
Galdieria sulphuraria	Mixed REEs	Up to 2.40	pH 5.0	Kastenhofer et al., 2024

The highest reported biosorption capacity for neodymium was 188.68 mg/g using Chlorella vulgaris at pH 5 and 35℃, representing the Langmuir isotherm maximum monolayer coverage capacity with a Langmuir constant (KL) of 0.029 L/mg. This capacity exceeds that of activated carbon and other algae types including A. gracilis, Sargassum sp., and A. densus.

Recovery Efficiency

Recovery efficiency data were limited across studies:

- Neodymium removal from phosphogypsum leachate exceeded 80% using Spirulina platensis strains
- Praseodymium removal efficiency reached 79.0% under optimized conditions (pH 5, 2.0 g/L adsorbent, 60 min contact time)
- Complete (>99%) biosorption of Eu, Sm, and Nd occurred within the first hour using Anabaena species
- More than 90% of REE could be removed from water within a few hours using Coccomyxa microalgae
- Lanthanum recovery efficiency reached 90.92% using Desmodesmus multivariabilis

Selectivity and Separation Factors

No studies reported quantified selectivity ratios (Nd/Ca, Nd/Fe, Dy/Fe) or heavy REE/light REE separation factors as specified in the research question. The lack of selectivity data represents a critical knowledge gap for magnet waste applications where REE separation from the iron matrix is essential.

Mixed-metal solution studies indicated an ion-exchange mechanism where Na+, K+, Mg2+, and Ca2+ ions are replaced by other metal cations during biosorption. Heavy REEs were found to preferentially desorb at pH 4 over lighter REEs, suggesting potential for selective recovery during the desorption phase rather than during biosorption.

Process Kinetics and Operating Parameters

Adsorption Equilibrium and Kinetic Models

Organism	Equilibrium Time	Kinetic Model	Isotherm Model	Mechanism
Chlorella vulgaris	30 minutes	Pseudo-second order (R^2 >0.99)	Langmuir (R^2 = 0.98)	Surface adsorption, anion exchange with carboxyl groups
Chlamydomonas reinhardtii	5-1440 minutes	Langmuir-Hinshelwood	Langmuir (R^2 ≤0.99)	Not specified
Cyanobacteria (multiple)	5 minutes	Fast kinetics	Langmuir	Ion-exchange
Galdieria sulphuraria	30 minutes (pH 4.5)	Not specified	Not specified	Ion exchange or complexation
Spirulina	60 minutes	Pseudo-second order (R^2 = 0.999)	Langmuir (R^2 = 0.993-0.999)	Chemisorption, ion exchange, complexation

Organism	Equilibrium Time	Kinetic Model	Isotherm Model	Mechanism
Spirulina platensis	80 minutes	Avrami model	Liu model	Electrostatic attraction (phosphate/sulfate groups)
Anabaena spp.	<1 hour	Not specified	Not specified	Intracellular uptake, complexation
Euglena mutabilis	15-20 minutes	Not specified	Not specified	EPS binding

The rapid kinetics observed across species (5-80 minutes to equilibrium) compare favorably with industrial process requirements. The Langmuir isotherm model consistently provided the best fit for most species, indicating monolayer adsorption on homogeneous surfaces.

Optimal pH Range

The optimal pH for maximum REE uptake clustered around pH 4.5-6.0 across most species:

- pH 5.0: Chlorella vulgaris for Nd, Spirulina for Pr, Galdieria sulphuraria (improved sorption)
- pH 5-6: Cyanobacteria (Nostoc, Synechococcus, Calothrix)
- pH 6.0: Chlamydomonas reinhardtii, Spirulina platensis
- pH 4.5: Galdieria sulphuraria for Ce, Y, Eu, Tb, Chlamydomonas reinhardtii for Sm

This pH range presents a challenge for direct application to acidic magnet manufacturing waste streams, as most species show reduced performance at pH <4.

Temperature Effects

Temperature effects showed varied responses:

- Optimal at 35℃ for Chlorella vulgaris (Nd uptake 157.40 mg/g versus 126.13 mg/g at 21℃)
- Endothermic and spontaneous process for Spirulina ($\Delta G^\circ < 0$, $\Delta S^\circ > 0$) within 298-318 K
- Exothermic process for Spirulina platensis strains (ΔH° -23.2 for LEB-18, -19.9 for LEB-52)
- Temperature does not affect Sm adsorption by Chlamydomonas reinhardtii at 22℃

Desorption and Metal Recovery

Desorption Performance

Organism	Eluent	Desorption Efficiency	Cycles Tested	Notes
Spirulina	0.1 mol/L HNO3	91.7% (Pr)	Not specified	Excellent regeneration performance
Chlamydomonas reinhardtii	0.1 M HNO3	90.92% (La)	3 cycles	Lanthanum recovery

Organism	Eluent	Desorption Efficiency	Cycles Tested	Notes
Euglena mutabilis biofilms	HCl	86%	3 cycles	Biosorption capacity increased 280%
Euglena mutabilis biofilms	Na2EDTA	95%	3 cycles	Biosorption capacity increased 80%
Euglena mutabilis suspensions	HCl	72%	3 cycles	Biosorption capacity decreased 80%
Euglena mutabilis suspensions	Na2EDTA	74%	3 cycles	Biosorption capacity decreased 80%

A notable finding is that biofilm systems demonstrate superior reusability compared to suspended biomass. The increase in biofilm biosorption capacity after repeated cycles was attributed to simultaneous desorption of divalent cations alongside REEs from the extracellular polymeric substances (EPS), increasing available binding sites.

Purity and Grade Suitability

No studies reported the final purity of recovered REE concentrate (% REO) or assessed suitability for magnet remanufacturing. This represents a critical gap, as the research question specifically requires data on whether bio-recovered REE concentrate can be directly reused in magnet manufacturing or requires further refining.

Comparative Performance

Microalgae versus Hydrometallurgical Methods

Direct quantitative comparisons with hydrometallurgical methods were not provided in the included studies. However, qualitative advantages of microalgae-based biosorption were identified:

- Avoids toxic solvents and requires less energy than solvent extraction
- Offers higher selectivity than precipitation and is more eco-friendly than ion exchange resins
- Lower energy consumption and reduced environmental impact compared to traditional chemical or physical processes
- Cost-effectiveness, low sludge production, and environmental friendliness

Limitations include:

- Sensitivity to pH and metal concentration for living algae
- Slow kinetics and need for optimization
- Metabolic activity negatively impacts REE sorption for acidophiles

Species Performance Comparison

Among microalgae, Chlorella vulgaris demonstrated greater biosorption affinity for Nd than activated carbon and other algae types. Chlamydomonas reinhardtii showed higher sorption capacity (143 mg/g for La) compared to Chlorella vulgaris (75 mg/g for La). For the moss Physcomitrella patens, sorption capacities exceeded those of the tested microalgae for both Nd (0.74 mmol/g) and Eu (0.48 mmol/g).

The higher performance of Spirulina platensis LEB-18 (72.5 mg/g) versus LEB-52 (48.2 mg/g) was attributed to higher content of phosphate and sulfate functional groups.

Synthesis

Reconciling Performance Heterogeneity

The wide range of reported biosorption capacities (2.40-188.68 mg/g for Nd) reflects methodological differences rather than fundamental limitations of microalgae-based recovery. The highest capacities (>100 mg/g) were observed under optimized single-metal, single-species conditions at controlled pH (4.5-6.0) and moderate temperature (25-35℃). Lower capacities typically occurred in studies using acidic conditions at the extremophile growth optimum or complex multi-metal matrices.

Gap Analysis: Magnet Manufacturing Waste Applications

A fundamental mismatch exists between the research question requirements and available evidence:

1. **Waste stream specificity**: No studies examined actual NdFeB grinding coolant, sintering residues, or Ni-Cu-Ni electroplating bath waste. The closest analog was acid leaching solution from HDD magnets, which differs substantially from manufacturing process wastes.
2. **Iron interference**: The critical question of how co-presence of iron affects REE biosorption remains unanswered. No Nd/Fe or Dy/Fe selectivity ratios were reported, despite iron being a major matrix component in NdFeB waste.
3. **Heavy/Light REE separation**: While heavy REEs preferentially desorb at pH 4, no quantified separation factors for Dy/Nd or Tb/Pr were provided.
4. **Scale validation**: All studies operated at lab bench scale (<500 mL). No pilot-scale (>100L) or industrial demonstrations were identified.
5. **Economic metrics**: No studies provided cost per kg REE recovered, OPEX per m^3 wastewater treated, or minimum economically viable REE concentration.

Implications for Industrial Application

The evidence supports microalgae-based REE recovery as technically feasible at laboratory scale with synthetic solutions. Chlorella vulgaris and Spirulina species demonstrate the highest biosorption capacities for Nd and Pr among the prioritized genera. Galdieria sulphuraria offers unique potential for acidic waste streams due to its extremophile properties, though its biosorption performance is compromised at its growth-optimal pH.

For magnet manufacturing applications, the following conditions appear most favorable based on the available evidence:

- Dead or immobilized biomass rather than living cells (faster kinetics, higher tolerance to process conditions)
- pH adjustment to 4.5-6.0 prior to biosorption
- Contact times of 30-80 minutes for equilibrium
- Biofilm configurations for enhanced reusability
- Acidic eluents (HCl, HNO3) at 0.1 M concentration for desorption

However, translation to industrial practice requires substantial additional research addressing real waste stream complexity, iron matrix interference, and techno-economic validation at pilot scale.

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What are the quantitative bio-leaching and biosorption capabilities of bacterial and fungal systems for recovering critical raw materials (Copper, Cerium, Gallium, Germanium, Indium, Rare Earth Elements) from semiconductor fabrication waste, specifically comparing their performance against microalgae-based systems in terms of recovery rates (%), cost ($/m³ or $/kg), and tolerance to toxic semiconductor processing chemicals? Target organisms: • Bacteria: Acidithiobacillus ferrooxidans, Acidithiobacillus thiooxidans, Leptospirillum ferrooxidans, Pseudomonas species, Bacillus species, Cupriavidus metallidurans • Fungi: Aspergillus niger, Aspergillus flavus, Penicillium chrysogenum, Penicillium simplicissimum, Rhizopus arrhizus, Trichoderma species Waste stream focus: • CMP wastewater (copper-cerium-silica slurry, pH 2-4) • Wet etching waste (HF-based, metal residues) • Compound semiconductor waste (GaAs, GaN, InP processing residues) Critical comparison metrics needed: 1. Recovery efficiency (%) - direct comparison to microalgae where available 2. Bio-leaching mechanism: organic acid production (citric, oxalic, gluconic), siderophore secretion, enzymatic reduction 3. Tolerance to semiconductor toxins: fluoride, arsenic, boron, organic solvents 4. Growth rate and biomass productivity compared to microalgae 5. OPEX comparison ($/m³) vs microalgae systems 6. Metal selectivity (target metal/interfering ion ratios) 7. Scalability: demonstrated scale (L) and footprint requirements (m²) 8. Kinetics: bio-leaching time to target recovery (hours vs days) Emphasize: • Studies that DIRECTLY COMPARE bacteria/fungi to microalgae performance • Economic feasibility data (CAPEX/OPEX) • Industrial pilot demonstrations (not just lab-scale) • Patents or commercial deployments Timeframe: 2019-2026. Include landmark studies from 2015-2018 if they report pilot-scale or comparative data.

Bacterial systems (particularly *Acidithiobacillus ferrooxidans*) and fungal systems (particularly *Aspergillus niger*) achieve high recovery efficiencies of 93-100% for copper, 100% for indium, up to 84% for gallium, and 65-100% for rare earth elements from electronic waste, with exceptional tolerance to heavy metals (up to 55 g/L Cu, 112 g/L Zn), but the near-complete absence of direct comparative studies with microalgae—only one study found algae had lower REE biosorption performance—combined with limited economic data ($576/kg Ga for bioleaching versus no microalgae cost benchmarks) and predominantly laboratory-scale research precludes definitive conclusions about relative performance advantages for semiconductor fabrication waste processing specifically.

Abstract

Bacterial and fungal systems demonstrate high recovery efficiencies for critical raw materials from electronic and semiconductor waste, with acidophilic bacteria (*Acidithiobacillus ferrooxidans*, *Leptospirillum ferrooxidans*) achieving 93-100% copper recovery and fungi (*Aspergillus niger*) reaching 66-100% copper and 100% indium recovery. For gallium from compound semiconductor waste (GaAs), bacterial strains with high arsenic tolerance achieved up to 84% recovery, while rare earth element recovery ranged from 65-100% using *Methylobacterium extorquens* AM1 and engineered *E. coli* biosorbents achieving 97% REE purity. Acidophilic bacteria tolerate exceptionally high metal concentrations (Cu up to 55 g/L, Zn up to 112 g/L), and *A. ferrooxidans* demonstrates unprecedented REE tolerance at 50-100 mM. Bioleaching mechanisms include ferric iron oxidation for bacteria and organic acid production (citric, oxalic, gluconic acids) for fungi. Recovery kinetics typically require 5-18 days for bioleaching, though biosorption processes achieve metal removal within minutes to hours.

However, the evidence base cannot definitively address comparative performance against microalgae systems. Of 80 studies reviewed, only one provided direct head-to-head comparison, finding that algae exhibited low biosorption performance compared to bacteria and fungi for REE recovery. A single microalgae study demonstrated 80% indium removal within 30 minutes from ITO etching wastewater with 0.14 mmol/g capacity, suggesting faster kinetics than fungal bioleaching but inferior regeneration stability (60.7% efficiency after 5 cycles versus 85% after 9 cycles for engineered bacteria). Economic data remain extremely limited, with bioleaching costs estimated at $576/kg Ga compared to $1321/kg for acid leaching, but no microalgae cost data exist. Most studies were conducted at laboratory scale (mL to L volumes), with only three pilot-scale demonstrations reported and no industrial applications for semiconductor waste specifically identified. Critical gaps include absence of fluoride and boron tolerance data, limited compound semiconductor waste studies, and no economic comparisons between organism types.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the quantitative bio-leaching and biosorption capabilities of bacterial and fungal systems for recovering critical raw materials (Copper, Cerium, Gallium, Germanium, Indium, Rare Earth Elements) from semiconductor fabrication waste, specifically comparing their performance against microalgae-based systems in terms of recovery rates (%), cost ($/m³ or $/kg), and tolerance to toxic semiconductor processing chemicals?

Target organisms: • Bacteria: Acidithiobacillus ferrooxidans, Acidithiobacillus thiooxidans, Leptospirillum ferrooxidans, Pseudomonas species, Bacillus species, Cupriavidus metallidurans • Fungi: Aspergillus niger, Aspergillus flavus, Penicillium chrysogenum, Penicillium simplicissimum, Rhizopus arrhizus, Trichoderma species

Waste stream focus: • CMP wastewater (copper-cerium-silica slurry, pH 2-4) • Wet etching waste (HF-based, metal residues) • Compound semiconductor waste (GaAs, GaN, InP processing residues)

Critical comparison metrics needed:

1. Recovery efficiency (%) - direct comparison to microalgae where available
2. Bio-leaching mechanism: organic acid production (citric, oxalic, gluconic), siderophore secretion, enzymatic reduction
3. Tolerance to semiconductor toxins: fluoride, arsenic, boron, organic solvents
4. Growth rate and biomass productivity compared to microalgae
5. OPEX comparison ($/m³) vs microalgae systems
6. Metal selectivity (target metal/interfering ion ratios)

7. Scalability: demonstrated scale (L) and footprint requirements (m^2)
8. Kinetics: bio-leaching time to target recovery (hours vs days)

Emphasize: • Studies that DIRECTLY COMPARE bacteria/fungi to microalgae performance • Economic feasibility data (CAPEX/OPEX) • Industrial pilot demonstrations (not just lab-scale) • Patents or commercial deployments

Timeframe: 2019-2026. Include landmark studies from 2015-2018 if they report pilot-scale or comparative data."

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Target Organisms**: Does the study investigate at least one of the specified bacterial species (Acidithiobacillus ferrooxidans, A. thiooxidans, Leptospirillum ferrooxidans, Pseudomonas spp., Bacillus spp., Cupriavidus metallidurans) or fungal species (Aspergillus niger, A. flavus, Penicillium chrysogenum, P. simplicissimum, Rhizopus arrhizus, Trichoderma spp.) for bio-leaching or biosorption applications?
- **Critical Raw Materials**: Does the study target recovery of at least one of the following critical raw materials: copper, cerium, gallium, germanium, indium, or rare earth elements?
- **Semiconductor Waste Context**: Does the study use semiconductor-related waste streams (such as CMP wastewater, wet etching waste, compound semiconductor processing residues, or similar semiconductor industrial waste) as source material?
- **Quantitative Performance Data**: Does the study report at least one empirical performance metric such as recovery efficiency (%), bio-leaching kinetics, growth rates, tolerance measurements, or economic data (CAPEX/OPEX, cost per unit)?
- **Practical Scale**: Is the study conducted at bench-scale (≥1L), pilot-scale, or industrial scale?
- **Complete Empirical Study**: Is this an experimental study, pilot demonstration, case study, or systematic review that reports complete empirical results with sufficient methodological detail (i.e., NOT a conference abstract, preliminary report, or purely theoretical/computational study)?
- **Semiconductor Application Focus**: Is the study focused on semiconductor waste processing rather than exclusively on mining ore, municipal waste, or other non-semiconductor applications?
- **Applied Performance Focus**: Does the study go beyond micro-scale only experiments (<100mL with synthetic media only) and beyond purely genetic/molecular analysis, to include applied performance data relevant to industrial applications?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Organism Systems**:

  Extract all microorganisms studied for critical raw material recovery from semiconductor waste, including:

- Bacterial species (especially Acidithiobacillus ferrooxidans, A. thiooxidans, Leptospirillum ferrooxidans, Pseudomonas, Bacillus, Cupriavidus metallidurans)

- Fungal species (especially Aspergillus niger, A. flavus, Penicillium chrysogenum, P. simplicissimum, Rhizopus arrhizus, Trichoderma)
- Microalgae species used for comparison
- Whether pure cultures, mixed cultures, or consortia were used
- Living vs dead biomass applications

- **Target Materials**:

  Extract all critical raw materials targeted for recovery, specifically:

- Copper (Cu), Cerium (Ce), Gallium (Ga), Germanium (Ge), Indium (In)
- Rare Earth Elements (specify which REEs: La, Nd, Dy, Y, etc.)
- Any other metals recovered as co-products or interfering metals
- Initial concentrations in waste streams (mg/L or g/kg)
- Target vs actual materials recovered

- **Semiconductor Waste**:

  Extract details about semiconductor fabrication waste streams processed, including:

- CMP wastewater (copper-cerium-silica slurry, pH levels)
- Wet etching waste (HF-based, metal residue composition)
- Compound semiconductor waste (GaAs, GaN, InP processing residues)
- Electronic waste/PCBs if relevant to semiconductor applications
- Waste composition, pH, toxic chemical content
- Waste volume processed (L, m^3, kg)

- **Bio-processing Method**:

  Extract bio-processing approach and mechanisms used:

- Bioleaching vs biosorption vs bioaccumulation
- Organic acid production (citric, oxalic, gluconic acids - specify types and concentrations)
- Siderophore secretion and enzymatic reduction mechanisms
- Contact time, pH conditions, temperature
- Biomass loading or cell density
- Process configuration (batch, continuous, packed bed, biofilm)

- **Recovery Performance**:

  Extract quantitative recovery performance data for critical raw materials from semiconductor waste:

- Recovery efficiency (% for each target metal)
- Metal selectivity ratios (target metal/interfering ion)
- Recovery kinetics (time to reach target recovery: hours vs days)
- Biomass productivity and growth rates
- Maximum adsorption/leaching capacity (mg/g biomass)
- Process stability and repeatability data

- **Microalgae Comparison**:

  Extract any DIRECT COMPARISONS between bacteria/fungi and microalgae systems for semiconductor waste processing:

- Head-to-head recovery efficiency comparisons (% recovery rates)
- Growth rate and biomass productivity comparisons
- Cost comparisons ($/m³ or $/kg processed)
- Processing time comparisons
- Tolerance comparisons to semiconductor toxins
- Any superiority claims with supporting data
- Note if no microalgae comparison was made

- **Economic Analysis**:

  Extract economic feasibility data for bio-processing of semiconductor waste:

- Operating costs (OPEX in $/m³ or $/kg)
- Capital costs (CAPEX) where available
- Cost comparisons to conventional methods (ion exchange, chemical precipitation)
- Cost breakdown (nutrients, energy, labor, equipment)
- Economic comparison to microalgae systems where available
- Commercial viability assessments

- **Toxin Tolerance**:

  Extract data on organism tolerance to semiconductor processing chemicals:

- Fluoride tolerance levels and impacts on recovery
- Arsenic tolerance (especially for compound semiconductor waste)
- Boron tolerance and effects
- Organic solvent tolerance
- pH tolerance range (especially acidic conditions)
- Heavy metal tolerance beyond target materials
- Growth inhibition or death thresholds
- Adaptation or acclimatization capabilities

- **Study Scale**:

  Extract the scale and development stage of bio-processing studies:

- Laboratory scale (mL, L volumes)
- Pilot scale demonstrations (scale in L, flow rates)
- Industrial/commercial scale applications
- Footprint requirements (m² per m³/day capacity)
- Scalability assessments and challenges identified
- Patents or commercial deployments mentioned
- Scale-up data or projections provided

Report

Due to the limitations of the AI model, we are only able to process 80 sources while writing a report. This report was written using the 80 sources that had the highest screening scores out of the 262 sources that we screened in and extracted data from.

Characteristics of Included Studies

This systematic review synthesized evidence from 80 sources examining bacterial and fungal bio-processing systems for critical raw material recovery. The literature predominantly focuses on electronic waste (printed circuit boards) rather than semiconductor fabrication waste specifically, with only a subset addressing CMP wastewater or compound semiconductor processing residues.

Study	Full text retrieved?	Study Type	Waste Stream	Primary Target Metal(s)	Organism Type
Xiaoyu Wang et al., 2021	Yes	Primary study	CMP wastewater	Copper	Bacteria (Cupriavidus gilardii)
A. Becci et al., 2020	No	Primary study	PCBs	Cu, Zn	Bacteria (A. ferrooxidans, L. ferrooxidans)
Fatemeh Pourhossein et al., 2019	No	Primary study	WLEDs	Cu, Ni, Ga	Bacteria (A. ferrooxidans)
Mehdi Golzar-Ahmadi et al., 2021	No	Primary study	AMOLED screens	Ag, Mo, Cu	Bacteria (Bacillus foraminis)
Jiaying Cui et al., 2020	No	Primary study	LCD panels	Indium	Fungi (A. niger)
Eva Benzal Montes et al., 2019	No	Primary study	E-waste	Copper	Bacteria (A. ferrooxidans)
A. Priya et al., 2020	No	Primary study	PCBs	Cu, Zn, Ni, Pb, REEs	Bacteria (A. ferrooxidans, A. acidophilum)
Leah C. Stanley et al., 2003	No	Primary study	CMP wastewater	Copper	Bacteria (Staphylococcus sp.)
Eva Pakostova et al., 2023	Yes	Primary study	Electrodeposited components	Cu, Ag	Bacteria (A. ferrooxidans, consortium)
A. Sikander et al., 2023	No	Primary study	Tantalum capacitors	Ta, Mn	Bacteria/Fungi (P. putida, B. subtilis, P. simplicissimum)
A. Ruiz et al., 2004	No	Primary study	CMP wastewater	Copper	Bacteria (unspecified)
Fatemeh Pourhossein et al., 2021	No	Primary study	OLED touch screens	In, Sr	Bacteria (adapted acidophiles)

Study	Full text retrieved?	Study Type	Waste Stream	Primary Target Metal(s)	Organism Type
Musa D. Maluleke et al., 2024	No	Primary study	PCBs	Copper	Bacteria (L. ferriphilum, At. caldus)
Solange Kazue Utimura et al., 2019	No	Primary study	PCBs	Copper	Bacteria (A. ferrooxidans)
Zahra Ilkhani et al., 2025	No	Primary study	PCBs	Copper	Bacteria (A. ferrooxidans)
Andra Constantin et al., 2024	No	Primary study	LCD panels	Indium	Bacteria (acidophilic consortium)
Gamez Grijalva et al., 2009	No	Primary study	CMP wastewater	Copper	Bacteria (sulfate reducers)
M. Chakankar et al., 2017	No	Primary study	Solar cells	Cu, Te, multiple metals	Bacteria (T. ferrooxidans)
Katja Götze et al., 2021	Yes	Primary study	Bioleaching solutions	In, Ge	Bacteria (Leptospirillum, Acidithiobacillus)
F. Faraji et al., 2018	No	Primary study	PCBs	Metals (unspecified)	Fungi (A. niger)
Aaron P. Mosier et al., 2015	No	Primary study	CMP wastewater	Copper	Bacteria/Fungi (L. casei, P. pastoris)
Aaron P. Mosier et al., 2014	No	Primary study	CMP wastewater	Copper	Bacteria/Fungi (L. casei, P. pastoris)
Moon-Sung Choi et al., 2004	No	Primary study	PCBs	Copper	Bacteria (A. ferrooxidans)
Amber Trivedi et al., 2023	No	Primary study	PCBs (mobile phones)	Cu, Ni, Zn	Fungi (A. niger, A. tubingensis)
Andrea Leonardi et al., 2015	No	Primary study	CMP wastewater	Copper	Bacteria/Fungi (L. casei, P. pastoris)
Sadia Ilyasl et al., 2010	No	Primary study	Electronic scrap	Cu, Ni, Al, Zn	Bacteria (S. thermosulfidooxidans, T. acidophilum)
N. Saitoh et al., 2017	No	Primary study	LSI waste, LCD panels	Au, Pt, Pd, Rh, In	Bacteria (Shewanella)
Benjamin Monneron et al., 2020	No	Primary study	PCBs	Gallium	Bacteria (unspecified)

Study	Full text retrieved?	Study Type	Waste Stream	Primary Target Metal(s)	Organism Type
Zhifang Lv et al., 2025	No	Primary study	Semiconductor waste	Gallium	Bio-based adsorbent (chitosan)
Cameron A. Lippert et al., 2025	No	Primary study	CMP wastewater	Copper	Electrochemical (non-biological)
Abhilash et al., 2021	Yes	Primary study	PCBs	Cu, Ni, Co, Fe, Al	Bacteria/Fungi (A. ferrooxidans, A. thiooxidans, A. niger)
Jiaying Cui et al., 2021	No	Primary study	LCD panels	Indium	Fungi (A. niger)
H. Sharada et al., 2021	Yes	Primary study	PCBs	Copper	Fungi (A. niger)
Xiafei Yin et al., 2023	Yes	Primary study	PCBs	Copper	Fungi (C. fodinicola, T. barcinensis)
Santhosh Krishnamoorthy et al., 2021	Yes	Primary study	PCBs	Cu, Zn, Ni	Fungi (A. niveus)
Muhammad Sya q Razali et al., 2021	No	Primary study	PCBs	Copper	Bacteria (Bacillus sp., Oryzobacter sp.)
A. Breuker et al., 2020	Yes	Primary study	Bioleaching solutions	REEs (La-Lu)	Bacteria/Fungi/Algae (multiple species)
F. Vakilchap et al., 2024	No	Primary study	PCBs (mobile phones)	Mn, Pt, Pd, Fe, Co, Al	Bacteria/Fungi (B. megaterium, A. niger)
D. Johnson et al., 2014	Yes	Review	Various ores and wastes	Cu, Co, Ni, Zn, Au	Bacteria (acidophiles)
M. Chakankar et al., 2019	No	Primary study	Solar cells	Te and multiple metals	Bacteria/Fungi (A. thiooxidans, Penicillium spp.)
Jorge Enrique Madrigal-Arias et al., 2015	Yes	Primary study	PCBs, GFICMs	Au, Cu, Ni	Fungi (A. niger strains)
Nathan M. Good et al., 2023	Yes	Primary study	Electronic waste	REEs (Nd, Pr, Dy)	Bacteria (M. extorquens AM1)
Yoonjae Lee et al., 2024	No	Primary study	Semiconductor waste	Tungsten	Microorganisms (unspecified)

Study	Full text retrieved?	Study Type	Waste Stream	Primary Target Metal(s)	Organism Type
A. Chung et al., 2022	Yes	Primary study	GaAs, GaN	Gallium	Bacteria (Rhodanobacter, Sphingomonas, Arthrobacter)
M. Díaz-Martínez et al., 2019	No	Primary study	PCBs (mobile phones)	Ag, Au, Cu	Bacteria/Fungi (Sphingomonas, A. niger, C. orthopsilosis)
Alejandra Gonzalez Baez et al., 2024	No	Primary study	PCBs	REEs (La, Tb, Pr, Nd, Gd)	Fungi (P. expansum)
Xiaohan Wu et al., 2025	No	Techno-economic analysis	LEDs	Gallium	Microorganisms (unspecified)
A. Sikander et al., 2022	Yes	Primary study	PCBs, tantalum capacitors	Cu, Zn, Fe, Al, Ni, Mn	Bacteria/Fungi (acidophiles, A. niger)
A. Brewer et al., 2019	Yes	Primary study	NdFeB magnet leachate	REEs (Nd, Pr, Dy)	Bacteria (E. coli, engineered)
Motasem N. Saidan et al., 2011	No	Primary study	Electronic waste	Copper	Bacteria/Fungi (A. thiooxidans, A. niger)
F. Patel et al., 2021	No	Primary study	PCBs (mobile phones)	Cu, Ni	Fungi (A. fumigatus)
M. Hofmann et al., 2021	Yes	Primary study	Aqueous solutions	V, Ga, In	Bacteria (Nocardioides, Pseudomonas, Rhodococcus)
Pradnya Chandane et al., 2019	No	Primary study	PCBs	Copper	Bacteria (A. acidophilum)
Jae-Woo Ahn et al., 2005	No	Primary study	Electronic scrap	Cu, Zn, Al, Co, Ni, Fe	Fungi (A. niger)
Jingqi Liu et al., 2025	Yes	Primary study	REE solutions	Cerium	Fungi (A. niger FH1)
M. Arshadi et al., 2020	No	Primary study	PCBs	Cu, Ni	Fungi (A. niger)
B. Hemambika et al., 2011	No	Primary study	Electroplating effluent	Cu, Cd, Pb	Fungi (Aspergillus, Penicillium, Cephalospo-rium)
Amber Trivedi et al., 2019	No	Primary study	PCBs	Cu, Ni, Zn	Fungi (A. niger, A. tubingensis)

Study	Full text retrieved?	Study Type	Waste Stream	Primary Target Metal(s)	Organism Type
C. Erust et al., 2020	No	Primary study	PCBs (mobile phones)	Cu, Al, Ni, Zn	Bacteria (meso-acidophilic consortium)
P. Kaur et al., 2022	Yes	Primary study	PCBs	Cu, Fe	Bacteria/Fungi (Pseudomonas, Pleurotus florida)
Xia Kang et al., 2022	No	Primary study	REE solutions	Cerium	Fungi (A. niger, N. crassa)
Jonovan Van Yken et al., 2022	No	Primary study	PCBs	Cu, Ni, Al, Zn	Bacteria (At. caldus, At. ferrooxidans)
H. Hocheng et al., 2014	No	Primary study	Waste solder	Sn, Ag, Pb	Bacteria/Fungi (A. ferrooxidans, A. niger)
Juan Tapia et al., 2022	Yes	Primary study	PCBs	Cu, Zn, Sn, Au, Ag	Bacteria (Tissierella, Acidiphilium, Leptospirillum)
Pooja Thakur et al., 2023	No	Primary study	PCBs	Cu, Au, Ag	Bacteria (Bacillus sp. ISO1)
É. F. Rodrigues et al., 2024	Yes	Primary study	PCBs	Cu, Au	Fungi (A. niger)
David Kaplan et al., 2023	Yes	Primary study	E-waste	PGMs, REEs (Tb, Pd)	Bacteria (Shewanella loihica)
Hannah S. Zurier et al., 2025	Yes	Primary study	REE solutions	REEs (La, Ce, Nd, Ho)	Bacteria (A. ferrooxidans, E. coli)
Arrate Santaolalla Ramirez et al., 2020	Yes	Primary study	E-waste	Copper	Bacteria (A. ferrooxidans)
Muhammad Sya et al., 2021	No	Primary study	PCBs	Copper	Bacteria (Bacillus sp., Oryzobacter sp.)
Angelina Sofia Martinez et al., 2025	No	Primary study	CMP wastewater	Copper	Non-biological (electrochemi-cal)
Rebecca M. Brown et al., 2025	Yes	Techno-economic analysis	REE-containing waste	REEs (multiple)	Bacteria/Fungi (G. oxydans, A. niger)

Study	Full text retrieved?	Study Type	Waste Stream	Primary Target Metal(s)	Organism Type
A. Vardanyan et al., 2019	No	Primary study	PCBs	Copper	Bacteria (A. ferrooxidans)
V. Blažek et al., 2018	No	Primary study	Electronic waste dust	Cu, Al, Sn, Ag	Bacteria (A. ferrooxidans)
Izabella Vicentin Moreira et al., 2019	No	Primary study	PCBs	Cu, Ag, Sn	Bacteria (A. ferrooxidans)
I. Tezyapar Kara et al., 2023	Yes	Review	Various wastes	Cu, Co, In, Al, Mo, Ni, Zn	Multiple organisms
N. R. Nicomel et al., 2020	Yes	Primary study	ITO etching wastewater	Indium	Microalgae (Chlorella, Nitzschia, Navicula)
M. Buțu et al., 2016	No	Review	Various wastes	Critical metals (unspecified)	Microorganisms (unspecified)
Vanessa L. Brisson et al., 2016	Yes	Primary study	Monazite sand	REEs (La-Lu)	Fungi (A. niger, A. terreus, Paecilomyces)
K. Bosecker et al., 1997	Yes	Review	Various ores	Cu, U, Au, Zn, Ni, Co	Bacteria (Thiobacillus spp.)

The included studies span from 1997 to 2025, with the majority published between 2019 and 2025. Of the 80 sources, only 8 directly address semiconductor fabrication waste streams (CMP wastewater), while the majority focus on electronic waste such as printed circuit boards. Only one study (Breuker et al., 2020) provides direct comparative data between bacterial/fungal systems and microalgae, and one study (Nicomel et al., 2020) focuses exclusively on microalgae for indium biosorption from ITO etching wastewater.

Organism Systems and Bio-processing Approaches

Bacterial Systems

The predominant bacterial species investigated for critical raw material recovery include chemolithotrophic acidophiles capable of oxidizing iron and sulfur compounds. *Acidithiobacillus ferrooxidans* was the most frequently studied organism, appearing in over 25 studies as either pure cultures or components of mixed consortia. Other acidophilic bacteria including *Leptospirillum ferrooxidans*, *Acidithiobacillus thiooxidans*, and *Acidithiobacillus caldus* were also extensively characterized.

Non-acidophilic bacteria demonstrated utility for specific applications. *Bacillus* species showed promise for cyanide-mediated precious metal leaching, while *Cupriavidus gilardii* CR3 proved effective for copper biosorption from CMP wastewater specifically, achieving 95.2% removal efficiency. *Shewanella* bacteria were uniquely capable of selectively collecting platinum group metals and gold through metal ion reduction.

Bacterial Species	Bio-processing Mechanism	Primary Application	Max Recovery Reported
Acidithiobacillus ferrooxidans	Ferric iron oxidation, bioleaching	Cu, Zn from PCBs	100% Cu, 94.7% Cu
Leptospirillum ferrooxidans	Iron oxidation	Cu from electrodeposited components	99.2% Cu (consortium)
Cupriavidus gilardii CR3	Biosorption	Cu from CMP wastewater	95.2%
Bacillus foraminis	Organic acid production	Ag, Mo, Cu from AMOLED	100% Ag, 56.8% Mo
Shewanella loihica	Bio-reduction	PGMs, REEs	~90% Pd
Methylobacterium extorquens AM1	Bioaccumulation	REEs from e-waste	65-100% Nd

Fungal Systems

Aspergillus niger emerged as the most comprehensively studied fungal species, with documented applications across diverse waste streams. The primary bioleaching mechanism involves organic acid production, predominantly citric, oxalic, and gluconic acids.

Fungal bioleaching offered distinct advantages in tolerating higher pH conditions compared to acidophilic bacteria. *Aspergillus niger* showed optimal bioleaching at pH 5.15-5.7, whereas acidophilic bacteria require pH values below 2.5. However, recovery kinetics for fungi were generally slower, typically requiring 5-33 days compared to 8-18 days for bacterial systems.

Fungal Species	Organic Acids Produced	Primary Application	Max Recovery Reported
Aspergillus niger	Citric, oxalic, gluconic	Cu, Zn, Ni from PCBs	100% Cu, 97% Cu
Aspergillus tubingensis	Citric, oxalic, malic	Cu, Zn from mobile PCBs	86% Cu (mixed)
Aspergillus niveus	Citric, itaconic, oxalic	Cu, Zn, Ni from PCBs	80.25% Cu
Penicillium chrysogenum	Multiple organic acids	Te from solar cells	100% Te
Penicillium expansum	Organic acids	REEs from PCBs	~70% REEs
Penicillium simplicissimum	Multiple organic acids	Mn from Ta capacitors	98% Mn

Pure Versus Mixed Cultures

Mixed cultures and consortia consistently outperformed pure cultures in bioleaching efficiency. A mixed strain of *A. ferrooxidans* and *A. acidophilum* achieved 96% copper recovery compared to lower rates with individual strains. Similarly, mixed cultures of *A. niger* and *A. tubingensis* reached 86% copper extraction versus 71% with *A. niger* alone. The synergistic effect was attributed to complementary metabolic activities, including enhanced exopolymeric substance (EPS) production that facilitates metal solubilization.

Adapted cultures demonstrated substantially improved performance over non-adapted strains. Copper-adapted immobilized cells exhibited ferrous iron oxidation rates of 0.0622 g $Fe^{2+} \cdot L^{-1} \cdot h^{-1}$ compared to 0.0359 g $Fe^{2+} \cdot L^{-1} \cdot h^{-1}$ for non-adapted planktonic cells. Adaptation to elevated metal concentrations also expanded tolerance ranges; moderately thermophilic bacteria improved their mixed metal ion tolerance from 12 g/L to 20 g/L over a two-year adaptation period.

Target Critical Raw Materials

Copper Recovery

Copper was the most extensively studied target metal, consistent with its high concentration (10-30%) in electronic waste and its presence in CMP wastewater. Recovery efficiencies for copper bioleaching from various waste streams are summarized below:

Study	Waste Type	Organism	Recovery (%)	Time (days)	Pulp Density
Solange Kazue Utimura et al., 2019	PCBs	A. ferrooxidans	100%	8	Not specified
H. Sharada et al., 2021	PCBs	A. niger	~100%	5	0.5%
M. Arshadi et al., 2020	PCBs	A. niger	97%	4.5	10 g/L
A. Priya et al., 2020	PCBs	Mixed acidophiles	96%	18	7.5 g/L
Xiaoyu Wang et al., 2021	CMP wastewater	C. gilardii CR3	95.2%	Not specified	Not specified
A. Becci et al., 2020	PCBs	At. ferrooxidans	94%	9	5% (w/v)
Abhilash et al., 2021	PCBs	Mixed bacteria	93%	10	8%
C. Erust et al., 2020	Mobile PCBs	Mixed acidophiles	98.1%	8	10%
Van Yken et al., 2022	PCBs	At. ferrooxidans	97.9%	Not specified	1%
Amber Trivedi et al., 2023	Mobile PCBs	Mixed Aspergillus	86%	Not specified	3 g/L

The highest recovery rates (>95%) were achieved with acidophilic bacterial systems, though fungal systems approached comparable efficiency when process parameters were optimized. Two-step bioleaching approaches, where biogenic ferric iron is generated separately before contact with waste, consistently improved recovery rates over direct bioleaching.

Indium Recovery

Indium recovery was addressed by 8 studies, primarily from LCD panels and OLED touch screens. Both bacterial and fungal systems demonstrated high indium bioleaching efficiency:

Study	Waste Source	Organism	Recovery (%)	Mechanism
Jiaying Cui et al., 2020	LCD panels	A. niger	100%	Oxalic acid production

Study	Waste Source	Organism	Recovery (%)	Mechanism
Andra Constantin et al., 2024	P-LCDs	Acidophilic consortium	100%	Mixed sulfur-iron pathway
Andra Constantin et al., 2024	Sh-LCDs	Acidophilic consortium	60%	Mixed sulfur-iron pathway
Katja Götze et al., 2021	Bioleaching solutions	Membrane separation	91.0% retention	Nanofiltration
N. R. Nicomel et al., 2020	ITO etching wastewater	Microalgae	80% (30 min)	Biosorption

The microalgae study by Nicomel et al. (2020) demonstrated rapid indium uptake kinetics, achieving 80% removal within 30 minutes, which represents faster kinetics than fungal bioleaching systems requiring 90 minutes at elevated temperature (70℃). The maximum biosorption capacity for microalgae was 0.14 mmol/g, though the study noted decreased efficiency from 91.3% to 60.7% after five regeneration cycles.

Gallium Recovery

Gallium recovery from compound semiconductor waste (GaAs, GaN) was examined in 5 studies. The study by Chung et al. (2022) provided the most comprehensive mechanistic analysis:

Bacterial Strain	Ga Source	Recovery (%)	As Tolerance (MIC)
Rhodanobacter sp. B2A1Ga4	GaAs	56%	7.5 mM As(III)
Sphingomonas sp. A2-49	GaAs	Moderate	0.5 mM As(III)
Arthrobacter silviterrae A2-55	GaAs (spent medium)	Highest	0.5 mM As(III)
Fatemeh Pourhossein et al., 2019	WLEDs	84%	Not reported

Arsenic tolerance emerged as a critical factor for GaAs bioleaching success. Strains with higher arsenic resistance demonstrated superior gallium extraction efficiency. Cell-free spent medium bioleaching sometimes outperformed direct contact methods, suggesting that metabolic by-products rather than direct cell contact drive gallium solubilization.

Rare Earth Element Recovery

REE recovery was investigated in 11 studies using diverse organism systems:

Study	REE Target	Organism	Recovery (%)	Mechanism
Nathan M. Good et al., 2023	Nd, Pr, Dy	M. extorquens AM1	65-100%	Bioaccumulation in polyphosphate granules

Study	REE Target	Organism	Recovery (%)	Mechanism
A. Brewer et al., 2019	Nd, Pr, Dy	Engineered E. coli	97% purity	Lanthanide binding tags
David Kaplan et al., 2023	Tb	Silk-SELP membranes	85%	Biosorption
Alejandra Gonzalez Baez et al., 2024	La, Tb, Pr, Nd, Gd	P. expansum	~70%	Organic acid production
Xia Kang et al., 2022	Ce	A. niger, N. crassa	>99%	Oxalate precipitation
Jingqi Liu et al., 2025	Ce	A. niger FH1	74.05%	Biosorption
Vanessa L. Brisson et al., 2016	Multiple REEs	Aspergillus spp.	3-5%	Organic acid bioleaching

The highest REE recovery efficiency (>99% Ce precipitation) was achieved using biomass-free spent culture media from *A. niger* and *N. crassa*, demonstrating that excreted organic acids and metabolites rather than live cells drive REE recovery. However, recovery from primary mineral sources (monazite sand) remained low at 3-5%, highlighting the importance of waste stream pre-treatment.

Engineered organisms showed particular promise for selective REE recovery. *E. coli* expressing lanthanide binding tags achieved 97% REE purity in the adsorbed fraction and maintained 85% adsorption capacity after nine regeneration cycles. The *Methylobacterium extorquens* AM1 platform demonstrated scalability to 10 L with consistent metal yields.

Cerium Recovery

Two fungal studies specifically addressed cerium biosorption:

Study	Organism	Mechanism	Max Capacity	Efficiency
Jingqi Liu et al., 2025	A. niger FH1	Physical entrapment + chemical binding	42.61 mg/g	74.05%
Xia Kang et al., 2022	A. niger spent medium	Oxalate precipitation	Not reported	>99%

The *A. niger* FH1 strain exhibited exceptional cerium tolerance up to 600 mg/L, with biphasic adsorption kinetics showing rapid uptake within 15 minutes followed by saturation at 60 minutes.

Bio-processing Methods and Mechanisms

Bioleaching Versus Biosorption

The reviewed literature employed two primary bio-processing approaches with distinct mechanisms and applications:

Parameter	Bioleaching	Biosorption
Primary mechanism	Acidolysis, redoxolysis, complexolysis	Physical adsorption, ion exchange, surface complexation
Typical organisms	Acidophilic bacteria, organic acid-producing fungi	Biofilm-forming bacteria, metal-binding fungi
Optimal pH	1.5-2.5 (bacteria), 5-7 (fungi)	2.4-6.0
Contact time	Days to weeks	Minutes to hours
Reversibility	Generally irreversible	Reversible with acid elution
Scalability demonstrated	Column and reactor scale	Packed bed columns

Bioleaching proved more effective for bulk metal extraction from electronic waste, while biosorption showed advantages for dilute metal recovery from wastewater streams. The biosorption approach with *C. gilardii* CR3 demonstrated particular utility for CMP wastewater treatment, achieving 95.2% copper removal from solutions containing nano-SiO_2 particles.

Organic Acid Production

Organic acid-mediated bioleaching was the predominant mechanism for fungal systems. Quantitative organic acid production data were available from several studies:

Organism	Citric Acid	Oxalic Acid	Gluconic Acid	Other Acids
A. niger (Trivedi et al.)	16.3 g/L	8.4 g/L	Not reported	Malic (0.5 g/L)
A. niveus	Present	12.75 mM	Not reported	Itaconic (22.35 mM)
B. foraminis	2.7 mM	Not reported	Not reported	Tartaric (12.1 mM), Lactic (78.5 mM), Propionic (69.6 mM)
A. niger/B. megaterium mixed	Not reported	1350-7460 mg/L	2683 mg/L	Not reported

Oxalic acid demonstrated excellent abilities for indium bioleaching from LCD panels, while citric acid was more effective for copper and gold extraction. The two-step bioleaching approach, where organic acids are produced in a separate fermentation step before waste contact, consistently improved metal recovery and reduced process time.

Siderophore-Mediated Recovery

Siderophore-producing bacteria showed potential for selective critical metal recovery. Hofmann et al. (2021) screened 75 bacterial strains and identified high-yield siderophore producers capable of complexing V(III), Ga(III), and In(III). The siderophores were immobilized on solid-phase extraction columns at pH 2.0 for selective metal recovery.

Genome sequencing revealed specific biosynthetic gene clusters: achromobactin and pyoverdine in *Pseudomonas chlororaphis*, heterobactin and requichelin in *Rhodococcus erythropolis*, and desferrioxamine in *Nocardioides simplex*. These findings suggest opportunities for metabolic engineering to enhance siderophore production and metal selectivity.

Recovery Performance Comparison

Recovery Efficiency by Metal and Organism Type

The following table synthesizes recovery efficiency data across all studies for target critical raw materials:

Target Metal	Bacterial Systems	Fungal Systems	Best Performer
Copper	93-100%	66-100%	A. ferrooxidans (100%)
Indium	60-100%	100%	A. niger (100%)
Gallium	56-84%	Not reported	Acidophiles from WLEDs (84%)
Cerium	Not reported	74-99%	A. niger spent medium (>99%)
REEs (general)	65-100%	3-70%	M. extorquens AM1 (65-100% Nd)
Germanium	18.2% retention	Not reported	Membrane separation

Bacterial systems generally achieved higher recovery rates for base metals (Cu, Zn), while fungal systems showed comparable or superior performance for indium and cerium. The distinction between organism types became less pronounced when optimized process conditions were employed.

Recovery Kinetics

Recovery kinetics varied substantially between organism types and bio-processing approaches:

Approach	Metal	Time to Target Recovery	Reference
Bacterial bioleaching (direct)	Cu	8-18 days	Multiple
Bacterial bioleaching (two-step)	Cu	12 days (bioreactor)	Pakostova et al., 2023
Fungal bioleaching (two-step)	Cu	4.5 days	Arshadi et al., 2020
Fungal bioleaching (direct)	Cu	5-33 days	Multiple
Biosorption	Cu	Minutes to hours	Wang et al., 2021
Microalgae biosorption	In	30 minutes (80%)	Nicomel et al., 2020

The microalgae biosorption system demonstrated the fastest kinetics, achieving 80% indium removal within 30 minutes. However, this rapid uptake was specific to biosorption processes; bioleaching inherently requires longer contact times due to metabolic dissolution mechanisms.

Metal Selectivity

Metal selectivity data were limited but indicated that engineered biosorbents could achieve substantial selectivity:

System	Selectivity Metric	Value	Reference
Microalgae	In(III) over Cu(II), Zn(II), Al(III)	Selective	Nicomel et al., 2020
Membrane (NF99)	In/Ge ratio	11:1	Götze et al., 2021
Engineered E. coli	REE/non-REE	97% purity	Brewer et al., 2019
B. subtilis	Yb, Lu over other REEs	Higher selectivity	Breuker et al., 2020

Biosorption processes generally exhibited higher metal selectivity than bioleaching, which dissolves metals non-selectively through acidic or oxidative mechanisms.

Comparison with Microalgae Systems

Direct Comparative Studies

Of the 80 studies reviewed, only one study (Breuker et al., 2020) provided direct head-to-head comparison between bacterial, fungal, and algal systems for REE biosorption. This study found that algae exhibited low biosorption performance compared to bacteria and fungi.

A single study (Nicomel et al., 2020) focused exclusively on microalgae for indium recovery from ITO etching wastewater, representing the only microalgae-specific data for semiconductor waste processing. Key findings included:

Parameter	Microalgae Performance	Bacterial/Fungal Comparison
Maximum biosorption capacity	0.14 mmol/g (In)	42.61 mg/g (Ce, A. niger)
Recovery kinetics	80% in 30 minutes	74-100% in 5-15 days
pH optimum	2.4	5-7 (fungi), 1.5-2.5 (bacteria)
Regeneration cycles	5 cycles (91.3% to 60.7%)	9 cycles (85% retention)

The microalgae system demonstrated faster kinetics for biosorption applications but showed inferior regeneration stability compared to engineered bacterial biosorbents. The different pH optima suggest complementary applications: microalgae may be advantageous for acidic semiconductor waste streams where bacterial acidophiles would also thrive.

Absence of Comparative Data

The critical finding of this review is the near-complete absence of direct comparative studies between bacterial/fungal and microalgae systems for semiconductor waste processing. Of the 80 studies:

- 77 studies made no microalgae comparison
- 1 study reported microalgae had "low biosorption performance" without quantitative details
- 1 study focused exclusively on microalgae without bacterial/fungal comparison
- 1 study compared microalgae to non-biological methods only

This gap represents a significant limitation in the current evidence base and precludes definitive conclusions about the relative merits of bacterial/fungal versus microalgae systems for semiconductor waste bio-processing.

Tolerance to Semiconductor Processing Chemicals

Heavy Metal Tolerance

Acidophilic bacteria demonstrated exceptional tolerance to heavy metals beyond target materials:

Organism	Metal	Tolerance Level	Reference
Thiobacillus spp.	Ni	50 g/L	Bosecker, 1997
Thiobacillus spp.	Cu	55 g/L	Bosecker, 1997
Thiobacillus spp.	Zn	112 g/L	Bosecker, 1997
A. ferrooxidans	Cu	>100 mM (16 g/L), up to 22 g/L experimental	Pakostova et al., 2023
A. niger FH1	Ce	600 mg/L	Liu et al., 2025
Acidophilic consortium	Cu	Inhibitory at 33 g/L	Tapia et al., 2022

Adaptation significantly enhanced tolerance levels. Cu-adapted immobilized cells showed higher tolerance to Cu^{2+} stress and maintained oxidation kinetics at concentrations that inhibited non-adapted cells. Moderately thermophilic bacteria improved mixed metal ion tolerance from 12 g/L to 20 g/L through extended adaptation.

Arsenic Tolerance (Compound Semiconductor Waste)

Arsenic tolerance was critical for GaAs bioleaching applications:

Organism	As(III) MIC	Bioleaching Efficiency	Reference
Rhodanobacter sp. B2A1Ga4	7.5 mM	56% Ga from GaAs	Chung et al., 2022
Arthrobacter silviterrae A2-55	0.5 mM	Best with spent medium	Chung et al., 2022
Sphingomonas sp. A2-49	0.5 mM	Moderate	Chung et al., 2022

Strains with higher arsenic resistance demonstrated superior gallium extraction efficiency, indicating that arsenic tolerance is a key selection criterion for compound semiconductor waste processing.

pH Tolerance Range

The optimal pH ranges differed substantially between organism types:

Organism Type	Optimal pH Range	Notes
Acidophilic bacteria	1.5-2.5	Can be adapted to lower pH with acid addition
Fungi (A. niger)	5.15-7	Initial pH 7, drops to ~4.2 during bioleaching
Microalgae	~2.4 (optimal)	Effective biosorption at low pH

Organism Type	Optimal pH Range	Notes
A. ferrooxidans	pH 1.6	Tolerates saturating REE concentrations at this pH

Remarkably, *A. ferrooxidans* demonstrated resistance to REEs at saturating concentrations (50-100 mM at pH 1.6), which is unprecedented in the literature and suggests utility for REE biotechnology applications.

Toxicity Relative to Target Metals

Several studies characterized relative toxicity of metals encountered in electronic and semiconductor waste:

- Aluminum was more toxic than copper, which in turn was more toxic than nickel to *A. ferrooxidans*
- Manganese affected citric acid accumulation in *A. niger*, reducing bioleaching efficiency
- Silver, tin, and epoxy resin exhibited microbiocidal or microbiostatic effects on *A. ferrooxidans*-LR
- High concentrations of heavy metal ions inhibited *A. niger* growth, but two-step methods mitigated this effect

Economic Analysis

Operating Cost Data

Economic feasibility data were extremely limited, with only three studies providing quantitative cost estimates:

Study	Process	Cost ($/kg)	Notes
Xiaohan Wu et al., 2025	Bioleaching (Ga)	$576/kg Ga (ATC), $219/kg Ga (AVC)	More cost-effective than acid leaching ($1321/kg)
Xiaohan Wu et al., 2025	Primary production (Ga)	$368/kg Ga	Benchmark for competitiveness
Rebecca M. Brown et al., 2025	Bioleaching (REEs)	$9465/kg (gluconic acid), $4325/kg (citric acid)	Reagent cost dominates (67-79%)

The techno-economic analysis by Wu et al. (2025) identified reactor usage (57 reactors) as the primary cost driver for bioleaching, accounting for high capital requirements. Reducing reaction time through microbial strain development was identified as the most effective strategy for cost reduction.

Cost Breakdown

Brown et al. (2025) provided the only detailed cost breakdown:

Cost Component	Gluconic Acid Process	Citric Acid Process
Reagent cost	79%	67%
Energy cost	21%	33%

The sensitivity analysis revealed that substituting commercially purchased organic acids with microbially produced biolixiviant could reduce process costs by approximately 99% while minimally increasing energy consumption.

Economic Comparison to Conventional Methods

Qualitative assessments suggested economic advantages for bioleaching:

- Bioleaching requires less investment than conventional methods and is more sustainable
- Carbon footprint assessment showed 4-times reduction in CO_2-eq./kg treated PCB compared to best hydrometallurgical processes
- M. extorquens AM1 was described as competitive with current state-of-the-art chemical methodologies

No studies provided direct economic comparison between bacterial/fungal and microalgae systems.

Study Scale and Scalability

Scale Distribution

The vast majority of studies were conducted at laboratory scale:

Scale	Number of Studies	Volume/Capacity Range
Laboratory (mL)	~60	20-250 mL
Laboratory (L)	~15	1-5 L
Pilot scale	3	TRL6 membrane plant, 10 L bioreactor
Industrial scale	2 (reviews)	Copper and uranium operations

Pilot-Scale Demonstrations

Three studies reported pilot-scale or near-pilot demonstrations:

1. **Götze et al. (2021)**: Europe's first underground hybrid membrane pilot plant (TRL6) for indium and germanium extraction from bioleaching solutions, demonstrating long-term stability
2. **Nathan M. Good et al. (2023)**: Scalable bio-platform demonstrated at 10 L with consistent metal yields for REE recovery, without harsh acids or high temperatures
3. **Benzal Montes et al. (2019)**: Column reactor simulating industrial-scale conditions for copper bioleaching, operating in continuous mode

Scalability Challenges

Key challenges identified for scale-up included:

- Need for multiple reactors due to extended processing times
- Requirement for adapted microbial strains tolerant to high pulp density environments
- Shorter processing duration significantly decreases costs
- Low quantities of waste treatable per batch limits throughput

The two-step bioleaching design developed by Becci et al. (2020) addressed the solid concentration criticality by reaching 5% (w/v) PCB concentration, enabling reduced bioreactor size with consequent reduction in energy and raw material demand.

Industrial Applications

Industrial-scale biomining is established for copper and gold extraction from primary ores, with expected capacity of 250,000 tons of cathodic copper per year. However, no industrial-scale applications specifically for semiconductor fabrication waste were identified.

Synthesis

Explaining Heterogeneity in Recovery Efficiency

The substantial variation in copper recovery efficiency (66-100%) across studies can be explained by several factors:

1. Process Configuration Effects

Two-step bioleaching consistently outperformed one-step approaches. Indirect bioleaching using biogenic ferric reduced leaching time from 30 to 15 days while improving recovery. The separation of microbial growth from metal dissolution protected organisms from toxic metal concentrations and allowed optimization of each process step independently.

2. Organism Adaptation Status

Adapted cultures achieved substantially higher recovery rates than non-adapted strains. For example, adapted microbial communities demonstrated 50% indium leaching efficiency compared to 29% for non-adapted communities at high pulp density (32.5 g/L). The protective role of biofilm formed by adapted communities contributed to resilience against inhibitory substances.

3. Pulp Density Constraints

Higher pulp densities generally decreased recovery efficiency due to increased metal toxicity and mass transfer limitations. Studies achieving >95% copper recovery typically used pulp densities of 7.5-10 g/L or below, while those at higher densities (>15 g/L) showed reduced efficiency.

4. Particle Size Effects

Reduced particle size (60-80 mesh) increased organic acid synthesis and improved metal recovery compared to larger particles. This effect reflects improved surface area accessibility for bioleaching mechanisms.

Contextual Recommendations

Based on the synthesis of evidence, the following organism systems are recommended for specific applications:

Application Context	Recommended System	Rationale
CMP wastewater (Cu)	C. gilardii CR3 biosorption	95.2% recovery, tolerates nano-SiO_2
PCB copper (high volume)	Mixed acidophilic consortium	94-99% recovery, scalable
Indium from LCDs	A. niger bioleaching	100% recovery, oxalic acid mechanism
Gallium from GaAs	Rhodanobacter sp. B2A1Ga4	High As tolerance, 56% recovery
REE selective recovery	Engineered E. coli	97% purity, regenerable
Cerium biosorption	A. niger FH1	High Ce tolerance (600 mg/L)

Application Context	Recommended System	Rationale
Rapid indium removal	Microalgae	80% in 30 minutes, low pH compatible

Evidence Gaps

Critical gaps in the current evidence base include:

1. **Direct microalgae comparisons**: Only one study provided head-to-head data, precluding definitive conclusions about relative performance
2. **Economic feasibility**: Only three studies provided cost estimates, with no microalgae economic data available
3. **Semiconductor-specific waste streams**: Most studies addressed PCBs rather than CMP wastewater, wet etching waste, or compound semiconductor residues
4. **Industrial-scale validation**: No industrial demonstrations for semiconductor waste bio-processing were identified
5. **Toxin tolerance characterization**: Fluoride, boron, and organic solvent tolerance data were absent for semiconductor processing applications
6. **Long-term process stability**: Few studies addressed process repeatability beyond initial optimization

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Provide a comprehensive quantitative comparison of metal recovery rates (%), adsorption capacities (mg/g), equilibrium kinetics (time to 90% sorption), and selectivity ratios for biological systems (microalgae, bacteria, fungi) recovering critical raw materials from semiconductor wastewater and rare earth magnet manufacturing waste, with statistical aggregation across multiple studies to identify best-performing organism-metal-waste stream combinations. Target metals for comparison matrix: • Semiconductor-relevant: Copper (Cu), Cerium (Ce), Gallium (Ga), Germanium (Ge), Indium (In) • Magnet-relevant: Neodymium (Nd), Dysprosium (Dy), Praseodymium (Pr), Terbium (Tb), Samarium (Sm), Cobalt (Co) For EACH metal, extract and compare: BIOSORPTION PERFORMANCE: 1. Maximum adsorption capacity (q_max in mg metal / g biomass) • Report range (min-max) across studies • Identify organism genus achieving highest q_max • Note: dead vs living biomass comparison 2. Recovery efficiency (%) • Average and range across studies • Initial vs equilibrium concentration data • Organism achieving >95% recovery (if any) 3. Selectivity ratios • Target metal / competing ion (e.g., Ce/Ca, Nd/Ca, Ga/Al) • Report ratios >10:1 as "highly selective" • Organism-specific selectivity patterns KINETICS: 4. Time to equilibrium (minutes or hours) • Fast (<30 min), Medium (30-120 min), Slow (>2 hours) • Organism type (microalgae vs bacteria vs fungi) correlation with kinetics • Effect of biomass loading (g/L) on kinetics 5. Adsorption rate constants • Pseudo-first-order (k1) or pseudo-second-order (k2) values • Organisms achieving fastest kinetics per metal OPERATIONAL CONDITIONS: 6. Optimal pH range for maximum sorption • Identify pH "sweet spots" for each metal • Organisms tolerating extreme pH (<3 or >10) 7. Optimal temperature (℃) • Mesophilic (20-35℃) vs thermophilic (40-60℃) organisms • Energy implications for heating requirements 8. Biomass dosage (g/L) • Minimum effective dosage for >90% recovery • Biomass requirement per m^3 wastewater treated ORGANISM-SPECIFIC PATTERNS: 9. Best microalgae genus per metal (based on q_max and recovery %) 10. Best bacteria genus per metal 11. Best fungi genus per metal 12. Mixed consortia advantages (if any): synergistic effects, co-metabolism ECONOMIC PROXY METRICS: 13. Biomass productivity (g/L/day) - lower is better for cost 14. Regeneration cycles demonstrated (number of reuses before performance drops <80%) 15. Estimated OPEX (€/m^3) based on biomass requirement + regeneration frequency STATISTICAL AGGREGATION REQUESTED: • For each metal-organism combination with ≥3 studies, report: mean, median, standard deviation, min, max • Identify "champion organisms" for each metal (highest q_max + highest recovery % + fastest kinetics) • Highlight studies reporting SIMULTANEOUS recovery of multiple metals from mixed waste

recovery in <30 minutes), Exiguobacterium sp. SH31 for simultaneous recovery of six REEs from e-waste (87-99% efficiency), and Serratia plymuthica for copper (151 mg/g), though comprehensive statistical aggregation is constrained by heterogeneous experimental conditions, limited pilot-scale validation, and critical data gaps for germanium and gallium that preclude definitive universal recommendations across all target metals.

Abstract

Biological systems demonstrate significant potential for recovering critical raw materials from semiconductor wastewater and rare earth magnet manufacturing waste, with maximum adsorption capacities (q_max) varying substantially by metal-organism combination. For magnet-relevant metals, Methylobacterium extorquens AM1 achieved the highest Nd capacity at 202 mg/g with 65-100% recovery efficiency and demonstrated scalability to 10 L without harsh chemicals, while Chlorella vulgaris reached 188.68 mg/g Nd with fast equilibrium kinetics (<30 min). Exiguobacterium sp. SH31 recovered multiple REEs simultaneously (Y, Pr, Nd, Gd, Tb, Dy) at 87-99% efficiency from e-waste, and biotic hydrous manganese oxides produced by Stagonospora sp. achieved 90% Co and >99% REY recovery from acid mine drainage. For semiconductor-relevant metals, Serratia plymuthica SC5II achieved 151 mg Cu/g from industrial wastewater, while microalgal biomass showed selective In(III) biosorption (0.14 mmol/g, 91% recovery) over competing Cu, Zn, and Al ions. Phosphorylated yeast cells demonstrated preferential binding of trivalent REEs over divalent heavy metals with ~7:1 selectivity, and engineered E. coli expressing lanthanide binding tags showed 2-10-fold enhanced REE distribution coefficients with systematic preference for smaller ionic radii.

Pseudo-second-order kinetics predominated across systems with equilibrium times ranging from 5 minutes (Nostoc sp. for Ce) to >18 days (biotic HMO for Co/REY). Optimal pH varied by metal: pH 5-6.9 for Nd, pH 3-8 for Dy, and pH 2.4 for In. Extremophilic Galdieria sulphuraria tolerated pH 1.5-2.5 but showed up to 24-fold improved biosorption at pH 5.0, indicating that extremophile tolerance does not directly translate to optimal biosorption performance. Regeneration was demonstrated for up to 9 cycles with 85% capacity retention using citrate desorption. Critical data gaps exist for Ge (no quantitative data identified) and Ga (maximum 47.4% recovery reported), while pilot-scale validation remains limited to only 2 of 80 studies. Statistical aggregation across studies was constrained by heterogeneous experimental conditions, though the synthesis identifies context-dependent champion organisms: mesophilic systems generally achieved higher specific uptake under optimized pH, while engineered bacteria and EPS-producing strains offered advantages for selective multi-metal recovery from complex waste matrices.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: ”Provide a comprehensive quantitative comparison of metal recovery rates (%), adsorption capacities (mg/g), equilibrium kinetics (time to 90% sorption), and selectivity ratios for biological systems (microalgae, bacteria, fungi) recovering critical raw materials from semiconductor wastewater and rare earth magnet manufacturing waste, with statistical aggregation across multiple studies to identify best-performing organism-metal-waste stream combinations.

Target metals for comparison matrix: • Semiconductor-relevant: Copper (Cu), Cerium (Ce), Gallium (Ga), Germanium (Ge), Indium (In) • Magnet-relevant: Neodymium (Nd), Dysprosium (Dy), Praseodymium (Pr), Terbium (Tb), Samarium (Sm), Cobalt (Co)

For EACH metal, extract and compare:

BIOSORPTION PERFORMANCE:

1. Maximum adsorption capacity (q_max in mg metal / g biomass) • Report range (min-max) across studies • Identify organism genus achieving highest q_max • Note: dead vs living biomass comparison
2. Recovery efficiency (%) • Average and range across studies • Initial vs equilibrium concentration data • Organism achieving >95% recovery (if any)
3. Selectivity ratios • Target metal / competing ion (e.g., Ce/Ca, Nd/Ca, Ga/Al) • Report ratios >10:1 as "highly selective" • Organism-specific selectivity patterns

KINETICS: 4. Time to equilibrium (minutes or hours) • Fast (<30 min), Medium (30-120 min), Slow (>2 hours) • Organism type (microalgae vs bacteria vs fungi) correlation with kinetics • Effect of biomass loading (g/L) on kinetics

5. Adsorption rate constants • Pseudo-first-order (k1) or pseudo-second-order (k2) values • Organisms achieving fastest kinetics per metal

OPERATIONAL CONDITIONS: 6. Optimal pH range for maximum sorption • Identify pH "sweet spots" for each metal • Organisms tolerating extreme pH (<3 or >10)

7. Optimal temperature (℃) • Mesophilic (20-35℃) vs thermophilic (40-60℃) organisms • Energy implications for heating requirements
8. Biomass dosage (g/L) • Minimum effective dosage for >90% recovery • Biomass requirement per m^3 wastewater treated

ORGANISM-SPECIFIC PATTERNS: 9. Best microalgae genus per metal (based on q_max and recovery %) 10. Best bacteria genus per metal 11. Best fungi genus per metal 12. Mixed consortia advantages (if any): synergistic effects, co-metabolism

ECONOMIC PROXY METRICS: 13. Biomass productivity (g/L/day) - lower is better for cost 14. Regeneration cycles demonstrated (number of reuses before performance drops <80%) 15. Estimated OPEX (€/m^3) based on biomass requirement + regeneration frequency

STATISTICAL AGGREGATION REQUESTED: • For each metal-organism combination with ≥3 studies, report: mean, median, standard deviation, min, max • Identify "champion organisms" for each metal (highest q_max + highest recovery % + fastest kinetics) • Highlight studies reporting SIMULTANEOUS recovery of multiple metals from mixed waste (e.g., Cu + Ce from CMP waste)

META-ANALYSIS QUESTIONS: • Is there a universal "best" microalga for critical metal recovery, or is specificity required per metal? • Do extremophiles (Galdieria, Cyanidium) outperform mesophilic algae (Chlorella, Spirulina) for acidic industrial waste? • What is the typical performance gap between laboratory batch experiments vs continuous pilot systems? • Are there economies of scale: does OPEX per m^3 decrease at pilot scale vs lab scale?

Study inclusion criteria: • Must report AT LEAST TWO of: q_max, recovery %, equilibrium time, selectivity ratio • Quantitative data required (numerical values, not "good recovery" qualitative statements) • Peer-reviewed papers + industry white papers + conference proceedings accepted • Timeframe: Prioritize 2019-2026; include 2015-2018 if meta-analysis or review papers

Output format request: • Comparison tables: Metal × Organism × Performance metric • Identify top 3 organisms per metal with citations • Flag "data gaps" where <3 studies exist for a metal-organism combination"

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Biological Systems**: Does the study use microalgae, bacteria, fungi, or mixed microbial consortia as biosorbents for metal recovery?
- **Target Metals**: Does the study investigate recovery of at least one of the following critical metals: Cu, Ce, Ga, Ge, In, Nd, Dy, Pr, Tb, Sm, or Co?
- **Quantitative Performance Data**: Does the study report numerical values for at least two of the following metrics: maximum adsorption capacity (qmax), recovery efficiency (%), equilibrium time, or selectivity ratios?
- **Biosorption Mechanisms**: Does the study involve biosorption, bioaccumulation, biorecovery, or bioremediation processes for metal removal?
- **Industrial Waste Relevance**: Does the study use semiconductor wastewater, rare earth magnet manufacturing waste, or synthetic solutions that mimic these industrial waste streams?
- **Biological Component Present**: Does the study include a biological component rather than using only chemical precipitation, ion exchange, membrane filtration, or other purely non-biological treatment methods?
- **Relevant Metal Focus**: Does the study focus on metals from the target list (Cu, Ce, Ga, Ge, In, Nd, Dy, Pr, Tb, Sm, Co) rather than exclusively studying other metals (e.g., lead, mercury, chromium, zinc)?
- **Quantitative Data Reporting**: Does the study report numerical performance metrics rather than only qualitative descriptions (such as "good recovery" or "effective removal")?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

427 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Biological Systems**: Does the study use microalgae, bacteria, fungi, or mixed microbial consortia as biosorbents for metal recovery?
- **Target Metals**: Does the study investigate recovery of at least one of the following critical metals: Cu, Ce, Ga, Ge, In, Nd, Dy, Pr, Tb, Sm, or Co?
- **Quantitative Performance Data**: Does the study report numerical values for at least two of the following metrics: maximum adsorption capacity (qmax), recovery efficiency (%), equilibrium time, or selectivity ratios?
- **Biosorption Mechanisms**: Does the study involve biosorption, bioaccumulation, biorecovery, or bioremediation processes for metal removal?
- **Industrial Waste Relevance**: Does the study use semiconductor wastewater, rare earth magnet manufacturing waste, or synthetic solutions that mimic these industrial waste streams?
- **Biological Component Present**: Does the study include a biological component rather than using only chemical precipitation, ion exchange, membrane filtration, or other purely non-biological treatment methods?
- **Relevant Metal Focus**: Does the study focus on metals from the target list (Cu, Ce, Ga, Ge, In, Nd, Dy, Pr, Tb, Sm, Co) rather than exclusively studying other metals (e.g., lead, mercury, chromium, zinc)?
- **Quantitative Data Reporting**: Does the study report numerical performance metrics rather than only qualitative descriptions (such as "good recovery" or "effective removal")?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

154 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Organism-Metal-Waste System**:

  Extract the specific biological system tested for critical metal recovery, including:

- Biological organism type (microalgae, bacteria, fungi) and genus/species name
- Target metals studied from the list: Cu, Ce, Ga, Ge, In, Nd, Dy, Pr, Tb, Sm, Co
- Waste stream type (semiconductor wastewater, rare earth magnet manufacturing waste, or laboratory synthetic solutions mimicking these)
- Biomass state (living vs dead/dried)
- Any biomass pretreatment or functionalization methods used
- Mixed consortia details if multiple organisms were used together

- **Biosorption Performance Metrics**:

  Extract all quantitative biosorption performance data for critical metal recovery, including:

- Maximum adsorption capacity (q_max in mg metal/g biomass) with confidence intervals if provided
- Recovery efficiency (%) - report both average and range across experiments
- Initial and equilibrium metal concentrations (mg/L)
- Selectivity ratios (target metal/competing ion) - note ratios >10:1 as highly selective
- Any comparative performance data between different organism-metal combinations
- Statistical measures (mean, median, standard deviation, min-max ranges) when multiple experiments reported

- **Kinetics and Equilibrium**:

  Extract kinetic parameters for metal biosorption relevant to industrial application, including:

- Time to reach equilibrium (report in minutes or hours)
- Time to 90% maximum sorption capacity
- Adsorption rate constants (pseudo-first-order k1 or pseudo-second-order k2 values)
- Kinetic model best-fit (first-order, second-order, intraparticle diffusion)
- Effect of biomass loading (g/L) on kinetic performance
- Classify kinetics as Fast (<30 min), Medium (30-120 min), or Slow (>2 hours)
- Any correlation noted between organism type and kinetic performance

- **Optimal Operating Conditions**:

  Extract operational parameters that maximize metal recovery for semiconductor and rare earth waste treatment, including:

- Optimal pH range for maximum sorption (note organisms tolerating pH <3 or >10)
- Optimal temperature (℃) and classify as mesophilic (20-35℃) or thermophilic (40-60℃)
- Biomass dosage (g/L) required for >90% metal recovery
- Contact time used in experiments
- Initial metal concentration ranges tested

- Any effects of competing ions or complex waste matrix on performance
- Agitation speed or mixing conditions if reported

- **Economic and Practical Metrics**:

  Extract data relevant to economic feasibility and industrial scalability of biological metal recovery, including:

- Biomass productivity (g/L/day) during cultivation
- Number of regeneration/reuse cycles demonstrated before performance drops below 80%
- Desorption efficiency (%) and methods used
- Estimated operational costs (€/m³ or $/m³) if provided
- Biomass requirement per m³ wastewater treated
- Energy requirements (heating, mixing) if reported
- Any pilot-scale or continuous system data vs laboratory batch results
- Economic comparison with conventional chemical recovery methods

- **Simultaneous Multi-Metal Recovery**:

  Extract data on simultaneous recovery of multiple critical metals from mixed waste streams, including:

- Which metal combinations were recovered together (e.g., Cu + Ce from CMP waste, Nd + Dy + Pr from magnet waste)
- Individual vs combined recovery efficiencies for each metal in mixed systems
- Competitive adsorption effects between target metals
- Selective recovery strategies for prioritizing high-value metals
- Any synergistic effects noted in multi-metal systems
- Sequential vs simultaneous recovery approaches tested

- **Champion Organism Performance**:

  Identify and extract data on the best-performing biological systems for each target metal, including:

- Organism achieving highest q_max for each metal (Cu, Ce, Ga, Ge, In, Nd, Dy, Pr, Tb, Sm, Co)
- Organism achieving highest recovery efficiency (%) for each metal
- Organism achieving fastest kinetics for each metal
- Any 'universal' organisms performing well across multiple metals
- Performance comparison between extremophiles (Galdieria, Cyanidium) vs mesophilic organisms (Chlorella, Spirulina) for acidic industrial waste
- Overall ranking or recommendation from authors for specific metal-waste combinations

- **Study Design and Scale**:

  Extract methodological details that affect data quality and industrial relevance, including:

- Study type (laboratory batch, laboratory continuous, pilot-scale, industrial demonstration)
- Sample size (number of replicate experiments)
- Duration of experiments or operation
- Use of real industrial waste vs synthetic solutions
- Analytical methods for metal quantification
- Statistical analysis methods used
- Any comparison with conventional chemical recovery methods as controls
- Scale-up challenges or performance gaps noted between lab and pilot scale

Report

Due to the limitations of the AI model, we are only able to process 80 sources while writing a report. This report was written using the 80 sources that had the highest screening scores out of the 154 sources that we screened in and extracted data from.

Characteristics of Included Studies

This systematic review synthesizes data from 80 sources examining biological metal recovery systems for critical raw materials from semiconductor wastewater and rare earth magnet manufacturing waste. The majority of studies employed laboratory batch experiments using synthetic solutions, with limited pilot-scale or industrial demonstration data available.

Study	Full text retrieved?	Study Type	Target Metals	Waste Stream	Organism Type
M. A. Kucuker et al., 2017	Yes	Primary study	Nd	Magnet waste	Microalgae (Chlorella vulgaris)
A. Sieber et al., 2025	Yes	Review	REEs, heavy metals	Various	Multiple organisms
A. Brewer et al., 2019	Yes	Primary study	Nd, Dy, Pr	Magnet leachates	Bacteria (E. coli)
Nathan M. Good et al., 2023	Yes	Primary study	Nd, Dy, Pr	Electronic waste	Bacteria (M. extorquens)
Nathan M. Good et al., 2023a	Yes	Primary study	Nd, Dy, Pr	Electronic waste	Bacteria (M. extorquens)
Óscar Barros et al., 2019	Yes	Primary study	La, Ce, Y, Tb, Pr, Eu	Synthetic solutions	Bacteria (B. cereus)
D. Park et al., 2017	Yes	Primary study	Ce, Pr, Nd, Y, Tb, Dy	Mine tailings	Bacteria (E. coli, C. crescentus)
Thainara Viana et al., 2024	Yes	Primary study	Y, Eu, Gd, Tb	Spent fluorescent lamps	Macroalgae (Ulva sp.)
Karem Gallardo et al., 2025	Yes	Primary study	Y, Pr, Nd, Gd, Tb, Dy	Mobile phone waste	Bacteria (Exiguobacterium sp.)
Mic Paper et al., 2023	Yes	Primary study	Ce, Nd, Tb, La	Industrial wastewater	Cyanobacteria
Zhenxiang Hu et al., 2025	Yes	Primary study	Pr	Synthetic solutions	Microalgae (Spirulina)
Genesis Serrano et al., 2023	Yes	Primary study	La, Eu, Gd, Sm	Synthetic solutions	Bacteria (Exiguobacterium sp.)

Study	Full text retrieved?	Study Type	Target Metals	Waste Stream	Organism Type
Roxana Alejandra Ramirez Moriano et al., 2022	Yes	Review	La, Ce, Eu	Aqueous solutions	Various organic adsorbents
Tashane J. Boothe-Lordon et al., 2025	Yes	Primary study	Co, Ni, REY	Acid mine drainage	Fungi
Thainara Viana et al., 2023	Yes	Primary study	Y, La, Ce, Eu, Gd, Tb	Industrial wastewater	Macroalgae (Ulva sp.)
N. R. Nicomel et al., 2020	Yes	Primary study	In	Synthetic solutions	Microalgae
Marcus Heilmann et al., 2021	Yes	Primary study	Nd, Eu, Cu	Synthetic solutions	Microalgae, moss
Mic Paper et al., 2023a	Yes	Primary study	Ce, Nd, Tb, La	Synthetic solutions	Cyanobacteria
João Pinto et al., 2023	Yes	Review	REEs	E-waste	Algae
Bita Deravian et al., 2025	Yes	Review	Cu, Pb, Zn, Co, Ni, Li	Various waste	Biosurfactants
Z. Birungi et al., 2018	Yes	Primary study	La, Tl	Industrial wastewater	Microalgae
M. Palmieri et al., 2022	Yes	Primary study	Y, Ce, Eu, Tb	Synthetic solutions	Microalgae (G. sulphuraria)
J. Dodson et al., 2015	Yes	Review	Pt, Au, In	Various waste	Yeast, fungi
Pro Danièle et al., 2016	Yes	Primary study	REEs	Waste matrices	Microalgae (Coccomyxa)
M. Hofmann et al., 2021	Yes	Primary study	V, Ga, In	Aqueous solutions	Bacteria (siderophore-producing)
A. Brewer et al., 2019a	Yes	Primary study	Tb	Geothermal fluids	Bacteria (E. coli)
Tashane Boothe et al., 2023	Yes	Primary study	Co, Ni, REY	Acid mine drainage	Fungi
I. Zinicovscaia et al., 2023	Yes	Primary study	Dy	Synthetic solutions	Cyanobacteria, yeast
A. Lucaci et al., 2024	Yes	Primary study	Zn, Cu, Co	Industrial wastewater	Marine algae
Jens Kastenhofer et al., 2024	Yes	Primary study	Cu, La, Nd, Dy	Synthetic solutions	Microalgae (G. sulphuraria)

Study	Full text retrieved?	Study Type	Target Metals	Waste Stream	Organism Type
Hari Lal Kharel et al., 2023	Yes	Review	Nd, Dy, La, Cu	Synthetic solutions	Cyanidiales
P. Cozma et al., 2024	Yes	Review	Cu, Co	Industrial waste	Bacteria, fungi
Yoshihiro Ojima et al., 2019	Yes	Primary study	Cu, Ce, Nd, Dy	Synthetic solutions	Yeast
A. Kosheleva et al., 2018	Yes	Primary study	Ce, Dy, Eu, La, Y, Nd	Synthetic solutions	Microalgae
Ni He et al., 2024	Yes	Primary study	La, Ce, Pr, Nd, Sm, Er, Y	Synthetic solutions	Bacteria, fungi
R. M. Pallares et al., 2022	Yes	Primary study	Lanthanides	Synthetic solutions	Engineered proteins
J. Urbina et al., 2019	Yes	Primary study	Cu	Synthetic solutions	Fungi
Hulqi Mila Haili et al., 2023	Yes	Primary study	Ce	Synthetic solutions	Biosorbent composite
Robson C. Oliveira et al., 2011	Yes	Review	Various metals	Various sources	Multiple organisms
S. Ilyas et al., 2017	Yes	Review	Y, Ce, La	Various waste	Bacteria, fungi
Weronika Urbańska et al., 2025	Yes	Primary study	Co, Cu, Li, Ni	Battery waste	Bacteria, fungi
Tímea Fóris et al., 2025	Yes	Primary study	Co	Battery effluents	Microalgae (C. vulgaris)
Fernanda Soto-Montandon et al., 2025	Yes	Review	Ti, Sc, V	Bauxite residue	Bacteria, fungi
Tímea Fóris et al., 2025a	Yes	Primary study	Co	Industrial wastewater	Microalgae (C. vulgaris)
Wadih Y. Rassy et al., 2023	Yes	Primary study	La, Nd, Eu, Dy, Yb, Y	Synthetic solutions	Bacteria (E. coli)
Paloma Martínez-Alesón García et al., 2024	Yes	Primary study	Sm	Synthetic solutions	Microalgae (C. reinhardtii)
J. Castillo et al., 2022	Yes	Book chapter	REEs	Various waste	Multiple organisms
David Kaplan et al., 2023	Yes	Primary study	Pd, Tb	E-waste leachate	Bacteria
Jingqi Liu et al., 2025	Yes	Primary study	Ce	Mining-related waste	Fungi (A. niger)

Study	Full text retrieved?	Study Type	Target Metals	Waste Stream	Organism Type
Carlos Vale et al., 2021	Yes	Primary study	Ce, Eu, Y, La, Nd	Synthetic solutions	Macroalgae
Byron Gonzalo Lapo Calderón et al.	Yes	Thesis	Nd, Eu, Y, Dy, Tb	Aqueous solutions	Lignocellulosic biomass
J. Benemann et al., 1991	Yes	Review	Heavy metals	Industrial wastewater	Microalgae
Patrick Diep et al., 2018	Yes	Review	Heavy metals	Wastewater	Engineered microorganisms
A. Breuker et al., 2020	Yes	Primary study	La, Ce, Nd, Gd, Dy, Er, Yb, Lu	Acidic solutions	Bacteria, fungi, algae
M. Iovinella et al., 2022	Yes	Primary study	Y, Ce, Eu, Tb	Synthetic solutions	Microalgae (G. sulphuraria)
Denys Villa Gomez et al., 2024	Yes	Review	Co, Ni, Re, Sn, W, V	Mining waste	Various organisms
A. Cheriti et al., 2011	Yes	Primary study	Cu	Synthetic solutions	Plant biomass
Shireen Ibrahim Hamadamin et al., 2022	Yes	Primary study	Cu	Synthetic solutions	Microalgae (Chlorella)
Xin-Yue Yang et al., 2025	Yes	Review	Cu, Cd, Pb, Cr	Wastewater	Microalgae
Diogo A. Ferreira-Filipe et al., 2025	Yes	Review	Cu	E-waste	Bacteria, fungi
Alina-Alexandra Ciobanu et al., 2024	Yes	Primary study	Cu, Zn, Co	Synthetic solutions	Marine algae
B. Arunraj et al., 2019	Yes	Primary study	Eu	Phosphor powder	Fungi (S. cerevisiae)
Lalropuia Lalropuia et al., 2024	Yes	Primary study	Li, Co, Ni, Mn, Al	Battery waste	Bacteria
Nicole Ferreira et al., 2025	Yes	Primary study	Y, La, Ce, Pr, Nd, Eu, Gd, Tb, Dy	Synthetic solutions	Biosorbent (hazelnut shells)
E. Wilde et al., 1991	Yes	Review	Heavy metals	Industrial waste	Various microorganisms
Josiane Ponou et al., 2014	Yes	Primary study	Dy	Aqueous solutions	Plant biomass
Niroshan Gajendra et al.	Yes	Primary study	Heavy metals, REEs	Mining wastewater	Aquatic plants, phytoplankton
E. Wilde et al., 1991a	Yes	Review	Heavy metals	Industrial waste	Various microorganisms

Study	Full text retrieved?	Study Type	Target Metals	Waste Stream	Organism Type
R. Gandolfi et al., 2022	Yes	Primary study	Cu	Galvanic wastewater	Bacteria
Yasunori Fukumoto et al., 2025	Yes	Primary study	La, Co, Ni, Cd	Synthetic solutions	Bacteria (E. coli)
A. Ю. Муравьева et al., 2020	Yes	Primary study	Cu, Co, Ni, Cr	Aqueous solutions	Yeast
Alexandre Kompalitch et al., 2023	Yes	Primary study	Co, Li, Mn, Ni	Battery bioleachates	Bacteria
E. El-Sheikh et al., 2013	Yes	Primary study	U, Th, REEs	Laboratory waste	Plant biomass
A. R. Carreira et al., 2023	Yes	Review	Various metals	Acid mine drainage	Non-living algae
M. Vítová et al., 2024	Yes	Review	REEs	Various waste	Microalgae
Katarzyna Staszak et al., 2024	Yes	Review	Cu, Pb, Cr, Zn, Ni, As	Wastewater	Various biosorbents
A. Aslam et al., 2025	Yes	Review	Heavy metals	Contaminated environments	Bacteria, fungi, algae
I. Tezyapar Kara et al., 2024	Yes	Primary study	Y, Ce, Nd, Li, Co, Cu, Zn, Mn, Al	Steelmaking dust	Bacteria
M. Buțu et al., 2019	Yes	Review	Ga, In, critical metals	Mine tailings	Bacteria
G. Gadd et al., 2009	Yes	Review	Various metals	Various sources	Various microorganisms

The included studies span from 1991 to 2025, with the majority published after 2019. Approximately 65% are primary experimental studies, while 35% are reviews or synthesis papers. Laboratory batch experiments predominate, with limited pilot-scale data available. Real industrial waste streams were used in approximately 30% of studies, while the remainder employed synthetic solutions to enable controlled parameter optimization.

Biosorption Performance by Metal

Neodymium (Nd) - Magnet-Relevant

Neodymium recovery represents the most extensively studied critical metal in the reviewed literature, reflecting its strategic importance in permanent magnets.

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Chlorella vulgaris	188.68	Not reported	Fast (<30 min)	5	Magnet waste	Kucuker et al., 2017
Methylobacterium extorquens AM1	202	65-100%	Medium	6.9	Electronic waste	Good et al., 2023a
E. coli (LBT-engineered)	2.64	91%	Fast (10 min)	4-6	Magnet leachates	Brewer et al., 2019
Nostoc sp. 20.02	84.2-91.5	Not reported	Fast (5 min)	5-6	Synthetic	Paper et al., 2023a
Banana rachis	100	Not reported	Fast (8 min)	4	Aqueous solutions	Lapo Calderón
ChiFer(III) beads	13.8	Not reported	Medium (20 min)	4	Synthetic	Lapo Calderón
Chitosan-magnetic composite	44.29	Not reported	Fast	4	Synthetic	Lapo Calderón

The highest maximum adsorption capacity for Nd was achieved by Methylobacterium extorquens AM1 at 202 mg Nd/g DW, followed closely by Chlorella vulgaris at 188.68 mg/g. The mesophilic bacterium M. extorquens demonstrated scalable performance up to 10 L bioreactor volumes without requiring harsh acids or high temperatures, representing a significant advantage for industrial implementation. Genetic engineering approaches, including overproduction of lanthanophores and pyrroloquinoline quinone, enhanced REE-specific bioleaching.

Cyanobacterial strains demonstrated moderate to high adsorption capacities, with Nostoc sp. 20.02 achieving 84.2-91.5 mg/g and exhibiting fast adsorption kinetics with equilibrium reached within 5 minutes. The involvement of hydroxyl and carboxyl functional groups during the adsorption process was confirmed through FT-IR analysis.

Dysprosium (Dy) - Magnet-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Saccharomyces cerevisiae	5.84	68%	Medium (60 min)	3.0	Synthetic	Zinicovscaia et al., 2023
Carbonized ginkgo leaves	25.64	86.40%	Fast (10 min)	3	Aqueous solutions	Ponou et al., 2014
Exiguobacterium sp. SH31	>35	87-99%	Not reported	8	Mobile phone waste	Gallardo et al., 2025
Bacillus subtilis	Selective for Yb, Lu	Up to 75%	Not reported	~2.5	Acidic solutions	Breuker et al., 2020

Dysprosium biosorption data reveals substantial variability in performance across organisms. The extremophilic bacterium Exiguobacterium sp. SH31 achieved recovery efficiencies of 87-99% at pH 8, demonstrating exceptional performance for e-waste-derived REE recovery. This strain produces extracellular polymeric substances (EPS) that facilitate metal binding through surface complexation rather than intracellular bioaccumulation.

Saccharomyces cerevisiae showed moderate performance with a maximum biosorption capacity of 5.84 mg/g and 68% recovery efficiency. The process followed pseudo-second-order kinetics and was exothermic for this yeast species.

Praseodymium (Pr) - Magnet-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Spirulina dry powder	51.10-55.31	79.0%	Medium (150 min)	5	Synthetic	Hu et al., 2025
Burkholderia cepacia	Not reported	100%	Fast (30 min)	Not reported	Synthetic	He et al., 2024
E. coli (LBT-displayed)	Not reported	>86%	Fast (30-60 min)	6.0	Mine tailings	Park et al., 2017

Spirulina dry powder biomass demonstrated a maximum adsorption capacity of 51.10-55.31 mg/g for Pr3+ with 79.0% removal efficiency. The adsorption followed the Langmuir isotherm model (R^2 = 0.993-0.999) with a pseudo-second-order kinetic rate constant of 0.054 g/(mg·min). Thermodynamic analysis confirmed the process was spontaneous and endothermic.

Under microgravity conditions, Burkholderia cepacia achieved 100% recovery for Pr, suggesting potential applications in space biomining contexts. This remarkable performance was attributed to altered microbial morphology and metabolite profiles under microgravity.

Terbium (Tb) - Magnet-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Komarekiella sp. 89.12	138.2	Not reported	Not reported	5	Industrial wastewater	Paper et al., 2023
E. coli (LBT-engineered)	28.3 ± 1.2	76%	Fast (30 min)	5-6	Geothermal fluids	Brewer et al., 2019a
Galdieria sulphuraria	621.16 ± 5.21 µmol/g	Not reported	Fast (30 min)	4.5	Synthetic	Palmieri et al., 2022
Silk-based membranes	Not reported	Up to 85%	Not reported	Low pH	E-waste leachate	Kaplan et al., 2023

Cyanobacterial EPS from Komarekiella sp. 89.12 achieved the highest adsorption capacity for Tb at 138.2 mg/g, with glucose-rich EPS produced under nitrogen-limited conditions showing significantly higher biosorption rates. The extremophilic microalga Galdieria sulphuraria demonstrated 621.16 ± 5.21 µmol/g Tb adsorption at pH 4.5, with equilibrium reached within 30 minutes.

Engineered E. coli expressing lanthanide binding tags (LBTs) achieved 28.3 ± 1.2 mg Tb/g dry cell weight with >90% desorption efficiency using 5 mM citrate.

Samarium (Sm) - Magnet-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Chlamydomonas reinhardtii (CSm4)	4.0 ± 0.4	41% increase per cell	Not reported	4.5	Synthetic	Martínez-Alesón García et al., 2024
Exiguobacterium sp. SH31	~23	>95%	Not reported	7-8	Synthetic	Serrano et al., 2023

Samarium recovery data is limited compared to other magnet-relevant REEs. Chlamydomonas reinhardtii subjected to settling selection demonstrated a 41% increase in Sm uptake per cell, achieving 4.0 ± 0.4 mg Sm/g wet biomass. TEM-EDX microanalyses revealed that approximately 70% of the Sm was bioaccumulated intracellularly near acidocalcisomes or autophagic vacuoles.

Cobalt (Co) - Magnet-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Chlorella vulgaris	Not reported	96.44-98.78%	Fast (30 min)	6.5	Battery effluents	Fóris et al., 2025
Stagonospora sp. (biotic HMO)	Not reported	90%	Slow (18 days)	7	Acid mine drainage	Boothe et al., 2023
Red algae (CC-RAB)	27.89	83-95%	Slow (120 min)	5.0	Synthetic	Ciobanu et al., 2024
Acidithiobacillus thiooxidans	Not reported	25.7%	Slow (7 days)	2.5	Battery waste	Urbańska et al., 2025
Saccharomyces cerevisiae	Not reported	90%	Not reported	Not reported	Aqueous solutions	Muravyeva et al., 2020

Chlorella vulgaris demonstrated exceptional cobalt recovery from battery effluents, achieving 96.44% removal within 30 minutes and 98.78% over 10 hours. The combination with amine-functionalized magnesium ferrite nanoparticles enabled rapid magnetic separation (94.9-99.2% efficiency within 60 seconds), representing a significant advancement for industrial-scale implementation.

Biotic hydrous manganese oxides (HMOs) produced by fungi achieved approximately 75% Co adsorption, substantially outperforming abiotic HMO which showed negligible Co adsorption. The biotic HMO remained stable and resistant to structural changes over 31 days.

Cerium (Ce) - Semiconductor-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Aspergillus niger FH1	42.61	74.05%	Fast (<30 min)	7.0	Mining waste	Liu et al., 2025
Nostoc sp. 20.02	84.2-91.5	Not reported	Fast (5 min)	5-6	Synthetic	Paper et al., 2023a
Ca-Alg/Fe3O4 composite	50.505	Not reported	Medium (3.5 h)	4	Synthetic	Haili et al., 2023
Burkholderia cepacia	Not reported	100%	Fast (30 min)	Not reported	Synthetic	He et al., 2024
Galdieria sulphuraria	214.03 ± 19.58 µmol/g	Not reported	Fast (30 min)	2.5-4.5	Synthetic	Palmieri et al., 2022

Aspergillus niger strain FH1 demonstrated remarkable Ce(III) tolerance (up to 600 mg/L) and achieved 74.05% adsorption efficiency through dual mechanisms: physical entrapment evidenced by Ce(III)-induced cellular invagination and chemical monolayer binding via extracellular functional group coordination. Transcriptomic analysis identified 3,733 differentially expressed genes under Ce(III) stress.

Cyanobacteria Nostoc sp. 20.02 showed the highest adsorption capacity among tested strains at 84.2-91.5 mg/g for cerium, with adsorption kinetics indicating equilibrium within 5 minutes.

Copper (Cu) - Semiconductor-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Serratia plymuthica SC5II	151.0 (mg Cu/g cells)	Not reported	Not reported	Not reported	Galvanic wastewater	Gandolfi et al., 2022
Red algae (CC-RAB)	81.25	83-95%	Medium (60 min)	5.0	Synthetic	Ciobanu et al., 2024
Acacia raddiana	82.63	86%	Slow (4 h)	5	Synthetic	Cheriti et al., 2011
Phosphorylated yeast	~1.0 mmol/g	97%	Fast (10 min)	Slightly acidic	Synthetic	Ojima et al., 2019
Chlorella sp.	0.24	59.7%	Slow (15 h)	8	Synthetic	Hamadamin et al., 2022
Ganoderma lucidum mycelium	Not reported	92-100%	Fast (30 min-72 h)	5.5	Synthetic	Urbina et al., 2019

Copper biosorption has been extensively studied across organism types. Serratia plymuthica strain SC5II achieved the highest reported capacity at 151.0 mg Cu(II)/g cells through EPS-mediated metal coordination. Phosphorylated dry baker's yeast cells demonstrated exceptional performance with 97% Cu recovery and fast kinetics (equilibrium within 10 minutes). The strong negative charges imparted by phosphorylation (zeta potential of −45 mV) enabled selective adsorption of trivalent rare earth ions over divalent heavy metals.

Fungal mycelium treated with metal-binding peptides containing chitin-binding domains achieved 92% Cu recovery within 30 minutes, increasing to 100% after 72 hours.

Indium (In) - Semiconductor-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Microalgal biomass	0.14 mmol/g	91.3% initially	Fast (80% in 30 min)	2.4 (final)	Synthetic	Nicomel et al., 2020
Siderophore-producing bacteria	Not reported	High in absence of Fe	Not reported	Not reported	Aqueous solutions	Hofmann et al., 2021

Indium recovery data is notably sparse in the literature. Unmodified microalgal biomass grown on wastewater achieved 0.14 mmol/g adsorption capacity with 91.3% initial recovery efficiency, decreasing to 60.7% over five regeneration cycles. The biomass demonstrated selectivity for In(III) over Cu(II), Zn(II), and Al(III).

Siderophore-producing bacteria (Nocardioides simplex, Rhodococcus erythropolis) showed promise for In(III) recovery when immobilized on C18 columns, though performance was strongly affected by Fe(III) competition.

Gallium (Ga) - Semiconductor-Relevant

Organism	q_max (mg/g)	Recovery (%)	Kinetics	Optimal pH	Waste Type	Reference
Methylacidiphilum fumariolicum SolV	Not reported	Up to 47.4%	Not reported	Not reported	Not specified	Sieber et al., 2025
Siderophore-producing bacteria	Not reported	Lower than In, higher than Fe	Not reported	Not reported	Aqueous solutions	Hofmann et al., 2021

Gallium recovery data represents a critical gap in the literature. The methanotrophic extremophile Methylacidiphilum fumariolicum strain SolV achieved up to 47.4% Ga recovery, though detailed mechanistic studies are lacking. Siderophore-functionalized columns showed strong preference for Fe(III) over Ga(III), indicating selectivity challenges for Ga recovery from complex waste streams.

Germanium (Ge) - Semiconductor-Relevant

No quantitative data on biological Ge recovery was identified in the reviewed literature. This represents a significant knowledge gap for semiconductor wastewater treatment applications.

Kinetics and Equilibrium Analysis

Adsorption kinetics varied substantially across organism types and metals, with implications for continuous process design.

Kinetics Class	Time Range	Organism Examples	Metals	Best-Fit Model
Fast	<30 min	Chlorella vulgaris, E. coli (engineered), Nostoc sp., Phosphorylated yeast	Nd, Pr, Ce, Cu	Pseudo-second-order
Medium	30-120 min	Spirulina, S. cerevisiae, Macroalgae	Pr, Dy, Cu, Co	Pseudo-second-order
Slow	>2 hours	Biotic HMO fungi, Chlorella (unmodified), Acidithiobacillus	Co, REY, Co	Various

The pseudo-second-order kinetic model provided the best fit for most biosorption systems, indicating chemisorption-controlled behavior. For Chlorella vulgaris biosorption of Nd, equilibrium was achieved within 30 minutes with $R^2 > 0.99$. Similarly, cyanobacterial strains demonstrated fast adsorption kinetics with equilibrium reached within 5 minutes for Ce.

The effect of biomass loading on kinetic performance was investigated in several studies. At low biosorbent dosage (0.5 g/L), Chlorella vulgaris yielded the highest Nd uptake, while higher doses did not proportionally increase removal rates for Galdieria sulphuraria. This inverse relationship between biomass concentration and specific uptake capacity reflects competition for limited metal ions in solution.

Optimal Operating Conditions

pH Optimization

Optimal pH ranges varied systematically by metal and organism type:

Metal	Optimal pH Range	Organisms Tolerating Extreme pH	Reference
Nd	5-6.9	Chlorella (pH 5), M. extorquens (pH 6.9)	Multiple
Dy	3-8	S. cerevisiae (pH 3), Exiguobacterium (pH 8)	Zinicovscaia, Gallardo
Ce	4-7	Ca-Alg composite (pH 4), A. niger (pH 7)	Haili, Liu
Co	2.5-6.5	A. thiooxidans (pH 2.5), C. vulgaris (pH 6.5)	Urbańska, Fóris
Cu	5-8	Various algae species	Ciobanu, Hamadamin
In	2.4	Microalgal biomass	Nicomel

Galdieria sulphuraria demonstrated biosorption at extremely low pH values (pH 1.5-2.5), though performance im-

proved up to 24-fold at pH 5.0 under phosphate-free conditions. This extremophile's tolerance to acidic conditions makes it particularly relevant for treating acidic industrial waste streams.

Temperature Effects

Most biosorption studies employed mesophilic conditions (20-35℃), with limited investigation of thermophilic ranges. Galdieria sulphuraria, a thermoacidophilic microalga, showed optimal growth at 42℃, though metabolic activity had a negative impact on REE sorption. For engineered E. coli, REE recovery efficiency and selectivity increased as a function of temperature up to ~70℃ due to thermodynamic properties of metal complexation.

Thermodynamic analysis across multiple studies confirmed that biosorption processes were generally spontaneous ($\Delta G^\circ < 0$) with endothermic characteristics ($\Delta H^\circ > 0$) for most metal-organism combinations, though exothermic behavior was observed for S. cerevisiae dysprosium biosorption.

Organism-Specific Performance Patterns

Microalgae Performance

Microalgae demonstrated broad applicability across multiple metals with consistent performance characteristics:

Genus/Species	Target Metals	q_max Range (mg/g)	Recovery Range (%)	Key Advantages
Chlorella vulgaris	Nd, La, Co	75-188.68	94-98.78%	Fast kinetics, scalable cultivation
Galdieria sulphuraria	Y, Ce, Eu, Tb, Nd, Dy	Up to 621 μmol/g	Strain/pH dependent	Extreme pH tolerance, thermophilic
Spirulina	Pr, Dy	3.24-55.31	50-79%	Commercially available biomass
Chlamydomonas reinhardtii	La, Sm	4.0 (Sm), 143 (La)	Variable	Selectable for improved uptake

Chlamydomonas reinhardtii demonstrated significant improvement in metal uptake through settling selection, with the derived CSm4 strain achieving 41% more Sm sequestration per cell compared to ancestral strains. This demonstrates the potential of directed evolution approaches for enhancing biosorption performance.

Bacteria Performance

Genus/Species	Target Metals	q_max Range (mg/g)	Recovery Range (%)	Key Advantages
E. coli (engineered)	Nd, Dy, Pr, Tb	2.64-28.3	86-91%	Genetic modification, fast kinetics
Methylobacterium extorquens	Nd, Dy, Pr	202	65-100%	Scalable (10 L), no harsh chemicals

Genus/Species	Target Metals	q_max Range (mg/g)	Recovery Range (%)	Key Advantages
Exiguobacterium sp. SH31	Y, Pr, Nd, Gd, Tb, Dy	>35	87-99%	EPS-mediated, extremophile
Bacillus subtilis	Yb, Lu, HREEs	Not reported	Up to 75%	HREE selectivity
Acidithiobacillus spp.	Co, Li, Ni	Not reported	Up to 100%	Bioleaching capability

Engineered bacteria expressing lanthanide binding tags (LBTs) on cell surfaces demonstrated 2-10-fold increase in distribution coefficients for individual REEs compared to non-engineered controls. The LBT-display systematically enhanced affinity for REEs as a function of decreasing atomic radius, providing a mechanism to separate high-value heavy REEs from more common light REEs.

Fungi Performance

Genus/Species	Target Metals	q_max Range (mg/g)	Recovery Range (%)	Key Advantages
Aspergillus niger	Ce, Co, Li	42.61	74.05%	High Ce tolerance, organic acid production
Saccharomyces cerevisiae	Dy, Eu, Cu	5.84-25.9	68-97.5%	Regenerable, selective for REEs
Paraphaeosphaeria sporulosa	Co, REY	Not reported	80% (Co), >99% (REY)	Biotic HMO production
Stagonospora sp.	Co, REY	Not reported	90% (Co), >99% (REY)	High Mn oxidation rate

Fungi producing biotic hydrous manganese oxides (HMOs) demonstrated exceptional performance for simultaneous REY and Co recovery. Stagonospora sp. oxidized 99% of Mn in solution after 14 days while achieving 90% Co reduction, and the biotic HMO remained stable and resistant to structural changes over 31 days. This contrasts sharply with abiotic HMO, which showed negligible Co and Ni adsorption and underwent structural transformation reducing adsorption capacity.

Selectivity and Multi-Metal Recovery

Selectivity Patterns

Organism	Preferential Binding	Selectivity Ratio	Competing Ions	Reference
Phosphorylated yeast	REE^{3+} over heavy $metals^{2+}$	~7:1 (Nd/Zn)	Cu, Pb, Zn, Cd	Ojima et al., 2019
E. coli (LBT)	REEs over non-REEs except Cu	2-10x enhanced	Al, Cu, Pb, U	Park et al., 2017

Organism	Preferential Binding	Selectivity Ratio	Competing Ions	Reference
Engineered S-layers	Lanthanides over other cations	Several orders of magnitude	Ca, Mg, Na, K	Pallares et al., 2022
B. subtilis	Yb, Lu over other REEs	High selectivity	Other REEs	Breuker et al., 2020
Microalgal biomass	In over Cu, Zn, Al	Highly selective	Cu, Zn, Al	Nicomel et al., 2020

Phosphorylated yeast cells selectively adsorbed trivalent rare earth ions (Nd^{3+}, Yb^{3+}) with ~70% removal efficiency compared to only ~10% for divalent heavy metal ions. This selectivity arises from preferential electrostatic interactions between the strongly negative phosphate groups and trivalent cations.

Engineered protein layers with hydroxypyridinonate chelators achieved several orders of magnitude higher selectivity for lanthanides compared to other cations commonly found in f-element sources. The binding affinity increased systematically for lanthanide ions with smaller ionic radius.

Simultaneous Multi-Metal Recovery

Several studies investigated recovery of multiple metals from mixed waste streams:

Study	Metal Combination	Waste Type	Key Findings	Reference
Brewer et al., 2019	Nd + Dy + Pr	Magnet leachates	97% REE purity upon breakthrough	
Gallardo et al., 2025	Y + Pr + Nd + Gd + Tb + Dy	Mobile phone waste	87-99% recovery for all six REEs	
Boothe et al., 2023	Co + Ni + REY	Acid mine drainage	90% Co, >99% REY simultaneously	
Barros et al., 2019	La + Ce + Y + Tb + Pr + Eu	Synthetic	La and Ce higher affinity in multi-component	
Lalropuia et al., 2024	Li + Co + Ni + Mn + Al	Battery waste	Up to 100% recovery for each metal	

In NdFeB magnet leachate processing, engineered microbe beads achieved 97% REE purity of the adsorbed fraction upon breakthrough, with a two-bed volume increase in breakthrough points for REEs compared to non-REE impurities. This selective extraction enabled separation of high-value REEs from Fe and other contaminants in a single-pass system.

Biotic HMO systems demonstrated exceptional capability for simultaneous Co and REY recovery from acid mine drainage, achieving near-complete REY removal (95-99%) while maintaining 75-90% Co adsorption. The fungal biomass itself contributed significantly to REY adsorption.

Economic and Practical Metrics

Regeneration and Reusability

Organism/System	Regeneration Cycles	Desorption Efficiency	Desorption Agent	Reference
E. coli microbe beads	9 cycles (85% retention)	94%	10 mM citrate	Brewer et al., 2019
S. cerevisiae-cellulose	4 cycles (>80%)	97.5% initial	Not specified	Arunraj et al., 2019
Microalgal biomass	5 cycles	87% (EDTA), 80% (HCl)	EDTA, HCl	Nicomel et al., 2020
Serratia plymuthica	8 cycles	Not reported	Not specified	Gandolfi et al., 2022
Silk-based membranes	4 cycles (70% retention)	100%	Low pH buffer	Kaplan et al., 2023
Spirulina	Not reported	91.7%	0.1 M HNO_3	Hu et al., 2025

Engineered E. coli microbe beads demonstrated excellent stability for reuse, retaining 85% of adsorption capacity after nine consecutive adsorption/desorption cycles. The citrate-based desorption achieved 94% efficiency, enabling REE recovery without biomass destruction.

Scale-Up Considerations

Limited pilot-scale data exists in the literature. Key findings include:

Methylobacterium extorquens AM1 demonstrated successful scale-up to 10 L bioreactor volumes with consistent metal yields and no requirement for harsh acids or high temperatures. Metal accumulation increased with leachate concentration and was highly specific for REEs.

Bioleaching processes face scale-up challenges related to permeability and aeration. Column studies with Acidithiobacillus ferrooxidans showed extraction efficiency reaching a plateau after 11 days, with variations in recirculation rates not significantly influencing extraction efficiency. Chemical leaching outperformed bioleaching for overall efficiency and ease of extraction in this system.

Estimated Operational Costs

Limited cost data was available in the reviewed literature:

Process	Estimated Cost	Comparison	Reference
A. thiooxidans bioleaching	$2500-3000/ton powder/cycle	vs $3000-9000 for citric acid	Urbańska et al., 2025
A. niger bioleaching	$1000-1400/ton powder/cycle	Most cost-effective biological option	Urbańska et al., 2025

Bioleaching with A. thiooxidans at 1% pulp density achieved comparable results to citric acid leaching at potentially lower cost ($2500-3000 vs $3000-9000 per ton of powder per reactor cycle).

Synthesis: Reconciling Heterogeneous Findings

Context-Dependent Performance

The substantial heterogeneity in reported performance metrics can be reconciled by considering systematic differences in experimental conditions:

pH Effects: Studies finding high adsorption capacities typically operated at organism-specific optimal pH ranges. For Nd biosorption by Chlorella vulgaris, maximum uptake occurred at pH 5, while Methylobacterium extorquens performed optimally at pH 6.9. Studies reporting lower performance often operated outside these optimal ranges.

Waste Stream Complexity: Studies using real industrial waste consistently reported lower recovery efficiencies than those using synthetic solutions. For example, Eu recovery from red phosphor powder achieved 86 ± 5.0%, while the same organism-system achieved only 55 ± 4.6% from leached fluorescent lamp waste containing competing ions (Y, La, Ca, Al, Ce).

Living vs Dead Biomass: Dead/dried biomass generally showed faster kinetics but required optimization of pretreatment conditions. Living biomass offered continuous self-regeneration of binding sites but was susceptible to metal toxicity. Metabolic activity in living Galdieria sulphuraria had a negative impact on REE sorption, challenging the feasibility of simultaneous sorption and biomass growth.

Study Quality Hierarchy

The highest-quality evidence comes from studies that:

1. Used real industrial waste streams
2. Demonstrated reproducibility through replicate experiments
3. Validated performance across multiple conditions
4. Reported scalability data

Studies using synthetic solutions at laboratory scale provide mechanistic insights but may overestimate performance in real applications.

Champion Organism Identification

Based on the synthesis of performance metrics across multiple studies:

Metal	Champion Organism	q_max (mg/g)	Recovery (%)	Key Advantage
Nd	M. extorquens AM1	202	65-100%	Scalable, no harsh chemicals
Nd (alternative)	C. vulgaris	188.68	-	Fast kinetics, established
Dy	Exiguobacterium sp. SH31	>35	87-99%	Multiple REE recovery
Pr	Spirulina	51-55	79%	Commercial availability
Tb	Komarekiella sp. EPS	138	-	Highest q_max

Metal	Champion Organism	q_max (mg/g)	Recovery (%)	Key Advantage
Ce	Nostoc sp. 20.02	84-92	-	Fast kinetics (5 min)
Co	C. vulgaris	-	96-99%	Exceptional recovery, fast
Cu	S. plymuthica SC5II	151	-	Industrial wastewater tested
In	Microalgal biomass	0.14 mmol/g	91%	Selective over Cu, Zn, Al

Data Gaps and Research Priorities

Critical data gaps identified in this review:

1. **Germanium (Ge)**: No quantitative biological recovery data identified
2. **Gallium (Ga)**: Limited data (47.4% maximum recovery)
3. **Pilot-scale validation**: Only 2 of 80 studies reported pilot-scale data
4. **Economic analysis**: Limited cost-benefit data for industrial implementation
5. **Continuous operation**: Most studies used batch systems; limited continuous flow data
6. **Multi-metal selectivity**: Quantitative selectivity ratios rarely reported

Extremophile vs Mesophile Performance

The comparison of extremophilic (Galdieria, Cyanidium) versus mesophilic (Chlorella, Spirulina) organisms for acidic industrial waste treatment reveals context-dependent advantages:

Galdieria sulphuraria demonstrated superior tolerance to acidic conditions (pH 1.5-2.5 growth optimum) and operated effectively at pH 2.5-4.5 for REE biosorption. However, biosorption at the organism's pH growth optimum (pH 1.5-2.5) was poor, improving up to 24-fold at pH 5.0. This indicates that extremophile tolerance does not directly translate to optimal biosorption performance.

Cyanidiales species showed removal efficiencies exceeding 90% at pH 2.5 for Nd, Dy, La, and Cu, with Galdieria sulphuraria identified as particularly effective for acidic environments. However, mesophilic Chlorella vulgaris achieved comparable or superior q_max values (188.68 mg/g Nd) at moderate pH (pH 5).

The recommendation is to use extremophiles when waste stream pH cannot be adjusted economically, but mesophilic organisms when pH adjustment is feasible, as the latter typically achieve higher specific uptake capacities under optimized conditions.

Performance Gap: Laboratory vs Industrial Scale

The limited pilot-scale data suggests significant performance gaps:

- M. extorquens maintained consistent yields at 10 L scale
- Column bioleaching showed efficiency plateau after 11 days with permeability challenges
- Continuous flow systems for microalgal In biosorption showed potential for scale-up
- Living hydrogels for Pd recovery maintained ~90% efficiency under semi-batch conditions

Process engineering challenges identified include biomass separation, continuous operation design, and maintaining microbial stability under industrial conditions.

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What are the reported CAPEX (capital expenditure), OPEX (operating expenditure), and break-even economic models for industrial-scale biological recovery of critical raw materials from semiconductor manufacturing wastewater and rare earth magnet production waste, with emphasis on pilot-scale demonstrations (>100L), commercial deployments, techno-economic analyses (TEA), and life cycle assessments (LCA)? Focus on: • INDUSTRIAL CASE STUDIES: real-world deployments at semiconductor fabs, magnet manufacturers, or industrial waste treatment facilities • PILOT-SCALE VALIDATIONS: systems ≥100L continuous operation, preferably ≥1000L • TECHNO-ECONOMIC MODELS: published TEA studies with cost breakdowns and sensitivity analyses • COMPARATIVE COST ANALYSIS: biological vs conventional technologies (ion exchange, solvent extraction, chemical precipitation, membrane filtration) CAPEX DATA REQUIRED (in € or $ per m^3/day capacity OR per tonne/year metal recovery): 1. Bioreactor system: photobioreactor (PBR), stirred tank, column reactor, airlift, immobilized bed 2. Pre-treatment equipment: pH adjustment, filtration, chelator destruction 3. Harvesting/separation: centrifuge, filtration, sedimentation 4. Metal recovery/elution: acid desorption, electrowinning, precipitation 5. Instrumentation: pH, DO, ORP sensors, online metal analyzers 6. Utilities connection: electricity, water, compressed air, waste discharge 7. Total installed CAPEX with breakdown by component (%) OPEX DATA REQUIRED (in € or $ per m^3 treated OR per kg metal recovered): 1. Biomass production costs: growth medium, nutrients (N, P), CO2 (if required), inoculum 2. Energy: pumping, aeration, mixing, heating/cooling, lighting (for photoautotrophs) 3. Chemicals: pH adjustment (acid/base), desorption eluent (HCl, H2SO4, EDTA), flocculants 4. Labor: operators, maintenance, laboratory analysis 5. Consumables: filters, membranes, sensor calibration 6. Waste disposal: spent biomass, residual sludge 7. Maintenance: equipment replacement, bioreactor cleaning 8. Total OPEX with breakdown by category (%) ECONOMIC PERFORMANCE METRICS: 9. Cost per m^3 wastewater treated (€/m^3 or $/$m^3$) 10. Cost per kg metal recovered (€/kg or $/kg) - specify which metal (Cu, Ce, Nd, Dy, etc.) 11. Revenue from recovered metals: market price ($/kg) × purity (%) × recovery yield (%) 12. Net economic value: Revenue - OPEX (€/m^3 or €/year) 13. Payback period (years) based on CAPEX and net annual value 14. NPV (Net Present Value) and IRR (Internal Rate of Return) if reported 15. Sensitivity analysis: impact of metal price volatility (±30%) on economics 16. Break-even metal concentration (ppm): minimum concentration for positive economics SCALE-UP FACTORS: 17. Pilot scale: reactor volume (L), throughput (m^3/day), duration of operation (months) 18. Scaling

techno-economic analyses address adjacent applications (lithium-ion battery recycling, phosphogypsum processing, acid mine drainage) with reported CAPEX of $7-60 million per facility, OPEX dominated by carbon source costs (44-98% of nutrients), break-even REE concentrations of 0.5 wt%, IRRs of 15-44%, and payback periods of 1-11 years, but these findings cannot be directly extrapolated to the target applications without validation.

Abstract

Techno-economic analyses of biological critical raw material recovery reveal substantial heterogeneity in reported economics, with CAPEX ranging from $44 million for 10,000 t/y lithium-ion battery bioleaching facilities to EUR 20 million for 100 m^3/h sulfate recycling systems. OPEX is dominated by carbon/nutrient costs, which constitute 44-98% of operational expenditure in bioleaching operations, while energy for membrane fouling control accounts for 80-90% of costs in membrane bioreactor configurations. Reported economic performance metrics show NPVs of $5.78 million to $18.4 billion with IRRs of 15-44% and payback periods of 1-11 years, though these figures predominantly derive from lithium-ion battery recycling, phosphogypsum processing, and acid mine drainage treatment rather than the specific target applications. Break-even analyses indicate minimum REE concentrations of 0.5 wt% for economic viability, with dilute sources (0.02-0.1 wt%) remaining unviable under current conditions. Biological methods demonstrate cost advantages over conventional chemical treatment, with fungal bioleaching achieving $1,000-1,400/ton compared to $3,000-9,000/ton for citric acid treatment.

Critically, the reviewed literature contains no direct CAPEX, OPEX, or break-even data specifically for semiconductor manufacturing wastewater or rare earth magnet production waste applications. Pilot-scale demonstrations exceeding 100L are limited, with the largest identified system operating at 11.7 m^3 for 6 months for microalgae-based wastewater treatment rather than metal recovery. Commercial deployments remain sparse, with only Inotec technology documented at full scale achieving 25-50% cost reductions. LCA data indicates biological recovery achieves 70-85% lower environmental factors than conventional acid leaching and 93-96% of conventional REO production impacts for ecosystem quality and resource depletion, though human health impacts may be higher. The value proposition for biological recovery currently depends more on compliance-driven waste disposal cost avoidance than metal recovery revenue, as REE prices would need to increase 26-fold from 2016 levels for bioleaching to break even on metal value alone.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the reported CAPEX (capital expenditure), OPEX (operating expenditure), and break-even economic models for industrial-scale biological recovery of critical raw materials from semiconductor manufacturing wastewater and rare earth magnet production waste, with emphasis on pilot-scale demonstrations (>100L), commercial deployments, techno-economic analyses (TEA), and life cycle assessments (LCA)?

Focus on: • INDUSTRIAL CASE STUDIES: real-world deployments at semiconductor fabs, magnet manufacturers, or industrial waste treatment facilities • PILOT-SCALE VALIDATIONS: systems ≥100L continuous operation, preferably ≥1000L • TECHNO-ECONOMIC MODELS: published TEA studies with cost breakdowns and sensitivity analyses • COMPARATIVE COST ANALYSIS: biological vs conventional technologies (ion exchange, solvent extraction, chemical precipitation, membrane filtration)

CAPEX DATA REQUIRED (in € or $ per m^3/day capacity OR per tonne/year metal recovery):

1. Bioreactor system: photobioreactor (PBR), stirred tank, column reactor, airlift, immobilized bed

2. Pre-treatment equipment: pH adjustment, filtration, chelator destruction
3. Harvesting/separation: centrifuge, filtration, sedimentation
4. Metal recovery/elution: acid desorption, electrowinning, precipitation
5. Instrumentation: pH, DO, ORP sensors, online metal analyzers
6. Utilities connection: electricity, water, compressed air, waste discharge
7. Total installed CAPEX with breakdown by component (%)

OPEX DATA REQUIRED (in € or $ per m^3 treated OR per kg metal recovered):

1. Biomass production costs: growth medium, nutrients (N, P), CO2 (if required), inoculum
2. Energy: pumping, aeration, mixing, heating/cooling, lighting (for photoautotrophs)
3. Chemicals: pH adjustment (acid/base), desorption eluent (HCl, H2SO4, EDTA), flocculants
4. Labor: operators, maintenance, laboratory analysis
5. Consumables: filters, membranes, sensor calibration
6. Waste disposal: spent biomass, residual sludge
7. Maintenance: equipment replacement, bioreactor cleaning
8. Total OPEX with breakdown by category (%)

ECONOMIC PERFORMANCE METRICS: 9. Cost per m^3 wastewater treated (€/m^3 or $/$m^3$) 10. Cost per kg metal recovered (€/kg or $/kg) - specify which metal (Cu, Ce, Nd, Dy, etc.) 11. Revenue from recovered metals: market price ($/kg) × purity (%) × recovery yield (%) 12. Net economic value: Revenue - OPEX (€/m^3 or €/year) 13. Payback period (years) based on CAPEX and net annual value 14. NPV (Net Present Value) and IRR (Internal Rate of Return) if reported 15. Sensitivity analysis: impact of metal price volatility (±30%) on economics 16. Break-even metal concentration (ppm): minimum concentration for positive economics

SCALE-UP FACTORS: 17. Pilot scale: reactor volume (L), throughput (m^3/day), duration of operation (months) 18. Scaling challenges reported: mass transfer limitations, biomass productivity decline, fouling, contamination 19. Footprint: land area required (m^2) per m^3/day capacity 20. Modular scalability: can capacity be increased by adding units (numbering-up) vs building larger single unit (scaling-up)?

COMPARATIVE COST ANALYSIS (vs conventional technologies): 21. Ion exchange resins: CAPEX (€/m^3/day), OPEX (€/m^3), resin replacement costs (€/kg), selectivity limitations 22. Chemical precipitation: chemical costs (€/m^3), sludge disposal (€/tonne), limited to >10 ppm concentrations 23. Solvent extraction: CAPEX (€/kg/day), OPEX ($/kg), solvent losses, environmental hazards 24. Membrane filtration: CAPEX (€/m^2), OPEX (€/m^3), fouling, limited by particle size 25. When is biological recovery economically superior? Identify crossover concentration (ppm) or waste volume (m^3/year)

LIFE CYCLE ASSESSMENT (LCA) DATA: 26. Carbon footprint: kg CO2-equivalent per m^3 treated or per kg metal recovered 27. Energy intensity: MJ/m^3 or kWh/kg metal 28. Water footprint: L water consumed per m^3 wastewater treated (net water balance) 29. Toxicity indicators: ecotoxicity, human toxicity potential 30. Circularity metrics: % of waste converted to valuable products, landfill diversion rate 31. Comparison to virgin mining: LCA of bio-recovered metal vs primary extraction

INDUSTRY PARTNERSHIPS & COMMERCIAL STATUS: 32. Companies or consortia: identify commercial entities deploying biological recovery (Phyco Systems, BlueBioTech, etc.) 33. Patents: key patents on biological metal recovery processes, organism strains, bioreactor designs 34. Regulatory approvals: EPA permits, EU REACH compliance, discharge permit compliance (<0.5 ppm targets) 35. Customer testimonials or case study results from industrial partners

CRITICAL DECISION-MAKING QUESTIONS: • What is the minimum wastewater flow rate (m^3/year) for a biological

system to be economically viable compared to chemical treatment? • Which metals have the strongest economic case for biological recovery based on recovery value vs treatment cost? • For semiconductor fabs: is the value proposition primarily COMPLIANCE (avoiding EPA fines) or REVENUE (recovered metal value)? • For magnet manufacturers: can bio-recovery compete with exporting scrap to Chinese refiners? • What are the top 3 cost drivers in OPEX that need optimization for commercial viability?

Study selection criteria: • MUST include quantitative economic data (CAPEX, OPEX, or break-even analysis) • Prioritize: Industrial case studies > Pilot demonstrations > TEA models > Lab-scale extrapolations • Timeframe: 2019-2026 strongly preferred • Include industry white papers, conference proceedings, and company technical reports (not just academic papers) • Papers from wastewater treatment companies (Veolia, Suez, Applied Materials) comparing biological to their conventional offerings are especially valuable"

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Target Materials**: Does the study focus on recovery of critical raw materials from semiconductor or rare earth magnet waste streams (e.g., Cu, Ce, Nd, Dy, or other materials specifically from these industries)?
- **Biological Recovery Method**: Is biological recovery the primary or co-primary intervention being studied (not just mentioned as a minor component)?
- **Economic Data**: Does the study report quantitative economic data such as CAPEX, OPEX, cost per m^3 treated, cost per kg recovered, payback period, or break-even analysis?
- **Scale and Modeling**: Does the study involve either: (1) industrial or pilot-scale systems ≥100L, OR (2) techno-economic analyses/life cycle assessments with realistic industrial parameters?
- **Laboratory Scale Limitation**: If the study is laboratory-scale (<100L), does it include economic extrapolation or modeling to industrial scale?
- **Beyond Technical Performance**: Does the study go beyond reporting only technical performance metrics (removal efficiency, kinetics, biological mechanisms) to include economic analysis?
- **Empirical Foundation**: If the study is theoretical or computational, does it include empirical validation or is based on validated experimental data?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

339 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Target Materials**: Does the study focus on recovery of critical raw materials from semiconductor or rare earth magnet waste streams (e.g., Cu, Ce, Nd, Dy, or other materials specifically from these industries)?
- **Biological Recovery Method**: Is biological recovery the primary or co-primary intervention being studied (not just mentioned as a minor component)?
- **Economic Data**: Does the study report quantitative economic data such as CAPEX, OPEX, cost per m^3 treated, cost per kg recovered, payback period, or break-even analysis?

- **Scale and Modeling**: Does the study involve either: (1) industrial or pilot-scale systems ≥100L, OR (2) techno-economic analyses/life cycle assessments with realistic industrial parameters?
- **Laboratory Scale Limitation**: If the study is laboratory-scale (<100L), does it include economic extrapolation or modeling to industrial scale?
- **Beyond Technical Performance**: Does the study go beyond reporting only technical performance metrics (removal efficiency, kinetics, biological mechanisms) to include economic analysis?
- **Empirical Foundation**: If the study is theoretical or computational, does it include empirical validation or is based on validated experimental data?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

97 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **System Scale & Configuration**:

  Extract details about the biological recovery system scale and configuration for critical raw material recovery from semiconductor/rare earth magnet wastewater, including:

- Reactor type and volume (L) - specify PBR, stirred tank, column, airlift, immobilized bed
- Throughput capacity (m^3/day wastewater or kg/day metal recovery)
- Operation duration (months) and continuous vs batch
- Target metals recovered (Cu, Ce, Nd, Dy, other REEs)
- Influent metal concentrations (ppm) and wastewater source (semiconductor fab, magnet manufacturer, etc.)
- Recovery efficiency (% metal recovered)
- System footprint (m^2 per m^3/day capacity)
- Modular vs single-unit scaling approach

- **CAPEX Breakdown**:

  Extract capital expenditure data for biological metal recovery systems from semiconductor/magnet production wastewater, including:

- Total CAPEX (€ or $ per m^3/day capacity OR per tonne/year metal recovery capacity)
- Component breakdown with percentages: bioreactor system, pre-treatment equipment, harvesting/separation, metal recovery/elution, instrumentation, utilities connection
- Specific equipment costs: photobioreactor, stirred tank, centrifuge, filtration, electrowinning, pH/DO/ORP sensors, online metal analyzers
- Installation and commissioning costs if reported
- Currency and year of cost estimates
- Source of cost data (actual deployment, vendor quotes, engineering estimates)

- **OPEX Analysis**:

  Extract operating expenditure data for biological recovery of critical raw materials, including:

- Total OPEX (€ or $ per m^3 wastewater treated OR per kg metal recovered)

- Cost breakdown by category with percentages: biomass production, energy, chemicals, labor, consumables, waste disposal, maintenance
- Specific cost drivers: growth medium, nutrients (N,P), CO2, pumping, aeration, pH adjustment chemicals, desorption eluent (HCl, H2SO4, EDTA), spent biomass disposal
- Energy consumption (kWh/m³ or kWh/kg metal) and associated costs
- Labor requirements (FTE operators per unit capacity)
- Consumable replacement frequency and costs (filters, membranes, sensors)

- **Economic Performance**:

  Extract economic performance metrics and break-even analysis for biological metal recovery from semiconductor/rare earth magnet wastewater, including:

- Cost per m³ wastewater treated (€/m³ or $/m³)
- Cost per kg metal recovered (€/kg or $/kg) - specify which metal
- Revenue from recovered metals: market price ($/kg) × purity (%) × recovery yield (%)
- Net economic value: Revenue minus OPEX (€/m³ or €/year)
- Payback period (years) based on CAPEX and net annual value
- NPV (Net Present Value) and IRR (Internal Rate of Return) if reported
- Break-even metal concentration (ppm): minimum concentration for positive economics
- Sensitivity analysis results: impact of ±30% metal price volatility on economics

- **Comparative Economics**:

  Extract cost comparisons between biological recovery and conventional technologies (ion exchange, chemical precipitation, solvent extraction, membrane filtration) for semiconductor/magnet wastewater treatment, including:

- Head-to-head cost comparisons: CAPEX and OPEX per unit capacity or throughput
- Crossover conditions: wastewater volume (m³/year) or metal concentration (ppm) where biological becomes economically superior
- Specific conventional technology costs: ion exchange resin replacement (€/kg), chemical precipitation reagents (€/m³), solvent losses, membrane fouling/replacement
- Performance limitations of conventional technologies: selectivity, minimum concentrations, environmental hazards
- Total cost of ownership comparisons over system lifetime
- Advantages/disadvantages affecting economic comparison (regulatory compliance, environmental impact, operational complexity)

- **Pilot & Commercial Status**:

  Extract information about pilot-scale demonstrations (≥100L) and commercial deployments of biological recovery for semiconductor/rare earth magnet wastewater, including:

- Pilot scale specifications: reactor volume (L), duration (months), throughput achieved
- Industrial case studies: semiconductor fabs, magnet manufacturers, waste treatment facilities involved
- Scaling challenges reported: mass transfer limitations, biomass productivity decline, fouling, contamination
- Commercial entities: companies deploying biological recovery (identify specific firms)
- Regulatory status: EPA permits, EU REACH compliance, discharge permit compliance targets
- Industry partnerships and customer testimonials
- Technology readiness level (TRL) assessment

- Patents on biological metal recovery processes, strains, bioreactor designs
- **Life Cycle Assessment**:

  Extract LCA data and environmental impact metrics for biological recovery of critical raw materials from semiconductor/magnet production wastewater, including:

- Carbon footprint: kg CO2-equivalent per m^3 treated or per kg metal recovered
- Energy intensity: MJ/m^3 or kWh/kg metal recovered
- Water footprint: L water consumed per m^3 wastewater treated (net water balance)
- Toxicity indicators: ecotoxicity, human toxicity potential
- Circularity metrics: % waste converted to valuable products, landfill diversion rate
- Comparison to virgin mining: LCA of bio-recovered metal vs primary extraction
- Environmental advantages over conventional treatment methods
- Waste streams generated and their environmental impact
- **Technical Performance**:

  Extract technical performance data for biological systems recovering critical raw materials from semiconductor/rare earth magnet wastewater, including:

- Metal recovery efficiency (%) for specific metals (Cu, Ce, Nd, Dy, other REEs)
- Selectivity: ability to recover target metals while rejecting interfering species
- Operating conditions: pH range, temperature, residence time, biomass concentration
- Microbial systems used: specific organisms, engineered strains, immobilized vs suspended
- Pre-treatment requirements: pH adjustment, chelator destruction, filtration needs
- Effluent quality achieved: final metal concentrations, discharge compliance (<0.5 ppm targets)
- System reliability: uptime, maintenance intervals, contamination resistance
- Process kinetics: metal uptake rates, biosorption capacity
- **Market Drivers**:

  Extract information about market drivers and business case factors for biological recovery of critical raw materials from semiconductor/rare earth magnet production, including:

- Value proposition: compliance-driven (avoiding EPA fines) vs revenue-driven (recovered metal value)
- Metal price volatility impact on economics: historical price ranges and trends
- Regulatory drivers: discharge limits, environmental regulations, circular economy policies
- Supply chain considerations: competition with traditional recycling (e.g., export to Chinese refiners for magnet scrap)
- Market size estimates: addressable wastewater volumes, metal recovery potential
- Customer requirements: semiconductor fab vs magnet manufacturer priorities
- Barriers to adoption: technical, economic, regulatory, operational
- Success factors for commercial viability

Report

Due to the limitations of the AI model, we are only able to process 80 sources while writing a report. This report was written using the 80 sources that had the highest screening scores out of the 97 sources that we screened in and extracted data from.

Characteristics of Included Studies

This systematic review synthesizes economic data from 80 sources examining biological recovery of critical raw materials from various waste streams. While the research question specifically targets semiconductor manufacturing wastewater and rare earth magnet production waste, the literature reveals that most techno-economic studies address adjacent applications including lithium-ion battery recycling, acid mine drainage treatment, fluidized catalytic cracking catalyst processing, and phosphogypsum valorization. These studies nonetheless provide transferable economic insights for the target applications.

Study	Full text retrieved?	Study Type	Waste Stream	Target Materials	Scale
S. Balchandani et al., 2023	Yes	TEA + LCA	Waste materials (CS, DP, NP)	REEs via gluconic acid	Lab-scale
Vicki S. Thompson et al., 2017	Yes	TEA	FCC catalysts	REEs	Continuous bioreactor
Adam Smerigan et al., 2025	Yes	TEA + LCA	Phosphogypsum	REEs	Modeling study
M. Alipanah et al., 2024	Yes	TEA + Supply chain	Lithium-ion batteries	Co, Li, Ni, Mn	900,000 t/y capacity
Mahugnon Samuel Ahossouhe et al., 2025	Yes	Extended CBA	E-waste	Metals	Scenario modeling
Dina Mohamed et al., 2025	Yes	Review	E-waste	Au, Ag, Pt, Cu	Literature synthesis
Weronika Urbańska et al., 2025	Yes	Comparative study	Spent LIBs	Co, Cu, Li, Ni	Lab-scale bioreactor
Nathan M. Good et al., 2023	Yes	Primary study	Electronic waste	Nd, REEs	10L bioreactor
Rhulani Shingwenyana et al., 2021	Yes	TEA	Acid mine drainage	Fe-minerals, Ca, Mg	Pilot plant
Xiangjun Huang et al., 2020	Yes	TEA	Municipal wastewater	N, P recovery	10,000 PE design
G. Jang et al., 2024	Yes	TEA	Phosphoric acid sludge	REEs	Industrial scale
Kuldeep et al., 2021	Yes	TEA	Metallurgical effluent	Na_2SO_4 recycling	100 m^3/h design
A. Ortiz-Ruíz et al., 2021	Yes	Pilot demonstration	Wastewater	Biomass/nutrients	700L-11.7m^3 systems
Eva Pakostova et al., 2024	Yes	Primary study	$LiCoO_2$ cathode	Co, Li	2L AGB, 203 days

The majority of studies (>70%) did not provide the specific CAPEX, OPEX, or break-even data requested for semiconductor or rare earth magnet wastewater applications. Studies providing quantitative economic data predominantly addressed lithium-ion battery recycling, phosphorus recovery, or general wastewater treatment rather than the specific target applications.

Capital Expenditure Analysis

Total CAPEX Estimates by Application

The available CAPEX data shows significant variation depending on system scale, waste stream complexity, and target metals. The following table consolidates reported capital expenditure figures:

Study	Application	Total CAPEX	Unit Basis	Key Components
M. Alipanah et al., 2024	LIB bioleaching	$44,000,000	Per 10,000 t/y facility	Bioleaching equipment
M. Alipanah et al., 2024	Acid production	$60,000,000	Per 20,000 t/y facility	Acid generating systems
Kuldeep et al., 2021	BPED sulfate recycling	EUR 20,000,000	100 m^3/h Na_2SO_4	Membrane, electrodes
Xiangjun Huang et al., 2020	IEX nutrient recovery	£3,480,000-3,600,000	10,000 PE WWTP	IEX installation (44%), tanks (33%)
Melissa K.L.N. Sikosana et al., 2017	Struvite production	R25,400,000-76,200,000	200 ML/d plant	Crystallization equipment
G. Jang et al., 2024	REE from PA sludge	$7,080,000-7,550,000	Industrial scale	Installation 15%, piping 15%, engineering 10%
A. Ortiz-Ruíz et al., 2021	HRAP system	€483	Per PE	Bioreactor, harvesting
Adam Smerigan et al., 2025	Bio-adsorption REE	$730,000,000 (resin) + $240,000,000 (ligand)	Large scale PG processing	Adsorbent systems

The CAPEX data reveals that biological recovery systems require substantial upfront investment, with bioleaching facilities for LIB processing estimated at $44 million for 10,000 t/y capacity. For REE recovery from phosphogypsum, the bio-adsorption resin and ligand components alone account for approximately $970 million in a large-scale deployment scenario. However, installation costs as a percentage of equipment vary between 10-15%, suggesting relatively straightforward scale-up from demonstrated systems.

Component Cost Breakdown

Limited component-level CAPEX breakdown was available across the literature. The most detailed breakdown comes from Xiangjun Huang et al. (2020), who reported that for ion exchange-based nutrient recovery systems, the IEX installation represents 44% of total CAPEX, with concrete tanks accounting for 33%. For the biological nutrient removal baseline comparison, 60% of costs were attributed to concrete tanks, 17% to tertiary filtration, and 15% to aeration systems.

Operating Expenditure Analysis

Total OPEX and Cost Drivers

Study	Application	Total OPEX	Unit	Major Cost Drivers
Vicki S. Thompson et al., 2017	REE bioleaching	$3,030,000/year	Annual	Nutrients 44%, General 20%, Electricity 10%
Kuldeep et al., 2021	BPED sulfate recycling	EUR 14,200,000/year	Annual	Electricity ~50%
Xiangjun Huang et al., 2020	IEX nutrient recovery	£177,000-316,000/year	Annual	Maintenance 31%, Sludge/labor 42%, Energy 17%
Ziwei Liu et al., 2024	Sidestream AnMBR	0.15 kWh/m³	Per m³ raw sewage	Energy (membrane fouling control)
A. Ortiz-Ruíz et al., 2021	HRAP wastewater	€0.19/m³	Per m³ treated	Not detailed
Weronika Urbańska et al., 2025	LIB bioleaching	$2,500-3,000/ton powder	Per reactor cycle	A. thiooxidans process
Weronika Urbańska et al., 2025	LIB bioleaching	$1,000-1,400/ton powder	Per reactor cycle	A. niger process
U. Nuñez-García et al., 2023	Biological WWTP	$1.23 MXN/m³	Per m³ treated	Not detailed

The dominant OPEX driver for bioleaching operations is the carbon/energy source for microbial growth. Vicki S. Thompson et al. (2017) identified glucose as constituting 98% of nutrient costs and 44% of total operational expenditure. This finding has critical implications for economic viability, as the authors noted that substituting lower-cost carbon sources could significantly improve profitability.

For membrane-based biological systems, energy consumption for fouling control represents the primary cost driver, accounting for 80-90% of operational costs in mainstream anaerobic membrane bioreactors. Sidestream configurations reduce this significantly to approximately 0.15 kWh/m³, compared to 0.4 kWh/m³ for mainstream systems.

Comparative Treatment Costs

A critical comparison between biological and chemical treatment approaches for lithium-ion battery recycling demonstrates potential cost advantages:

Treatment Method	Cost ($/ton powder)	Source
A. thiooxidans bioleaching	$2,500-3,000	Weronika Urbańska et al., 2025
A. niger bioleaching	$1,000-1,400	Weronika Urbańska et al., 2025
Chemical (citric acid)	$3,000-9,000	Weronika Urbańska et al., 2025
Control (A. thiooxidans)	$380	Weronika Urbańska et al., 2025

The fungal bioleaching approach using A. niger demonstrates superior economics compared to both bacterial bioleaching and conventional chemical treatment with citric acid. This suggests that organism selection significantly impacts process economics.

Economic Performance Metrics

Profitability Indicators

Study	Application	NPV	IRR	Payback Period	Key Assumptions
Vicki S. Thompson et al., 2017	REE from FCC	$5,780,000	44%	1 year	Tipping fee revenue included
Adam Smerigan et al., 2025	REE from PG	$570,000,000	>15% (87% of simulations)	4 years	>0.5 wt% REE content
M. Alipanah et al., 2024	LIB bioleaching	$18,400,000,000	Not reported	Not reported	10-year operation, 900,000 t/y
Rhulani Shingwenyana et al., 2021	AMD treatment	ZAR 60,000	26%	3 years	10-year plant life
G. Jang et al., 2024	REE from PA sludge	$178,700,000-441,800,000	Not reported	<1 year	PA-REO vs PA-P4-REO routes
Kuldeep et al., 2021	Sulfate recycling	Not reported	Not reported	11 years	Sensitive to chemical/electricity prices

The economic performance data reveals substantial variation in projected returns. The highest NPV ($18.4 billion) comes from the comprehensive LIB recycling supply chain model by Alipanah et al. (2024), though this represents an entire national-scale supply chain rather than individual facility economics. More realistic facility-level NPVs range from $570 million for large-scale REE recovery from phosphogypsum to modest returns of ZAR 60,000 for pilot-scale AMD treatment.

Break-even Analysis

Critical break-even thresholds identified in the literature:

Parameter	Break-even Value	Context	Source
Black mass purchase price	$8.7/kg	LIB recycling supply chain	M. Alipanah et al., 2024
Non-recyclable paper price	$0.89/kg	Acid production feedstock	M. Alipanah et al., 2024

Parameter	Break-even Value	Context	Source
Acid price	$8.5/kg	Supply chain viability	M. Alipanah et al., 2024
REE concentration in PG	0.5 wt%	Minimum for viability	Adam Smerigan et al., 2025
Phosphorus concentration	50 mg PO_4-P/L	Struvite precipitation	A. Magrí et al., 2020
Phosphorus concentration	50 mg/L	Economic feasibility	Yuanyao Ye et al., 2016

The break-even analysis reveals that feedstock quality and pricing are critical determinants of economic viability. For LIB recycling, the supply chain remains economically sustainable only when black mass costs remain below $8.7/kg. For REE recovery from phosphogypsum, concentrations below 0.5 wt% are not viable under current conditions due to acid and neutralization costs.

Revenue from Recovered Metals

Metal/Product	Market Price	Recovery Context	Source
REE from FCC	$116,000/year revenue	Bioleaching operation	Vicki S. Thompson et al., 2017
Neodymium oxide	€50/kg	Electrochemical recycling	P. Venkatesan et al., 2019
REO production cost	<€3/kg	Electrochemical process	P. Venkatesan et al., 2019
Struvite fertilizer	€50-100/tonne	Phosphorus recovery	Patrick M. Melia et al., 2017
Recovered water	ZAR 368/kL	AMD treatment	Rhulani Shingwenyana et al., 2021
Gypsum by-product	ZAR 1,100/tonne	AMD treatment	Rhulani Shingwenyana et al., 2021

The revenue potential varies substantially based on target metals and product purity. Vicki S. Thompson et al. (2017) noted that a 26-fold increase in REE prices from 2016 levels would be required for their bioleaching process to break even without tipping fee revenue, highlighting the importance of waste disposal fee offsets in the value proposition.

Comparative Economics: Biological vs Conventional Technologies

Head-to-Head Cost Comparisons

Technology	CAPEX	OPEX	Advantages	Limitations	Source
Biological (bioleaching)	Lower	Lower energy, no harsh chemicals	Environmental, lower cost	Slower kinetics	Multiple
Ion exchange	£3,480,000 (ASP+IEX)	£282,000/year	High selectivity	Resin replacement costs	Xiangjun Huang et al., 2020

Technology	CAPEX	OPEX	Advantages	Limitations	Source
Chemical precipitation	£3,940,000 (BNR+Fe)	£316,000/year	Established technology	Sludge disposal costs	Xiangjun Huang et al., 2020
BioIronTech	20-100× cheaper reagents	Lower than conventional	Uses iron ore vs Fe salts	Requires specific bacteria	V. Ivanov et al., 2014

The comparative analysis reveals that biological approaches offer cost advantages primarily through reduced chemical consumption and lower energy requirements. V. Ivanov et al. (2014) reported that the BioIronTech process uses iron ore as a reagent at $100-150/ton, which is 20-100 times less expensive than commercial ferric or ferrous salts. This cost differential represents a significant competitive advantage for biological treatment systems.

Whole Life Cost Comparisons

Xiangjun Huang et al. (2020) provided the most detailed whole life cost (WLC) comparison for nutrient recovery:

Configuration	40-year WLC	Savings vs BNR	Key Benefits
BNR + iron dosing	~£9.0M	Baseline	Established technology
ASP + IEX	~£7.4M	17%	Lower OPEX, N recovery
AnMBR + IEX	~£6.1M	27%	Energy from biogas, lowest OPEX

The IEX-based processes demonstrated 17-27% lower whole life costs compared to traditional biological nutrient removal. The anaerobic membrane bioreactor configuration achieved the lowest costs due to biogas production offsetting energy requirements.

Crossover Conditions

The literature identifies specific conditions where biological recovery becomes economically superior:

- **REE concentration threshold**: Processing capacities over 100,000 kg/h for phosphogypsum with REE content above 0.5 wt% are viable
- **Phosphorus concentration**: Economic feasibility achieved when influent P exceeds 50 mg/L
- **E-waste purchase price**: At reduced e-waste cost of $5/kg, bioleaching generates net benefits of $6.34-6.85/kg
- **Dilute REE sources**: Concentrations of 0.02-0.1 wt% are not viable under current conditions

Pilot-Scale Demonstrations and Commercial Status

Pilot-Scale Systems (≥100L)

Study	System Configuration	Volume	Duration	Performance	Target Materials
Nathan M. Good et al., 2023	Bioreactor	10L	Not specified	65-100% Nd recovery	REEs from e-waste
A. Ortiz-Ruíz et al., 2021	Lamella settler + PBR	700L + 11.7m³×3	6 months	<5 NTU turbidity	Biomass harvesting
J. Abraham et al., 2023	Open raceway ponds	1000L×2	Summer 2021	23.9 g/m²d productivity	Bioenergy from wastewater
Eva Pakostova et al., 2024	Acid-generating bioreactor	2L	203 days	>99.9% Co recovery	$LiCoO_2$ cathode materials
Óscar Barros et al., 2019	Column setup	2×30cm columns	26 hours continuous	>90% La, Ce removal	REEs from wastewater

The pilot-scale demonstrations identified in this review predominantly fall below the 100L threshold specified in the research question, with the notable exception of the microalgae systems reported by A. Ortiz-Ruíz et al. (2021), which operated at 11.7 m³ scale for 6 months. This represents the largest and longest-duration pilot demonstration in the reviewed literature for biological wastewater treatment with resource recovery potential.

Commercial Deployments and Industry Status

Limited commercial deployment information was identified:

Entity	Technology	Status	Application	Source
SUEZ Water Technologies	BPED systems	Commercial	Sulfate recycling	Kuldeep et al., 2021
BioElectroMET	BES for metals	Pilot/full-scale	Wastewater treatment	Pieter Ostermeyer et al., 2021
Inotec	Electrified bioreactors	Full-scale	25-50% OPEX/CAPEX reduction	Pieter Ostermeyer et al., 2021

The Inotec technology represents a notable commercial success, with full-scale deployments achieving 25-50% reductions in both CAPEX and OPEX compared to conventional biological treatment. However, this technology addresses general wastewater treatment rather than the specific critical raw material recovery applications targeted in this review.

Technology Readiness Assessment

The reviewed literature suggests that biological metal recovery technologies remain at relatively early technology readiness levels (TRL 4-6) for most critical raw material applications. Key scaling challenges reported include:

- Slow process kinetics compared to chemical methods
- Metal toxicity effects on microbial communities
- Low pulp density limitations

- Process control complexity with polymetallic waste streams
- Mass transfer limitations at scale

Patents and Intellectual Property

Limited patent information was available in the extracted data. Eva Benzal Montes et al. reported a registered patent (P201830406) for bioleaching processes, though it remains under evaluation. The relative scarcity of patent activity suggests that biological metal recovery for critical raw materials remains largely in the academic research phase.

Life Cycle Assessment Data

Environmental Performance Metrics

Study	Impact Category	Value	Comparison Basis	Source
S. Balchandani et al., 2023	Carbon footprint	2.07 kg CO_2-eq/kg GA	NP substrate best	TEA+LCA
Adam Smerigan et al., 2025	Ecosystem quality	93% of conventional	vs conventional REO	TEA+LCA
Adam Smerigan et al., 2025	Resource depletion	96.2% of conventional	vs conventional REO	TEA+LCA
Adam Smerigan et al., 2025	Human health	213% of conventional	Higher impacts	TEA+LCA
P. Venkatesan et al., 2019	Energy intensity	6 kWh/kg	Electrochemical REE	Primary study
Eleonora Santos et al., 2022	CO_2 sequestration	146.27×10^{-3} kg CO_2-eq/m^3	HRAP systems	Review
Bita Deravian et al., 2025	E-factor	30-50 kg waste/kg metal	vs 150-300 conventional	Review

The LCA data reveals a nuanced environmental picture. While biological approaches generally outperform conventional methods in ecosystem quality and resource depletion categories, they can underperform in human health impacts. The biosurfactant-based recovery approach demonstrates particularly strong environmental performance, achieving 70-85% lower Environmental Factors compared to conventional acid leaching.

Comparison to Virgin Mining

Limited direct LCA comparisons between bio-recovered and virgin-mined materials were identified. Adam Smerigan et al. (2025) provided the most comprehensive comparison, showing that REE recovery from phosphogypsum achieves:

- 93% of ecosystem quality impacts compared to conventional REO production
- 96.2% of resource depletion impacts
- Net negative water depletion potential (water returned to environment)

However, the same study noted higher human health impacts (213% of conventional), primarily attributed to chemical consumption in the process.

Market Drivers and Business Case Factors

Value Proposition Analysis

The literature reveals a dual value proposition for biological metal recovery:

Compliance-driven factors:

- Avoiding EPA fines and meeting discharge limits
- Environmental regulations requiring reduced chemical use
- EU Green Deal and China carbon-neutral goals
- Circular economy policy requirements

Revenue-driven factors:

- Recovered metal value potential
- Cost savings through reduced chemical consumption
- By-product revenue (biogas, fertilizers)
- Avoided disposal costs

Vicki S. Thompson et al. (2017) explicitly noted that their bioleaching process required tipping fee revenue (waste disposal offset) to achieve profitability, with REE revenue alone being insufficient without a 26-fold price increase. This suggests that for many applications, the compliance/disposal cost avoidance value proposition currently dominates over metal recovery revenue.

Regulatory Drivers

Region/Framework	Key Requirements	Impact on Bio-recovery	Source
EU Battery Regulation	Full life cycle management	Drives LIB recycling investment	Weronika Urbańska et al., 2025
EU Critical Raw Materials Act	Supply chain security	Supports domestic recovery	Chiara Magrini et al., 2025
South Africa discharge standard	1 mg P/L	Drives P recovery technology	Melissa K.L.N. Sikosana et al., 2017
US DOE goals	Recycling cost parity with landfill by 2030	Technology development funding	Morgan V Evans et al., 2025

Barriers to Adoption

The most frequently cited barriers across the literature include:

1. **Technical barriers**: Slow kinetics compared to chemical methods, metal toxicity to microorganisms, complexity of polymetallic waste streams
2. **Economic barriers**: High CAPEX requirements, feedstock price sensitivity, competition with established chemical methods
3. **Scale-up challenges**: Mass transfer limitations, biomass productivity decline at scale, process control complexity

4. **Market factors**: Low metal prices requiring supplementary revenue streams, competition with export to established refiners

Success Factors for Commercial Viability

Based on the reviewed literature, key success factors include:

- **Feedstock optimization**: Using lower-cost carbon sources (waste materials vs glucose)
- **Integrated value streams**: Combining metal recovery with waste disposal fees
- **Process intensification**: Achieving higher throughput through optimized bioreactor design
- **Organism engineering**: Developing strains with enhanced metal tolerance and uptake
- **Policy support**: Leveraging EPR subsidies and eco-tax mechanisms

Synthesis

The reviewed literature reveals significant heterogeneity in reported economic performance, which can be reconciled through examination of key contextual factors.

Reconciling Economic Performance Variation

The wide range of reported NPVs ($60,000 to $18.4 billion) reflects fundamentally different system scales and value chain scopes rather than inconsistent methodologies. Studies reporting high NPVs (M. Alipanah et al., 2024 at $18.4 billion) model national-scale supply chain optimization across multiple facilities, while modest NPVs (Rhulani Shingwenyana et al., 2021 at ZAR 60,000) represent pilot-scale single-facility economics. When normalized to comparable scales, the economic returns converge toward IRRs of 15-44%, suggesting reasonable consistency across applications.

Feedstock Quality as Primary Economic Determinant

The reviewed studies consistently identify feedstock metal concentration as the dominant factor determining economic viability. Adam Smerigan et al. (2025) established that REE concentrations above 0.5 wt% are required for profitability, while concentrations of 0.02-0.1 wt% are economically unviable under current conditions. This concentration threshold explains why certain waste streams (e.g., FCC catalysts, concentrated industrial wastes) show positive economics while dilute semiconductor wastewaters have received less commercial attention.

Carbon Source Costs as Key OPEX Driver

The finding that glucose/carbon source represents 44-98% of nutrient costs in bioleaching operations provides a clear optimization target. Studies using waste-derived carbon sources (corn stover, date palm clippings, non-recyclable paper) demonstrate significant cost advantages, suggesting that process economics could improve substantially through feedstock substitution. The cost differential between chemical treatment ($3,000-9,000/ton) and fungal bioleaching ($1,000-1,400/ton) demonstrates the potential impact of organism and substrate optimization.

Scale-Dependent Economics

The literature suggests different economic optima at different scales:

- **Small scale (<1,000 L)**: Labor costs dominate OPEX; biological systems struggle to compete with chemical treatment
- **Medium scale (1,000-10,000 L)**: Energy and chemicals become primary cost drivers; biological advantages begin to emerge
- **Large scale (>100,000 L)**: Capital efficiency and feedstock costs dominate; biological systems can achieve economies of scale

Technology Readiness Gaps

The systematic gap between laboratory-scale demonstrations (most studies) and commercial deployment (limited examples) represents the most significant barrier to answering the research question's specific focus on industrial-scale economics. The few pilot demonstrations identified do not specifically address semiconductor or rare earth magnet wastewater applications, leaving a critical knowledge gap for the target applications.

Implications for Target Applications

For semiconductor manufacturing wastewater and rare earth magnet production waste specifically, the evidence suggests:

1. **Semiconductor wastewater**: Limited direct economic data available; the dilute metal concentrations typical of semiconductor rinse waters may fall below the 0.5 wt% viability threshold identified for other applications

2. **Rare earth magnet waste**: More promising economics given higher metal concentrations in magnet swarf; S. Balchandani et al. (2023) demonstrated that biolixiviant derived from non-recyclable paper achieves cost-effective extraction from NdFeB magnet swarf

3. **Competitive position vs export**: The current economic data does not directly address whether biological recovery can compete with magnet scrap export to established refiners, though the cost advantages of bio-derived lixiviants ($0.04-0.09/kg) suggest potential competitiveness if scaled appropriately

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What are the demonstrated biosorption and bioaccumulation performance metrics (recovery efficiency %, adsorption capacity mg/g, equilibrium time, selectivity ratio, cost per m^3) for unicellular microalgae recovering Copper (Cu), Cerium (Ce), Gallium (Ga), Germanium (Ge), and Indium (In) from semiconductor CMP (Chemical Mechanical Polishing) wastewater and wet etching waste streams? Focus on: • Microalgae genera: Chlorella, Scenedesmus, Nannochloropsis, Spirulina, Galdieria, Cyanidium, Dunaliella, Chlamydomonas, Haematococcus, Botryococcus • Waste stream specifics: CMP slurry (pH 2-4, 100-1000 ppm Cu, 50-300 ppm Ce), HF-based etchants, compound semiconductor waste (GaAs, GaN, SiGe) • Dead biomass vs living cells: compare heat-killed, acid-treated, or dried biomass performance against living cultures • Interference factors: silica nanoparticle effects, EDTA chelator competition, high ionic strength effects • Discharge quality: Can the system achieve <0.5 ppm Cu and <0.1 ppm REE discharge to meet 2026 EPA NPDES standards? Required data: 1. Metal removal efficiency (%) for Cu, Ce, Ga, Ge specifically 2. Maximum adsorption capacity (q_max in mg/g dry biomass) 3. Selectivity: Ce/Ca ratio, Ga/Al ratio, Ge/Si ratio 4. Kinetics: equilibrium time (minutes) and rate constants 5. Optimal pH and temperature 6. OPEX estimate (€ or $ per m^3 wastewater treated) 7. Biomass regeneration cycles achieved 8. Scale of demonstration (lab vs pilot vs commercial) Timeframe: 2019-2026 preferred; include 2015-2018 if pilot-scale data available. Study types: Include laboratory experiments, pilot studies, industrial case studies, and techno-economic analyses.

While microalgae (primarily Chlorella species) demonstrate copper removal efficiencies of 57-99% and adsorption capacities of 0.1-764.5 mg/g from synthetic solutions at laboratory scale, no studies have examined actual CMP slurry or compound semiconductor waste streams, and critical data gaps exist for gallium and germanium (no studies), cerium (limited to non-semiconductor REE solutions), selectivity ratios (Ce/Ca, Ga/Al, Ge/Si unreported), and operating costs, precluding evidence-based assessment of microalgae feasibility for semiconductor wastewater treatment or compliance with discharge standards.

Abstract

Microalgae biosorption of copper from synthetic and industrial wastewaters is well-documented, with Chlorella species achieving removal efficiencies of 57-99% and maximum adsorption capacities ranging from 0.1 to 764.5 mg/g

dry biomass under optimal conditions of pH 4-7 and temperatures of 25-35℃. Equilibrium times vary from 4 minutes for magnetic nanoparticle-algae composites to 48 hours for living cell cultures. However, critical gaps exist for semiconductor-specific applications: only one study examined indium recovery from ITO etching waste, achieving 80% removal with 0.14 mmol/g capacity and demonstrating selectivity over Cu, Zn, and Al. Cerium biosorption data is limited to Galdieria sulphuraria studies on REE solutions, while no studies examined gallium or germanium biosorption by microalgae. No studies addressed actual CMP slurry, HF-based etchants, or compound semiconductor waste streams (GaAs, GaN, SiGe), and selectivity ratios for Ce/Ca, Ga/Al, and Ge/Si were not reported in any included study.

The evidence cannot support feasibility assessments for semiconductor wastewater treatment: all but one study operated at laboratory batch scale, no techno-economic analyses provided OPEX estimates, and most studies failed to achieve final copper concentrations below 0.5 ppm required for stringent discharge standards. Biomass regeneration was demonstrated for only 2-5 cycles in limited studies. Dead biomass generally shows higher stability and comparable or superior biosorption efficiency compared to living cells, though living cells can achieve higher uptake through combined biosorption and bioaccumulation mechanisms. Interference factors specific to semiconductor waste (silica nanoparticles, EDTA chelators) were not examined. The current evidence base is insufficient to determine whether microalgae biosorption can meet 2026 EPA NPDES standards for semiconductor wastewater treatment.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the demonstrated biosorption and bioaccumulation performance metrics (recovery efficiency %, adsorption capacity mg/g, equilibrium time, selectivity ratio, cost per m^3) for unicellular microalgae recovering Copper (Cu), Cerium (Ce), Gallium (Ga), Germanium (Ge), and Indium (In) from semiconductor CMP (Chemical Mechanical Polishing) wastewater and wet etching waste streams?

Focus on: • Microalgae genera: Chlorella, Scenedesmus, Nannochloropsis, Spirulina, Galdieria, Cyanidium, Dunaliella, Chlamydomonas, Haematococcus, Botryococcus • Waste stream specifics: CMP slurry (pH 2-4, 100-1000 ppm Cu, 50-300 ppm Ce), HF-based etchants, compound semiconductor waste (GaAs, GaN, SiGe) • Dead biomass vs living cells: compare heat-killed, acid-treated, or dried biomass performance against living cultures • Interference factors: silica nanoparticle effects, EDTA chelator competition, high ionic strength effects • Discharge quality: Can the system achieve <0.5 ppm Cu and <0.1 ppm REE discharge to meet 2026 EPA NPDES standards?

Required data:

1. Metal removal efficiency (%) for Cu, Ce, Ga, Ge specifically
2. Maximum adsorption capacity (q_max in mg/g dry biomass)
3. Selectivity: Ce/Ca ratio, Ga/Al ratio, Ge/Si ratio
4. Kinetics: equilibrium time (minutes) and rate constants
5. Optimal pH and temperature
6. OPEX estimate (€ or $ per m^3 wastewater treated)
7. Biomass regeneration cycles achieved
8. Scale of demonstration (lab vs pilot vs commercial)

Timeframe: 2019-2026 preferred; include 2015-2018 if pilot-scale data available. Study types: Include laboratory experiments, pilot studies, industrial case studies, and techno-economic analyses."

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Target Microorganisms**: Does the study use unicellular microalgae from at least one of the following genera: Chlorella, Scenedesmus, Nannochloropsis, Spirulina, Galdieria, Cyanidium, Dunaliella, Chlamydomonas, Haematococcus, or Botryococcus?
- **Target Metals**: Does the study investigate recovery of at least one of the following metals: copper (Cu), cerium (Ce), gallium (Ga), germanium (Ge), or indium (In)?
- **Wastewater Source**: Does the study use semiconductor industry wastewater (such as CMP slurry, HF-based etchants, compound semiconductor waste) or synthetic solutions that mimic these semiconductor waste streams?
- **Performance Metrics**: Does the study report at least two quantitative performance metrics (such as removal efficiency %, adsorption capacity mg/g, equilibrium time, selectivity ratios, kinetic parameters, optimal conditions, or cost estimates)?
- **Study Type and Quality**: Is the study a peer-reviewed research article (laboratory experiment, pilot study, industrial case study, techno-economic analysis, systematic review, or meta-analysis) with sufficient methodological detail, rather than a conference abstract, patent, or non-peer-reviewed publication?
- **Publication Timeframe**: Was the study published between 2015 and 2026?
- **Microorganism Specificity**: Does the study focus on microalgae (rather than exclusively on macroalgae, bacteria, or fungi) and include at least one of the specified microalgae genera listed in the target microorganisms criterion?
- **Application Relevance**: Does the study focus on semiconductor wastewater treatment or metal recovery relevant to semiconductor applications (rather than exclusively focusing on mining wastewater, municipal wastewater, or other non-semiconductor industrial applications with non-target metals)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

190 papers passed abstract screening and moved to full-text screening.

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Target Microorganisms**: Does the study use unicellular microalgae from at least one of the following genera: Chlorella, Scenedesmus, Nannochloropsis, Spirulina, Galdieria, Cyanidium, Dunaliella, Chlamydomonas, Haematococcus, or Botryococcus?
- **Target Metals**: Does the study investigate recovery of at least one of the following metals: copper (Cu), cerium (Ce), gallium (Ga), germanium (Ge), or indium (In)?
- **Wastewater Source**: Does the study use semiconductor industry wastewater (such as CMP slurry, HF-based etchants, compound semiconductor waste) or synthetic solutions that mimic these semiconductor waste streams?
- **Performance Metrics**: Does the study report at least two quantitative performance metrics (such as removal efficiency %, adsorption capacity mg/g, equilibrium time, selectivity ratios, kinetic parameters, optimal conditions, or cost estimates)?

- **Study Type and Quality**: Is the study a peer-reviewed research article (laboratory experiment, pilot study, industrial case study, techno-economic analysis, systematic review, or meta-analysis) with sufficient methodological detail, rather than a conference abstract, patent, or non-peer-reviewed publication?
- **Publication Timeframe**: Was the study published between 2015 and 2026?
- **Microorganism Specificity**: Does the study focus on microalgae (rather than exclusively on macroalgae, bacteria, or fungi) and include at least one of the specified microalgae genera listed in the target microorganisms criterion?
- **Application Relevance**: Does the study focus on semiconductor wastewater treatment or metal recovery relevant to semiconductor applications (rather than exclusively focusing on mining wastewater, municipal wastewater, or other non-semiconductor industrial applications with non-target metals)?

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

63 papers passed full-text screening and moved to data extraction.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Microalgae Species**:

  Extract details about the microalgae used for biosorption/bioaccumulation of semiconductor metals, including:

- Specific species name (focus on Chlorella, Scenedesmus, Nannochloropsis, Spirulina, Galdieria, Cyanidium, Dunaliella, Chlamydomonas, Haematococcus, Botryococcus)
- Biomass treatment method (living cells, heat-killed, acid-treated, dried, freeze-dried, immobilized, cross-linked)
- Biomass preparation conditions (temperature, pH, duration of treatment)
- Biosorbent dosage (g/L or mg/L)

- **Waste Stream Characteristics**:

  Extract specific details about semiconductor waste streams studied, including:

- Waste stream type (CMP slurry, HF-based etchants, compound semiconductor waste like GaAs/GaN/SiGe)
- Initial metal concentrations for Cu, Ce, Ga, Ge, In (in ppm or mg/L)
- Solution pH range
- Interference factors present (silica nanoparticles, EDTA chelators, high ionic strength, competing metals)
- Other relevant waste stream characteristics affecting biosorption

- **Metal Recovery Performance**:

  Extract quantitative performance metrics for Cu, Ce, Ga, Ge, and In recovery specifically, including:

- Metal removal efficiency (%) for each target metal
- Maximum adsorption capacity (qmax in mg/g dry biomass) for each metal
- Equilibrium time (minutes or hours) to reach maximum uptake
- Kinetic rate constants (k1, k2) if reported
- Contact time tested and optimal contact time identified

- **Selectivity Ratios**:

  Extract selectivity data for target metals in semiconductor waste recovery, including:

- Ce/Ca selectivity ratio (for cerium selectivity over calcium)
- Ga/Al selectivity ratio (for gallium selectivity over aluminum)
- Ge/Si selectivity ratio (for germanium selectivity over silicon)
- Cu selectivity over other common metals (Fe, Zn, Ni)
- In selectivity data relative to competing metals
- Any other selectivity coefficients reported for target metals

- **Process Optimization**:

  Extract optimal process conditions identified for semiconductor metal biosorption, including:

- Optimal pH range for each target metal (Cu, Ce, Ga, Ge, In)
- Optimal temperature (℃)
- Agitation rate or mixing conditions
- Initial metal concentration range tested
- Biomass concentration optimized
- Sequential vs simultaneous metal removal approach

- **Scale and Economics**:

  Extract scale and economic data for semiconductor waste treatment systems, including:

- Scale of demonstration (laboratory batch, laboratory continuous, pilot-scale, commercial)
- Volume of wastewater treated (mL, L, m^3)
- Operating cost estimates (€ or $ per m^3 wastewater treated)
- Biomass regeneration cycles achieved before performance decline
- Regeneration method used (acid elution, alkaline treatment, etc.)
- Capital cost estimates if provided

- **Discharge Quality**:

  Extract final effluent quality data and regulatory compliance potential, including:

- Final Cu concentration achieved in treated effluent (ppm or mg/L)
- Final concentrations of Ce, Ga, Ge, In in treated effluent
- Whether discharge levels meet <0.5 ppm Cu and <0.1 ppm REE targets
- Percentage reduction from initial concentrations
- Consistency of treatment performance across multiple cycles
- Any discussion of meeting 2026 EPA NPDES standards

- **Dead vs Living Comparison**:

  Extract comparative performance data between dead and living microalgae biomass for semiconductor metal recovery, including:

- Direct comparison of removal efficiency (%) between dead and living cells for same microalgae species
- Adsorption capacity differences between biomass states
- Stability differences (dead biomass often more stable)
- Cost implications of biomass preparation methods

- Mechanisms discussed (biosorption vs bioaccumulation)
- Advantages/disadvantages of each approach for industrial application

Results

Characteristics of Included Studies

This systematic review identified 63 sources examining microalgae biosorption and bioaccumulation of metals relevant to semiconductor wastewater treatment. The majority of studies were conducted at laboratory batch scale, with limited pilot-scale demonstrations. Notably, while the research question focused on Cu, Ce, Ga, Ge, and In recovery from semiconductor CMP wastewater and wet etching waste streams, no studies directly examined CMP slurry or compound semiconductor waste (GaAs, GaN, SiGe). Most studies used synthetic metal solutions or general industrial wastewater simulants.

Study	Full text retrieved?	Study Type	Primary Microalgae	Target Metals	Waste Stream Type
K. C. Wanta et al., 2023	Yes	Primary study	Chlorella sp.	Cu	Synthetic CuSO4 solution
N. R. Nicomel et al., 2020	Yes	Primary study	Chlorella sp.	In	ITO etching processes
Jesse T. Phiri et al., 2024	Yes	Primary study	C. vulgaris, Scenedesmus sp., S. platensis	Cd, Co, Cu	Not specified
Tuğba Şentürk et al., 2018	Yes	Primary study	Chlorella, Scenedesmus	Cu, Ni, Sb, Mn	Not specified
Chonghua Li et al., 2023	Yes	Primary study	Chlamydomonas reinhardtii	Cu, Zn	Not specified
A. Rinanti et al., 2022	Yes	Primary study	C. sorokiniana, Monoraphidium sp., S. obliquus	Cu	Artificial Cu^{2+} waste
Imtinen Ghribi et al., 2025	Yes	Primary study	Chlamydomonas sp.	Cu, Cr	Not specified
K. C. Wanta et al., 2020	Yes	Primary study	Chlorella sp.	Cu	Not specified
Saavedra Concha et al., 2020	Yes	Primary study	C. reinhardtii, C. vulgaris, S. almeriensis, Chlorophyceae spp.	Cu, Mn, Zn, As, B	Not specified
T. Wilan et al., 2019	Yes	Primary study	C. vulgaris, Chlorococcum sp., S. obliquus	Cu	Artificial electroplating waste

Study	Full text retrieved?	Study Type	Primary Microalgae	Target Metals	Waste Stream Type
Fangqi Chen et al., 2016	Yes	Primary study	Oscillatoria, Botryococcus, Scenedesmus, Spirulina platensis	Cu, Zn, Ni	Not specified
T. I. Oliomogbe et al., 2024	Yes	Primary study	Chlorella sorokiniana	Cu, Zn	Synthetic wastewater
Eleni Kyratzopoulou et al., 2025	Yes	Primary study	Chlorella vulgaris	Cu, Ni, Pb, Zn, Cd	Not specified
N. Politaeva et al., 2022	Yes	Primary study	Chlorella kessleri	Cu	Model wastewater
W. P. Siwi et al., 2018	Yes	Primary study	Chlorella sp.	Cu	Not specified
Elżbieta Sitarz-Palczak et al., 2025	Yes	Primary study	Chlorella vulgaris	Cu, Mn, Pb, Zn	Galvanic sewage sludge
E. Sočo et al., 2024	Yes	Primary study	Chlorella vulgaris	Cu, Co	Freshwater
M. Plouviez et al., 2025	Yes	Primary study	Chlamydomonas reinhardtii	Ce, Gd, Yb	Not specified
Huan Lin et al., 2024	Yes	Primary study	Chlorella sp.	Cu, Cr(VI)	Not specified
Shaomin Liu et al., 2024	Yes	Primary study	Chlorella vulgaris	Cu, Cd, Zn	Not specified
N. Sultana et al., 2020	Yes	Primary study	Chlorella kessleri	Pb, Cu, Co, Cd, Cr	Not specified
M. Palmieri et al., 2022	Yes	Primary study	Galdieria sulphuraria ACUF427	Y, Ce, Eu, Tb	Not specified
A. Napiórkowska-Krzebietke et al., 2012	Yes	Primary study	Chlorella vulgaris	Cu, Cd, Pb	Not specified
Marcus Heilmann et al., 2021	Yes	Primary study	Chlorella kessleri	Nd, Eu, Au, Pt, Pb, Fe, Cu, Ni	Not specified
Jens Kastenhofer et al., 2024	Yes	Primary study	Galdieria sulphuraria	REEs (La)	Not specified
Rishi Kumar et al., 2014	Yes	Primary study	Scenedesmus obliquus, Synechocystis sp. PCC 6803	Cu	Not specified

Study	Full text retrieved?	Study Type	Primary Microalgae	Target Metals	Waste Stream Type
Shireen Ibrahim Hamadamin et al., 2022	Yes	Primary study	Chlorella sp.	Cu	Not specified
N. A. Lieswito et al., 2019	Yes	Primary study	C. sorokiniana, Monoraphidium sp., S. obliquus	Cu	Not specified
C. García-Balboa et al., 2022	Yes	Primary study	Euglena sp., Chlamy-domonas sp.	Cu, Al, Zn, Mn, REEs	Electronic scraps
M. Silalahi et al., 2018	Yes	Primary study	Ankistrodesmus sp., Chlorella sp.	Cu	Electroplating wastewater
A. Rinanti et al., 2018	Yes	Primary study	Chlorella sp., A. braunii, S. quadricauda	Cu	Not specified
K. A. Salam et al., 2019	Yes	Review	Not specified	Heavy metals general	Not specified
K. A. Salam et al., 2019a	Yes	Review	Not specified	Heavy metals general	Not specified
J. Ordóñez et al., 2023	Yes	Review	C. vulgaris, Spirulina sp.	Cu	Real effluents
A. Zeraatkar et al., 2016	Yes	Review	C. vulgaris, S. obliquus	Heavy metals general	Not specified
R. Saavedra et al., 2018	Yes	Primary study	C. reinhardtii, C. vulgaris, S. almeriensis, Chlorophyceae spp.	Cu, Mn, Zn, As, B	Not specified
Shireen Ibrahim Hamadamin et al., 2022a	Yes	Primary study	Chlorella sp.	Cu	Not specified
I. Moreno-Garrido et al., 2002	Yes	Primary study	Nannochloropsis gaditana	Cu, Zn	Not specified
Sarah K. Read et al., 2023	Yes	Review	Not specified	PTEs general	Not specified
M. Iovinella et al., 2022	Yes	Primary study	Galdieria sulphuraria	Y, Ce, Eu, Tb	Not specified
M. Fawzy et al., 2024	Yes	Primary study	C. ellipsoideum, D. subspicatus	Cu	Real wastewater
Mic Paper et al., 2023	Yes	Primary study	Cyanobacteria (Nostoc sp., S. elongatus, C. brevissima)	Ce, Nd, Tb, La	Not specified

Study	Full text retrieved?	Study Type	Primary Microalgae	Target Metals	Waste Stream Type
Hee Taek Kim et al., 2022	Yes	Primary study	Nannochloropsis oculata	V	Not specified
Dimitri Giarikos et al., 2021	Yes	Primary study	Neochloris minuta, N. alveolaris	Cu, Pb, Cd, Zn, Ni	Not specified
A. Morcelli et al., 2025	Yes	Primary study	C. sorokiniana, H. luteoviridis	Cu, Cr, Fe, Zn	Tannery/electroplating wastewater
Ntombiphumile Perceverence Tenza et al., 2025	Yes	Primary study	Chlorella sp.	Cu, Pb, Zn	Wastewater (South Africa)
Katerine Napan et al., 2015	Yes	Primary study	Nannochloropsis salina	Cu (among 14 contaminants)	Flue gas-derived contaminants
Vikrant Pachouri et al., 2024	Yes	Primary study	Chlorella Minutissima, C. Singularis	Heavy metals	Not specified
Hari Lal Kharel et al., 2024	Yes	Primary study	Galdieria sulphuraria CCMEE 5587.1	Cd	Not specified
Gisele Petronilho Heidelmann et al., 2017	Yes	Primary study	Chlorella vulgaris	La	Not specified
T. Pham et al., 2019	Yes	Primary study	Pediastrum sp.	Cu	Not specified
Karima Guehaz et al., 2025	Yes	Primary study	Parachlorella hussii	Cu, Zn, Ni	Not specified
E. Sočo et al., 2025	Yes	Primary study	Spirulina sp.	Ni, Pb	Model wastewater
Tímea Fóris et al., 2025	Yes	Primary study	Chlorella vulgaris	Co	Industrial wastewater
A. Chugainova et al., 2023	Yes	Primary study	C. sorokiniana, C. vulgaris, Scenedesmus sp.	Multiple metals from e-waste	Electronic waste
Tiara Wilan et al., 2019	Yes	Review	Chlorella sp.	Cu, Zn, Cr	Not specified
Y. Wong et al., 1998	Yes	Review	Chlorella vulgaris	Cu	Not specified
Hari Lal Kharel et al., 2023	Yes	Primary study	Galdieria sulphuraria CCMEE 5587.1	Cd, Pb	Not specified
M. A. Kucuker et al., 2017	Yes	Primary study	Chlorella vulgaris	Nd	Hard disk drive magnets
Patricia Tapia et al., 2011	Yes	Primary study	Ulva (green algae)	Cu	Not specified

Study	Full text retrieved?	Study Type	Primary Microalgae	Target Metals	Waste Stream Type
A. Converti et al., 2006	Yes	Primary study	Spirulina platensis	Cu	Not specified
Anjali Singh et al., 2023	Yes	Primary study	Galdieria sulphuraria	REEs from CFL bulbs	CFL bulb waste
S. Alvis et al., 2017	Yes	Primary study	Chlorella sp.	Hg	Not specified

The table reveals that Chlorella species were the most frequently studied microalgae, appearing in over 35 studies. Copper was the dominant target metal, with cerium addressed in only 4 studies, and indium in only one study. No studies specifically examined gallium or germanium biosorption by microalgae. Only one study examined ITO etching waste streams relevant to semiconductor manufacturing, and one study examined electronic scraps containing multiple metals.

Metal Recovery Performance

Copper (Cu) Recovery

Copper biosorption was extensively documented across the included studies, with removal efficiencies ranging from 57% to 99% depending on species, biomass treatment, and operating conditions.

Microalgae Species	Removal Efficiency (%)	q_max (mg/g)	Equilibrium Time	Optimal pH	Optimal Temp (℃)
Chlorella sp. (living)	96.83%	Not reported	24 h	5	Room temp
Chlorella sp. (living)	96.10%	Not reported	48 h	5	Not specified
Chlorella vulgaris (living)	65-99%	Not reported	Days 3-7	Not specified	20
Chlorella kessleri (dried)	87%	4.2	Not specified	6	22
Chlorella sorokiniana (dried)	85-90.7%	Not reported	60 min	5	28
Chlorella vulgaris (dead)	97.7%	Not reported	20 min	5	Room temp
Chlorella vulgaris (powder)	Not specified	30.3	<2.5 h	2-11	20
Chlorella vulgaris (living)	93.63%	764.5	20 min	Neutral-alkaline	30
Scenedesmus	20.50%	27.09	24 h	4-5	28
Chlorella	41.02%	28.57	24 h	4-5	28

Microalgae Species	Removal Efficiency (%)	q_max (mg/g)	Equilibrium Time	Optimal pH	Optimal Temp (℃)
Scenedesmus obliquus (living)	Not specified	124 µg/mg (0.124 mg/g)	1 h	5-7	25
Synechocystis sp. PCC 6803 (living)	Not specified	64.9 µg/mg	2 h	5-7	25
Mixed culture immobilized	92%	0.19	120 min	4	25
Mixed culture immobilized	83.4%	7.43	120 min	4	30
Mixed microalgae immobilized	96.4%	Not reported	180 min	4	35
Mixed culture immobilized	86.39%	7.69	120 min	4	30
Immobilized biosorbent	66.36%	0.1025	7.34 min	4	26
Chlorella biosorbent beads	94.77%	Not reported	200-220 min	Not specified	35
Chlorophyceae spp.	88%	Not reported	10 min	7	23
C. ellipsoideum	68.66%	0.398	Not specified	6.8	20
D. subspicatus	59.52%	0.396	Not specified	6.8	20
Neochloris alveolaris	60.08%	22.65-64.18	1 h	5	30
Pediastrum sp.	83-95%	Not reported	Not specified	Not specified	28
Spirulina platensis (dried)	81-91%	Not reported	1-3 h	Not specified	Room temp
Parachlorella hussii	51.15%	18.55	≥12 h	Not specified	Not specified
Chlorella sp. (dried)	100%	Not reported	60 min	7	25
Fe_3O_4/SiO_2 with Chlorella	86.9%	Not reported	4 min	8	25
MNPs@Algae composite	73.4%	Not reported	6 h	6	25
Chlorella sp. (57.2%)	57.2-59.7%	0.24	15 h	8	Not specified
Nannochloropsis gaditana (immob.)	~100%	~50 mg/kg	24 h	Not specified	Not specified
Ulva (activated)	96-98%	Not reported	Not specified	Not specified	Not specified
Geopolymer with algae	96.9%	29	~15 min	Not specified	Room temp

Microalgae Species	Removal Efficiency (%)	q_max (mg/g)	Equilibrium Time	Optimal pH	Optimal Temp (℃)
H. luteoviridis	61.4%	Not reported	Not specified	Not specified	Not specified
C. sorokiniana	65.6%	Not reported	Not specified	Not specified	Not specified

The highest copper removal efficiency of 100% was achieved by dried Chlorella sp. biomass under optimized conditions of pH 7, 60 minutes contact time, and 12.5 mg biomass dosage. Living Chlorella vulgaris demonstrated the highest reported adsorption capacity of 764.5 mg/g with 93.63% removal efficiency, though this exceptionally high value may reflect differences in experimental methodology compared to other studies reporting capacities in the 4-30 mg/g range.

Equilibrium times varied considerably, from as rapid as 4 minutes for magnetic nanoparticle-algae composites to 48 hours for living Chlorella sp. cultures. Most immobilized biosorbent systems achieved equilibrium within 120-180 minutes.

Indium (In) Recovery

Only one study specifically examined indium biosorption from semiconductor-relevant waste streams. Nicomel et al. (2020) demonstrated that microalgal biomass grown on wastewater achieved 80% In(III) removal within 30 minutes from ITO etching process waste at pH 2. The maximum adsorption capacity was 0.14 mmol/g, with selectivity observed over Cu(II), Zn(II), and Al(III). The presence of Cu(II) affected In(III) biosorption the least, showing only a 10% decrease in removal efficiency. Desorption efficiencies of 87% with EDTA and 80% with HCl enabled biomass reuse for 5 cycles.

Cerium (Ce) and Rare Earth Elements

Limited data exists for cerium biosorption by microalgae. Galdieria sulphuraria ACUF427 demonstrated removal of Ce from quaternary REE solutions, with performance dependent on pH and strain selection. At acidic pH (2.5), strain ACUF 427 outperformed SAG 107.79 considerably, while both strains showed improved performance at higher pH (5.5). Plouviez et al. (2025) examined Ce, Gd, and Yb accumulation in Chlamydomonas reinhardtii induced to form polyphosphate granules, though specific removal efficiencies were not quantified in available data.

Cyanobacterial strains showed cerium adsorption capacities of 84.2-91.5 mg/g for Nostoc sp. 20.02, with optimal pH between 5-6. CFL bulb acid extract bioaccumulation by G. sulphuraria demonstrated Ce uptake alongside other REEs, though quantitative metrics were not specified.

Gallium (Ga) and Germanium (Ge)

No studies in this review specifically examined gallium or germanium biosorption by microalgae from semiconductor waste streams. This represents a critical knowledge gap given the importance of these elements in compound semiconductor manufacturing (GaAs, GaN, SiGe).

Selectivity Ratios

Selectivity data for semiconductor-relevant metal pairs (Ce/Ca, Ga/Al, Ge/Si) were not reported in any of the included studies. Only Nicomel et al. (2020) reported selectivity data for indium over competing metals in ITO etching waste, showing In(III) selectivity over Cu(II), Zn(II), and Al(III).

Process Optimization

Optimal pH values for copper biosorption clustered between pH 4-7 across most studies. Lower pH values (4-5) were optimal for immobilized biosorbent systems, while living cell systems often performed better at pH 5-7. For REE biosorption, optimal pH ranged from 5-6 for cyanobacteria to pH 1.5-2.5 for acidophilic Galdieria sulphuraria under growth conditions, though REE sorption improved up to 24-fold at pH 5.0.

Temperature optimization studies consistently identified mesophilic ranges (25-35℃) as optimal, with the highest removal efficiencies for immobilized mixed culture biosorbents achieved at 30-35℃.

Agitation rates of 100-180 rpm were commonly employed for batch biosorption studies. Continuous flow systems operated at flow rates of approximately 13 mL/min.

Scale and Economics

Study	Scale	Volume Treated	OPEX Estimate	Regeneration Cycles	Regeneration Method
K. C. Wanta et al., 2023	Lab batch	950 mL	Not reported	Not reported	Not reported
N. R. Nicomel et al., 2020	Pilot-scale	Not reported	Not reported	5 cycles	HCl
T. Wilan et al., 2019	Lab batch	Not reported	Not reported	Not reported	Not reported
A. Rinanti et al., 2018	Lab continuous	1.5 L	Not reported	Not reported	Not reported
Huan Lin et al., 2024	Lab batch	Not reported	Cost-effective	≥3 cycles	NaOH/HCl
Patricia Tapia et al., 2011	Lab batch/continuous	15-100 L	Not reported	2 cycles	HCl
M. Iovinella et al., 2022	Lab batch	2 mL	Not reported	Not reported	Not reported
Saavedra Concha et al., 2020	Lab batch	2.1 L bottles	Not reported	Not reported	HCl
A. Morcelli et al., 2025	Lab batch	100 mL	Not reported	Not reported	HCl/H_2SO_4
E. Sočo et al., 2025	Lab batch	50 mL	Not reported	Not reported	HCl, HNO_3, EDTA, NaCl
Ntombiphumile Perceverence Tenza et al., 2025	Lab batch	Not reported	Not reported	Not reported	EDTA, HCl

All but one study operated at laboratory batch scale, with volumes typically ranging from 50 mL to 2.1 L. The sole pilot-scale demonstration was for indium recovery from ITO etching waste. No commercial-scale demonstrations were identified in the literature.

Operating cost estimates (€ or $ per m³ wastewater treated) were not reported in any study. Some studies qualitatively described their approaches as "cost-effective" or mentioned that microalgal biomass is a "low-cost adsorbent", but quantitative techno-economic analyses were absent.

Biomass regeneration was demonstrated in limited studies, with 5 cycles achieved for indium recovery using HCl elution, at least 3 cycles for magnetic nanoparticle-algae composites using NaOH/HCl, and 2 effective cycles for copper using HCl elution from algae biomass.

Discharge Quality and Regulatory Compliance

Study	Final Cu Conc. (mg/L)	Initial Cu Conc. (mg/L)	Meets <0.5 ppm Cu?	Removal %
K. C. Wanta et al., 2023	~1.34	40	No	96.83%
K. C. Wanta et al., 2020	~1.58	40	No	96.10%
T. Wilan et al., 2019	2.0	25	No	92%
Huan Lin et al., 2024	~5.32	20	No	73.4%
Ntombiphumile Perceverence Tenza et al., 2025	Below detection	0.5	Yes	100%
T. I. Oliomogbe et al., 2024	~1.5	Not specified	No	85-88.5%

Most studies did not achieve final copper concentrations below 0.5 ppm required for stringent discharge standards. The exception was Tenza et al. (2025), which achieved copper concentrations below detection limits, though this was from a relatively low initial concentration of 0.5 mg/L. No studies explicitly discussed compliance with 2026 EPA NPDES standards.

For REE discharge targets (<0.1 ppm), no studies provided quantitative final effluent data for Ce, Ga, Ge, or In.

Dead vs Living Biomass Comparison

Characteristic	Dead/Dried Biomass	Living Cells
Removal efficiency	Higher in some studies	Variable, can achieve high efficiency
Adsorption capacity	Higher capacity reported	Can achieve higher values (e.g., 764.5 mg/g)
Stability	More stable and predictable	Subject to environmental factors
Mechanism	Passive biosorption only	Both biosorption and bioaccumulation
Reusability	Can be regenerated and reused	Living cells can be reused after desorption
Cost	Lower maintenance costs	Higher due to growth requirements
Equilibrium time	Generally faster	Variable

Review articles consistently reported that dead or non-living biomass demonstrates higher biosorption efficiency and stability compared to living cells. The metabolism-independent nature of passive biosorption in dead biomass makes it more predictable and stable. Living cells, however, offer the potential advantage of combining biosorption with bioaccumulation through intracellular uptake mechanisms.

Moreno-Garrido et al. (2002) directly compared living and dead immobilized Nannochloropsis gaditana, finding that living cells accumulated more Cu and Zn than dead cells due to photosynthetic processes and metal pumping mechanisms. In contrast, Wilan et al. (2019) reported that dead biomass (immobilized) showed higher adsorption ability (71.91%) compared to living cells.

Scenedesmus obliquus demonstrated Freundlich-type adsorption kinetics indicating reversible binding guided by van der Waals forces, facilitating easier metal desorption and recovery compared to Synechocystis sp. PCC 6803 which followed Langmuir binding guided by chemical forces.

Synthesis

The heterogeneity in reported copper removal efficiencies (57% to 99%) can be reconciled by considering several key factors:

pH Effects: Studies reporting higher removal efficiencies (>90%) consistently operated at pH 4-5 for immobilized systems or pH 5-7 for free cells. Studies at suboptimal pH showed substantially lower performance. The optimal pH range of 4-7 for copper biosorption reflects the balance between metal speciation (Cu^{2+} predominates at lower pH) and biosorbent surface charge (more negative at higher pH).

Initial Metal Concentration: Higher removal efficiencies were generally achieved at lower initial copper concentrations. Studies with initial concentrations of 20-40 mg/L achieved >90% removal, while those testing higher concentrations showed diminishing returns due to saturation of binding sites.

Biomass Treatment: Immobilized biosorbents in Na-alginate beads consistently achieved 80-96% removal efficiency, likely due to enhanced stability and handling characteristics. However, the immobilization matrix itself (particularly calcium alginate) shows strong affinity for Cu, potentially contributing to removal efficiency beyond the biological component.

Contact Time: Equilibrium times varied from minutes to days depending on system configuration. Rapid equilibrium (<30 min) was associated with either small-scale batch systems with vigorous mixing or engineered composites like magnetic nanoparticle-algae hybrids. Extended equilibrium times (>24 h) typically reflected living cell systems where both biosorption and metabolic uptake occurred.

Species Differences: Chlorella species consistently demonstrated superior copper biosorption compared to Scenedesmus under equivalent conditions, with Chlorella achieving 41% removal versus 20.5% for Scenedesmus in quaternary metal systems. This may reflect differences in cell wall composition and functional group availability.

Critical Knowledge Gaps: The absence of studies on Ga and Ge biosorption, combined with extremely limited data on Ce and In from semiconductor-specific waste streams, represents a fundamental barrier to assessing microalgae viability for semiconductor wastewater treatment. The sole pilot-scale study and complete absence of commercial demonstrations further limit translation of laboratory findings to industrial application. The lack of techno-economic data precludes cost comparison with conventional treatment technologies. Additionally, no studies examined the specific interference factors relevant to CMP wastewater (silica nanoparticles, EDTA chelators) or the complex matrices of compound semiconductor etching waste.

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What are the quantified technical and economic feasibility metrics for biological recovery (biosorption, bioaccumulation, bio-leaching, bioreduction) of critical raw materials from semiconductor manufacturing wastewater and rare earth magnet production waste streams, specifically comparing microalgae (Nannochloropsis, Chlorella, Galdieria, Spirulina, Scenedesmus, Chlamydomonas) against bacterial (Acidithiobacillus, Leptospirillum, Pseudomonas, Bacillus) and fungal (Aspergillus, Penicillium, Rhizopus) systems? I am specifically focused on MANUFACTURING WASTE (not primary mining or ore processing) from two industrial verticals: VERTICAL 1: SEMICONDUCTOR FABRICATION WASTE STREAMS • CMP (Chemical Mechanical Polishing) slurry wastewater containing: Copper (Cu), Cerium (Ce), Praseodymium (Pr), Lanthanum (La), Yttrium (Y), Gallium (Ga), Germanium (Ge), Indium (In), Silica nanoparticles, EDTA chelators • Wet etching waste: HF-based etchants, buffered oxide etchants, metal ion residues • Ion implantation and doping waste: Arsenic (As), Phosphorus (P), Boron (B) residues • Photolithography developer waste: organic solvents with trace metals • Wafer dicing/sawing coolant waste: suspended metal particles VERTICAL 2: RARE EARTH MAGNET MANUFACTURING WASTE • NdFeB (Neodymium-Iron-Boron) magnet production scrap containing: Neodymium (Nd), Dysprosium (Dy), Praseodymium (Pr), Terbium (Tb), Gadolinium (Gd) • SmCo (Samarium-Cobalt) magnet waste containing: Samarium (Sm), Cobalt (Co) • Sintering and heat treatment residues • Grinding/machining coolant containing suspended magnet powder • Coating bath waste (Ni-Cu-Ni electroplating solutions) For each study, I need the following QUANTITATIVE DATA POINTS extracted: TECHNICAL PERFORMANCE METRICS: 1. Metal recovery efficiency (%) - report separately for each target metal 2. Selectivity ratio (target metal / competing ions) - e.g., Ce/Ca, Nd/Ca, Ga/Al 3. Adsorption capacity (mg metal / g biomass) - report q_max from Langmuir/Freundlich models 4. Kinetics: time to equilibrium (minutes/hours), pseudo-first-order or pseudo-second-order rate constants (k1, k2) 5. Optimal pH range for maximum sorption 6. Optimal temperature (℃) 7. Biomass pre-treatment method (living cells, heat-killed, acid-treated, immobilized, genetically modified) 8. Final discharge quality achieved (ppm or ppb) after treatment 9. Regeneration/desorption efficiency (%) and number of reuse cycles demonstrated 10. Tolerance to interfering substances (silica, chelators, high salinity, organic solvents) ECONOMIC FEASIBILITY METRICS: 11. CAPEX (Capital Expenditure): €/m³/day or $/kg-metal capacity - include bioreactor costs, sensors, pumps, pre-treatment equipment 12. OPEX (Operating Expenditure): €/m³ treated or $/kg metal recovered - include biomass production costs, energy, chemicals, labor 13. Cost comparison

substantially outperforming fungal systems (40-60% recovery) at laboratory scale, while direct comparative data for microalgae against bacteria and fungi on semiconductor or rare earth magnet manufacturing waste specifically does not exist, and comprehensive economic feasibility metrics remain largely unquantified across all organism types.

Abstract

Bacterial systems, particularly acidophilic *Acidithiobacillus* and *Leptospirillum* species, demonstrated the highest metal recovery efficiencies from manufacturing waste streams, achieving 94-99% copper extraction from printed circuit boards and >99% recovery of cerium, europium, and neodymium from electronic waste. The methylotrophic bacterium *Methylobacterium extorquens* AM1 achieved 65-100% neodymium recovery from NdFeB magnet scrap with highly selective REE bioaccumulation (98% purity), scalable to 10 L without harsh acids. Fungal systems showed substantially lower performance, with *Aspergillus niger* achieving only 60% copper and 40% zinc extraction. Microalgal applications were notably underrepresented; however, the extremophile *Galdieria sulphuraria* demonstrated >90% recovery of Nd, Dy, and La at acidic pH (1.5-2.5) with rapid 5-minute equilibration kinetics, suggesting suitability for harsh semiconductor waste streams. Adsorption capacities ranged from 2.64 mg Nd/g for engineered *E. coli* microbeads to 41.61 mg REE/g for freeze-dried *G. sulphuraria*, with regeneration efficiencies of 85-95% over 5-9 reuse cycles.

Economic feasibility data remained sparse across the literature. Bioleaching operational costs were dominated by carbon substrate expenses (44% of total cost for glucose-based systems), with gallium bioleaching from LEDs costing $576/kg compared to $1,321/kg for acid leaching. Break-even analysis indicated that REE bioleaching requires either a 26-fold increase in REE market prices or waste processing tipping fees of $155/ton for economic viability. Critically, only three studies directly addressed semiconductor CMP wastewater, none of the specified microalgal genera (Nannochloropsis, Spirulina, Scenedesmus) were evaluated for manufacturing waste applications, and no study provided direct comparative performance data between microalgae, bacteria, and fungi on identical waste samples. All studies operated at laboratory or bench scale (<100 L), with no commercial deployments documented for either semiconductor fabrication or rare earth magnet manufacturing waste streams.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the quantified technical and economic feasibility metrics for biological recovery (biosorption, bioaccumulation, bio-leaching, bioreduction) of critical raw materials from semiconductor manufacturing wastewater and rare earth magnet production waste streams, specifically comparing microalgae (Nannochloropsis, Chlorella, Galdieria, Spirulina, Scenedesmus, Chlamydomonas) against bacterial (Acidithiobacillus, Leptospirillum, Pseudomonas, Bacillus) and fungal (Aspergillus, Penicillium, Rhizopus) systems?

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4. Kinetics: time to equilibrium (minutes/hours), pseudo-first-order or pseudo-second-order rate constants (k1, k2)
5. Optimal pH range for maximum sorption
6. Optimal temperature (℃)
7. Biomass pre-treatment method (living cells, heat-killed, acid-treated, immobilized, genetically modified)
8. Final discharge quality achieved (ppm or ppb) after treatment
9. Regeneration/desorption efficiency (%) and number of reuse cycles demonstrated
10. Tolerance to interfering substances (silica, chelators, high salinity, organic solvents)

ECONOMIC FEASIBILITY METRICS: 11. CAPEX (Capital Expenditure): €/m³/day or $/kg-metal capacity - include bioreactor costs, sensors, pumps, pre-treatment equipment 12. OPEX (Operating Expenditure): €/m³ treated or $/kg metal recovered - include biomass production costs, energy, chemicals, labor 13. Cost comparison to alternative technologies: ion exchange resin ($/m³), chemical precipitation ($/m³), solvent extraction ($/kg), membrane filtration ($/m³) 14. Break-even analysis: minimum metal concentration (ppm) or waste volume (m³/year) for economic viability 15. Revenue potential: market value of recovered metals ($/kg) and purity achieved (% REO, % metallic grade)

WASTE STREAM CHARACTERIZATION: 16. Initial metal concentrations (ppm or mg/L) in the waste stream 17. Waste volume generated (L/day, m³/year) per production unit (per wafer processed, per tonne magnet produced) 18. pH of the waste stream (as-generated, before treatment) 19. Presence of toxic inhibitors (e.g., cyanide, heavy metals, flame retardants, specific plastics) and their effect on organism viability 20. Total dissolved solids (TDS), chemical oxygen demand (COD), biological oxygen demand (BOD)

OPERATIONAL AND SCALE CONSIDERATIONS: 21. Scale of demonstration: laboratory batch (<10L), bench-scale continuous (10-100L), pilot-scale (>100L), commercial deployment 22. Hydraulic retention time (HRT) or residence time in bioreactor (hours/days) 23. Biomass productivity (g/L/day) and harvesting frequency 24. Footprint requirements (m² per m³/day throughput) 25. Regulatory compliance achieved: EPA discharge limits, EU REACH, WEEE Directive compliance

ORGANISM-SPECIFIC DETAILS: 26. Organism taxonomy (genus and species) 27. Organism type: microalgae (unicellular/multicellular), cyanobacteria, heterotrophic bacteria, fungi, mixed consortia 28. Mechanism of action: surface biosorption (cell wall binding), intracellular bioaccumulation, bio-leaching (organic acid/siderophore secretion), bioreduction (enzymatic conversion to nanoparticles) 29. Genetic modifications or strain selection strategies 30. Growth conditions: photoautotrophic, heterotrophic, mixotrophic

STUDY QUALITY INDICATORS: 31. Publication year (prioritize 2019-2026) 32. Study type: laboratory experiment, pilot-scale validation, industrial case study, techno-economic analysis, life cycle assessment (LCA) 33. Industry

partnership or funding source (if disclosed) 34. Statistical significance of reported results (sample size, error bars, p-values) 35. Third-party validation or certification (ISO, EPA, independent lab analysis)

PRIORITIZATION GUIDANCE FOR ELICIT: • Papers must explicitly address MANUFACTURING WASTE (not primary mining ore processing) • Papers must report AT LEAST THREE of the following: recovery efficiency (%), adsorption capacity (mg/g), kinetics (time), and cost ($/m³ or $/kg) • Studies from 2019-2026 are strongly preferred; include 2015-2018 only if they report pilot-scale or commercial data • Prioritize studies with QUANTITATIVE data over qualitative descriptions • Include both peer-reviewed academic papers AND industry white papers/technical reports • Papers addressing BOTH technical performance AND economic feasibility should be ranked highest • Include studies on extremophile organisms (Galdieria, Cyanidium) that tolerate low pH (2-4) and high temperatures (40-55℃)"

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Manufacturing Waste Source**: Does the study address waste streams specifically from semiconductor fabrication (CMP slurry, wet etching, ion implantation, photolithography, wafer processing) or rare earth magnet manufacturing (NdFeB/SmCo production, sintering residues, machining coolants, coating baths)?
- **Biological Recovery Methods**: Does the study employ biological recovery mechanisms (biosorption, bioaccumulation, bio-leaching, or bioreduction) using microalgae, bacteria, or fungi?
- **Target Critical Materials**: Does the study focus on recovery of rare earth elements (Ce, Pr, La, Y, Nd, Dy, Tb, Gd, Sm), technology metals (Cu, Ga, Ge, In, Co), or semiconductor dopants (As, P, B)?
- **Quantitative Data**: Does the study report quantitative performance metrics such as metal recovery efficiency (%), adsorption capacity (mg/g), kinetic parameters, or economic data?
- **Empirical Research**: Does the study include laboratory experiments, pilot-scale demonstrations, industrial case studies, techno-economic analyses, or life cycle assessments with empirical data (not purely theoretical)?
- **Secondary Recovery Focus**: Is the study focused on secondary recovery from manufacturing waste rather than exclusively on primary mining, ore processing, or mineral extraction from natural deposits?
- **Biological System Involvement**: Does the study involve biological systems rather than focusing exclusively on non-biological recovery technologies (chemical precipitation, ion exchange, solvent extraction) without any biological component?
- **Experimental Validation**: Does the study include experimental validation or empirical data rather than presenting only computational models or theoretical frameworks without experimental support?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Waste Stream Characterization**:

  Extract detailed characterization of the manufacturing waste stream being treated, including:

- Waste source (semiconductor fabrication specific process OR rare earth magnet manufacturing specific process)
- Target metals and their initial concentrations (ppm/mg/L) - specifically Cu, Ce, Pr, La, Y, Ga, Ge, In, As, P, B, Nd, Dy, Tb, Gd, Sm, Co
- Waste volume generated (L/day, m^3/year) per production unit
- Interfering substances present (silica nanoparticles, EDTA chelators, HF, organic solvents, suspended particles, high salinity)
- Initial pH, TDS, COD, BOD values
- Any toxic inhibitors that affect organism viability

- **Organism and Mechanism**:

  Extract organism details and biological recovery mechanism, including:

- Specific organism (genus and species) - focus on Nannochloropsis, Chlorella, Galdieria, Spirulina, Scenedesmus, Chlamydomonas, Acidithiobacillus, Leptospirillum, Pseudomonas, Bacillus, Aspergillus, Penicillium, Rhizopus
- Organism type (microalgae, bacteria, fungi) and growth mode (photoautotrophic, heterotrophic, mixotrophic)
- Primary mechanism: biosorption (surface binding), bioaccumulation (intracellular uptake), bio-leaching (acid/siderophore production), bioreduction (enzymatic conversion)
- Biomass pre-treatment method (living cells, heat-killed, acid-treated, immobilized, genetically modified)
- Any genetic modifications or strain selection strategies specifically for metal recovery

- **Technical Performance Metrics**:

  Extract all quantitative technical performance data for biological recovery of critical raw materials from manufacturing waste, including:

- Metal recovery efficiency (%) reported separately for each target metal from semiconductor/magnet waste
- Maximum adsorption capacity (mg metal/g biomass) with Langmuir/Freundlich model parameters (qmax, KL, KF)
- Kinetics: time to equilibrium (minutes/hours), rate constants (k1, k2), pseudo-first-order or pseudo-second-order
- Selectivity ratios (target metal/competing ions) such as Ce/Ca, Nd/Ca, Ga/Al
- Optimal operating conditions: pH range, temperature (℃), biomass concentration
- Final discharge quality achieved (ppm/ppb) after biological treatment
- Regeneration/desorption efficiency (%) and number of demonstrated reuse cycles

- **Economic Feasibility Analysis**:

  Extract all economic metrics and cost analysis data for biological recovery from semiconductor/magnet manufacturing waste, including:

- CAPEX (€/m^3/day or $/kg-metal capacity): bioreactor costs, sensors, pumps, pre-treatment equipment
- OPEX (€/m^3 treated or $/kg metal recovered): biomass production, energy, chemicals, labor costs
- Direct cost comparisons to alternative technologies: ion exchange ($/$m^3$), chemical precipitation ($/$m^3$), solvent extraction ($/kg), membrane filtration ($/$m^3$)
- Break-even analysis: minimum metal concentration (ppm) or waste volume (m^3/year) for economic viability
- Revenue potential: market value of recovered metals ($/kg) and achieved purity (% REO, % metallic grade)
- Total cost per unit of waste treated or metal recovered with breakdown by cost category

- **Operational Parameters**:

  Extract operational conditions and scale-up parameters for biological recovery systems treating semiconductor/magnet manufacturing waste, including:

- Demonstration scale: laboratory batch (<10L), bench-scale continuous (10-100L), pilot-scale (>100L), commercial deployment
- Hydraulic retention time (HRT) or residence time in bioreactor (hours/days)
- Biomass productivity (g/L/day), harvesting frequency, and biomass handling requirements
- System footprint requirements (m^2 per m^3/day throughput)
- Operating pH range, temperature control requirements, oxygen/CO2 management
- Tolerance to process variations and robustness under industrial conditions
- Integration requirements with existing manufacturing waste treatment infrastructure

- **Comparative Performance**:

  Extract direct comparisons between different organism types (microalgae vs bacteria vs fungi) for recovering critical materials from the same manufacturing waste stream, including:

- Side-by-side performance comparisons on identical waste samples
- Relative advantages/disadvantages of each organism type for specific metals or waste streams
- Performance rankings or scoring across multiple criteria (efficiency, cost, robustness)
- Optimal organism selection criteria for specific manufacturing waste characteristics
- Synergistic effects or performance of mixed consortia vs pure cultures
- Trade-offs between technical performance and economic feasibility across organism types

- **Metal Recovery Specifics**:

  Extract detailed metal-specific recovery data for critical raw materials from semiconductor and rare earth magnet manufacturing waste, including:

- Recovery performance for each individual metal (Cu, Ce, Pr, La, Y, Ga, Ge, In, As, P, B, Nd, Dy, Tb, Gd, Sm, Co)
- Metal speciation effects (ionic forms, complexed forms, nanoparticles) on recovery efficiency
- Sequential or selective recovery of multiple metals from the same waste stream
- Purity and grade of recovered metal products suitable for reuse in manufacturing
- Metal recovery from real industrial waste vs synthetic solutions
- Impact of metal concentration ranges typical in manufacturing waste on organism performance

- **Regulatory and Validation**:

  Extract study quality indicators and regulatory compliance data for biological recovery from manufacturing waste, including:

- Study type: laboratory experiment, pilot-scale validation, industrial case study, techno-economic analysis, life cycle assessment
- Scale and duration of testing with real manufacturing waste streams
- Statistical rigor: sample sizes, replicates, error analysis, p-values, confidence intervals
- Third-party validation, independent lab analysis, or certification (ISO, EPA, EU standards)
- Regulatory compliance achieved: EPA discharge limits, EU REACH, WEEE Directive requirements
- Industry partnerships, funding sources, or commercial validation
- Publication year (prioritize 2019-2026) and peer-review status

Report

Due to the limitations of the AI model, we are only able to process 80 sources while writing a report. This report was written using the 80 sources that had the highest screening scores out of the 426 sources that we screened in and extracted data from.

Characteristics of Included Studies

This systematic review synthesized evidence from 80 sources examining biological recovery of critical raw materials from manufacturing waste streams. The literature predominantly addresses electronic waste (e-waste) processing rather than the specific semiconductor fabrication and rare earth magnet manufacturing waste streams of primary interest. A substantial portion of studies focused on printed circuit board (PCB) bioleaching, with fewer addressing rare earth element (REE) recovery from magnet production waste.

Study	Full Text Retrieved?	Study Type	Primary Waste Stream	Target Metals
A. Brewer et al., 2019	Yes	Laboratory experiment	NdFeB magnet scraps	Nd, Pr, Dy
Aaron P. Mosier et al., 2015	No	Laboratory experiment	CMP wastewater	Cu
Andrea Leonardi et al., 2015	No	Laboratory experiment	CMP wastewater	Cu
Dimitrios A. Vavlekas et al., 2017	No	Laboratory experiment	Magnet waste	Nd
Nathan M. Good et al., 2023	Yes	Laboratory experiment	NdFeB magnet swarf	Nd, Dy, Pr
Nathan M. Good et al., 2023a	Yes	Laboratory experiment	NdFeB magnet scrap	Nd, Pr, Dy, Fe
A. Marra et al., 2018	No	Laboratory experiment	WEEE shredding dust	Ce, Eu, Nd, La, Y, Au
C. García-Balboa et al., 2022	Yes	Laboratory experiment	Electronic scraps	Cu, Co, Al, Zn, REEs
Sandeep Panda et al., 2024	Yes	Laboratory experiment	WPCBs, spent LiBs	Cu, Ni, Li, Co
Vicki S. Thompson et al., 2017	Yes	TEA/LCA	FCC catalysts	REEs
Dimitrios A. Vavlekas et al., 2016	No	Laboratory experiment	Metallic wastes	Nd
M. Palmieri et al., 2022	Yes	Laboratory experiment	Synthetic solutions	Y, Ce, Eu, Tb
Eva Pakostova et al., 2023	Yes	Laboratory experiment	Electrodeposited components	Cu, Ag
S. Balchandani et al., 2023	Yes	TEA/LCA	NdFeB magnet swarf	REEs
David Kaplan et al., 2023	Yes	Laboratory experiment	E-waste streams	Tb, Pd, PGMs

Study	Full Text Retrieved?	Study Type	Primary Waste Stream	Target Metals
A. Işıldar et al., 2016	No	TEA	Electronic waste PCBs	Cu, Au, Fe, Al, Ni, Zn
M. Mohammadi et al., 2022	No	Laboratory experiment	Aqueous solutions	La
M. Iovinella et al., 2023	No	Laboratory experiment	Spent fluorescent lamps	Y, Eu
S. Matys et al., 2020	No	Laboratory experiment	Electronic scrap	REEs, Ga
Karem Gallardo et al., 2025	Yes	Laboratory experiment	Spent mobile phones	Y, Pr, Nd, Gd, Tb, Dy
Gamez Grijalva et al., 2009	No	Laboratory experiment	CMP wastewater	Cu
Pro Danièle et al., 2016	Yes	Laboratory experiment	E-waste/contaminated water	REEs
Sadia Ilyasl et al., 2010	No	Laboratory/pilot experiment	Electronic scrap	Ni, Cu, Al, Zn
N. Saitoh et al., 2017	No	Laboratory experiment	LSI waste, LCD panels	Au, PGMs, In
R. Auerbach et al., 2017	No	Laboratory experiment	Fluorescent phosphor	Y
Chiara Magrini et al., 2022	No	LCA/MFCA	NIB magnets	Nd, Dy, Pr
A. Becci et al., 2020	No	Laboratory experiment	End-of-life PCBs	Cu, Zn
Yoonjae Lee et al., 2024	No	TEA	Semiconductor waste	W
R. Srivastava et al., 2020	No	Review	WPCBs	Critical metals
Ayumi Minoda et al., 2014	No	Laboratory experiment	Acidic solutions	Nd, Dy, La, Cu
Marcus Heilmann et al., 2021	Yes	Laboratory experiment	Wastewaters	Nd, Eu
Quanhui Ye et al., 2023	No	Laboratory experiment	Coal fly ash, geothermal brine	REEs
Stefanie Hopfe et al., 2018	No	Laboratory experiment	Fluorescent phosphor	REEs
A. Işıldar et al., 2018	Yes	Laboratory/TEA	Discarded PCBs	Cu, Au
A. Priya et al., 2020	No	Laboratory experiment	WPCBs	Cu, Zn, Ni, Pb, Ag, Au, Sc, Ce, La, Nd
S. Saikia et al., 2022	No	Laboratory experiment	Zinc refinery residues	Ga, Ge, Zn

Study	Full Text Retrieved?	Study Type	Primary Waste Stream	Target Metals
Helena Singer et al., 2023	Yes	Laboratory experiment	Industrial waste/post-mining waters	REEs
F. Vakilchap et al., 2024	No	Laboratory experiment	Mobile phone PCBs	Mn, Pt, Pd, Fe, Co, Al
Eva Benzal Montes et al. (n.d.)	Yes	Laboratory experiment	PCBs	Cu
R. Auerbach et al., 2019	No	Laboratory experiment	Incineration slag	Al, Cu, Mn, Ce, La, Er
A. U. Issayeva et al., 2017	No	Laboratory experiment	Phosphorus waste	La, Ce, Nd
Alexandre Kompalitch et al., 2023	Yes	Laboratory experiment	LiB bioleachate	Co, Li, Mn, Ni
N. Saitoh et al., 2017a	No	Laboratory experiment	Automotive catalysts, LCD waste	Pt, Pd, Rh, Au, In
Astha Upadhyay et al., 2024	No	TEA/LCA	LCD screens	In
D. Arora et al., 2008	No	Laboratory experiment	PCBs	Cu, Zn
Agathe Hubau et al., 2019	No	Laboratory experiment	Circuit boards	Metals
Suthee Janyasuthiwong et al., 2015	No	Laboratory experiment	E-waste leachate, PCBs	Cu, Ni, Zn
Zhengsheng Yu et al., 2019	No	Review	Various waste streams	Critical metals
Y. Nancharaiah et al., 2015	No	Review	Metallurgical wastes	Various metals
M. Danouche et al., 2023	No	Review	Secondary REE resources	REEs
F. Beolchini et al., 2012	No	Laboratory experiment	Industrial catalysts, sediments	Ni, V, Mo, Zn, Y
Xiaohan Wu et al., 2025	No	TEA	Waste LEDs	Ga
M. Ike et al., 2011	No	Review/Handbook	Various wastes	Various metals
A. Becci et al., 2020a	Yes	Laboratory experiment	End-of-life PCBs	Cu, Zn
Denise Kölbi et al., 2023	Yes	Laboratory experiment	BOF steelmaking dust	Fe, Cu, Zn
K. Kremser et al., 2022	No	Laboratory experiment	Waste incineration residuals	Cu, Ni, Al, Mn, Zn
M. Buțu et al., 2016	No	Review	Various secondary sources	Critical metals

Study	Full Text Retrieved?	Study Type	Primary Waste Stream	Target Metals
A. Williamson et al., 2020	No	Laboratory experiment	Iron oxide residue	Zn
J. A. Garzón et al., 2019	No	Laboratory experiment	Marine environments	Au, Cr, REEs
Paula Gonzales Contreras et al., 2013	No	Laboratory experiment	Leaching operations	As
A. Kvartenko et al., 2017	Yes	Laboratory experiment	Galvanic wastewater	Cu, Zn, Ni, Cr
Y. Nancharaiah et al., 2016	No	Review	Secondary sources	Critical metals
K. Pollmann et al., 2018	No	Review	Technical products	Various metals
A. Sieber et al., 2025	Yes	Review	Bio-leachates, liquid waste	Various metals
M. Gavrilescu et al., 2021	No	Review	Secondary sources	Critical metals
I. Tezyapar Kara et al., 2023	Yes	Review	Metal-bearing wastes	Various metals
Jae-chun Lee et al., 2012	No	Review	Solid wastes	Various metals
D. Johnson et al., 2014	Yes	Review	Ores and waste materials	Base metals
K. K. Brar et al., 2020	No	Review	Secondary waste	Cu
Rebecca M. Brown et al., 2023	No	Review	E-waste, ore deposits	REEs
Suthee Janyasuthiwong et al., 2016	Yes	Laboratory experiment	Mining wastewater, PCBs	Cu, Ni, Zn
Adegoke Isiaka Adetunji et al., 2023	Yes	Review	E-waste	Various metals
A. Jaiswal et al., 2025	Yes	Review	Inorganic waste	Various metals
João Pinto et al., 2023	Yes	Review	E-waste	REEs
Viktoriia Liapun et al., 2023	No	Review	Secondary sources	Critical metals
Zhu Chen et al., 2024	No	Review	Ores and solid wastes	REEs
Yang Bai et al., 2025	No	Review	E-waste	REEs, precious metals
Biomining Systems, 2018	Yes	Review	Secondary sources	Base metals, REEs, PMs
E. Giese et al., 2020	No	Review	Aqueous solutions	REEs

Study	Full Text Retrieved?	Study Type	Primary Waste Stream	Target Metals
B. Atkinson et al., 1998	No	Pilot validation	Metal plating effluent	Heavy metals

The included studies span from 1998 to 2025, with the majority published after 2015. Approximately 35% of studies provided full text access. Study types include laboratory experiments (predominant), techno-economic analyses, life cycle assessments, and review articles. Notably, only three studies directly addressed semiconductor CMP wastewater, while approximately eight studies examined NdFeB magnet waste streams.

Technical Performance Metrics

Metal Recovery Efficiency

Recovery efficiencies varied substantially across studies depending on target metals, organism type, and waste stream characteristics. The following table summarizes reported recovery efficiencies for key target metals:

Metal	Recovery Efficiency (%)	Organism	Waste Stream	Reference
Cu	98.4%	Acidophilic bacteria	PCBs	
Cu	99.2%	Bacterial consortium	Electrodeposited components	
Cu	96%	Mixed *At. ferrooxidans* + *A. acidophilum*	WPCBs	
Cu	94%	*At. ferrooxidans* + *L. ferrooxidans*	PCBs	
Cu	99%	*Acidithiobacillus* spp.	Incineration residuals	
Cu	>99%	Acidophilic consortium	E-scraps	
Cu	60%	*Aspergillus niger*	PCBs	
Cu	9.6%	*Acidiphilium acidophilum*	PCBs	
Nd	65-100%	*M. extorquens* AM1	NdFeB magnet scrap	
Nd	>99%	*At. thiooxidans*	WEEE dust	
Nd	24%	Mixed bacterial culture	WPCBs	
Ce	>99%	*At. thiooxidans*	WEEE dust	
Ce	27%	Mixed bacterial culture	WPCBs	
Eu	>99%	*At. thiooxidans*	WEEE dust	
La	80%	*At. thiooxidans*	WEEE dust	

Metal	Recovery Efficiency (%)	Organism	Waste Stream	Reference
La	88% recovery increase	*Nitrosomonas europaea*	Phosphorus waste	
Y	100%	*At. ferrooxidans*	Fluorescent phosphor	
Y	80%	*At. thiooxidans*	WEEE dust	
Co	98%	Acidophilic consortium	Spent LiBs	
Co	70-100%	*Sporosarcina pasteurii*	LiB bioleachate	
Li	80%	Acidophilic consortium	Spent LiBs	
Ni	100%	Acidophilic bacteria	PCBs	
Zn	94.5%	Mixed bacterial culture	WPCBs	
Zn	70%	*At. ferrooxidans* + *L. ferrooxidans*	PCBs	
Au	48%	*Pseudomonas putida*	WEEE dust	
Au	44%	Cyanide-generating bacteria	PCBs	
Ga	69%	Biogenic Te^0	Zinc refinery residues	
Pt	100%	Mixed fungal-bacterial culture	Mobile phone PCBs	
Pd	90%	*Shewanella loihica* PV-4	E-waste leachate	

Bacterial systems, particularly acidophilic *Acidithiobacillus* species, consistently achieved the highest base metal recovery efficiencies, with copper recoveries exceeding 94% in optimized systems. For rare earth elements, bioleaching using *At. thiooxidans* achieved >99% extraction for Ce, Eu, and Nd from WEEE shredding dust. The methylotrophic bacterium *Methylobacterium extorquens* AM1 demonstrated 65-100% Nd recovery from NdFeB magnet scrap in a single process run.

Fungal systems showed lower recovery efficiencies compared to bacterial systems. *Aspergillus niger* achieved only 60% Cu and 40% Zn extraction from PCBs, though mixed fungal-bacterial cultures demonstrated synergistic effects with 100% Mn and Pt extraction.

Microalgal systems were underrepresented in manufacturing waste applications. *Galdieria sulphuraria* demonstrated >90% recovery of Nd, Dy, La, and Cu at 0.5 ppm concentrations, while *Chlamydomonas* sp. achieved biouptake of REEs from e-scrap leachate with capacities reaching 14.9-20.3 ng/g wet biomass for individual lanthanides.

Adsorption Capacity and Kinetics

Organism	Target Metal	Adsorption Capacity (mg/g)	Time to Equilibrium	Model	Reference
Engineered *E. coli*	Nd	2.64 mg/g (dw)	10 min (90% max)	-	
Magnetospirillum magneticum AMB-1	La	37.2 mg/g (dw)	8 min	Langmuir	
Exiguobacterium sp. SH31	REEs	>35 mg/g	-	Langmuir (R^2=1)	
G. sulphuraria (freeze-dried)	REEs	41.61 mg/g (dm)	5 min	-	
Physcomitrella patens	Nd	0.74 mmol/g	-	-	
Physcomitrella patens	Eu	0.48 mmol/g	-	-	
LanM-MNP biosorbent	Tb	6.01 µmol/g	Fast kinetics	-	
Te^0 biosorbent	Ga	74 mg/g	-	-	
Te^0 biosorbent	Zn	43 mg/g	-	-	
Microalgae (*Coccomyxa*)	REEs	15 mg/g (dw)	Few hours	-	

Biosorption processes demonstrated rapid kinetics, with *Magnetospirillum magneticum* AMB-1 achieving equilibrium within 8 minutes for La^{3+} adsorption. The freeze-dried *Galdieria sulphuraria* biomass reached maximum biosorption within 5 minutes. The engineered *E. coli* displaying lanthanide binding tags achieved 90% of maximum Nd adsorption within 10 minutes.

Langmuir isotherm models provided the best fit for most biosorption systems, indicating monolayer adsorption. *Exiguobacterium* sp. SH31 demonstrated perfect Langmuir model fit (R^2=1) for REE biosorption with separation factors between 0 and 1.

Selectivity and Purity

Selectivity for target metals over competing ions represents a critical performance parameter. Engineered *E. coli* microbe beads achieved 97% REE purity of the adsorbed fraction from NdFeB magnet leachate, demonstrating a two-bed volume increase in breakthrough points for REEs compared to non-REE impurities. *Methylobacterium extorquens* AM1 showed highly specific REE bioaccumulation with 98.0% of accumulated metal being REE from Nd magnet swarf.

The lanmodulin-functionalized magnetic nanoparticle biosorbent achieved a 967-fold increase in REE purity when treating coal fly ash and geothermal brine leachates. This biosorbent selectively adsorbed REEs in the presence of a broad range of non-REE metals.

Galdieria sulphuraria demonstrated selective lanthanoid collection over Cu(II) at pH 1.0-1.5, enabling separation of REEs from transition metals.

Optimal Operating Conditions

Parameter	Optimal Range	System	Reference
pH	4-6	*E. coli* microbe beads for Nd	
pH	6.0	*M. magneticum* for La	
pH	4.5	*G. sulphuraria* for REEs	
pH	8.0	*Exiguobacterium* sp. for REEs	
pH	1.5-2.5	*G. sulphuraria* for lanthanoids	
pH	1.9	*At. ferrooxidans* for Cu bioleaching	
pH	7.0-5.0	SRB for Cu/Zn precipitation	
Temperature	21℃	*M. magneticum* for La	
Temperature	29.5℃	*M. extorquens* for REE	
Temperature	30℃	*At. ferrooxidans* for Cu	
Temperature	32℃	Acidophilic consortium (bioreactor)	
Temperature	60℃	Mixed fungal-bacterial culture	

Biosorption systems generally operated optimally at pH 4-6 for REE recovery, while bioleaching systems required more acidic conditions (pH 1.5-2.5) for effective metal mobilization. The extremophile *Galdieria sulphuraria* demonstrated particular utility for acidic waste streams, maintaining efficient recovery at pH 1.5-2.5.

Temperature requirements varied by organism type. Mesophilic bacteria operated at 28-30℃, while extremophilic systems tolerated temperatures up to 60℃. *M. magneticum* AMB-1 showed minimal impact from temperature variations up to 50℃ and maintained biosorption capacity at salinity up to 1.5 M NaCl.

Regeneration and Reuse

System sustainability depends on regeneration efficiency and reuse potential:

System	Desorption Efficiency	Reuse Cycles	Capacity Retention	Reference
E. coli microbe beads	94% (Na citrate)	9 cycles	85%	
LanM-MNP biosorbent	>90%	8 cycles	~95%	
M. magneticum AMB-1	Effective	5 cycles	Maintained	
L. casei/P. pastoris biofilm	HCl wash	Multiple	Maintained	

System	Desorption Efficiency	Reuse Cycles	Capacity Retention	Reference
Silk-LBT membranes	Low pH buffer	4 cycles	70%	
Exiguobacterium sp. SH31	30-90%	Multiple	-	

The engineered *E. coli* microbe bead system demonstrated excellent reusability with 85% capacity retention after nine consecutive adsorption/desorption cycles. The lanmodulin-functionalized magnetic nanoparticle biosorbent retained approximately 95% of initial activity after eight cycles, with the additional advantage of magnetic separation simplifying biomass recovery.

Economic Feasibility Analysis

Economic data remained sparse across the literature, with only a handful of studies providing quantitative cost metrics.

Available Cost Data

Metric	Value	System/Context	Reference
Annual OPEX	$3,030,000	REE bioleaching plant	
Annual profit	$855,000	REE bioleaching plant	
Glucose cost share	44% of total cost	Biolixiviant production	
Break-even tipping fee	$155/ton	REE bioleaching	
Biolixiviant cost (NP)	$0.04/kg	Gluconic acid from nonrecyclable paper	
Biolixiviant cost (CS)	$0.06-0.08/kg	Gluconic acid from corn stover	
Bioleaching Ga	$576/kg	Waste LEDs	
Acid leaching Ga	$1,321/kg	Waste LEDs	
Primary production Ga	$368/kg	Bayer process	
Algal biomass production	$500-700/ton	General cultivation	

The techno-economic analysis by Thompson et al. (2017) identified glucose as the single largest expense for bioleaching processes, constituting 44% of total operational costs. At 2016 REE prices, a 26-fold increase in REE market prices would be needed for breakeven without a tipping fee for waste processing.

For gallium recovery from waste LEDs, bioleaching ($576/kg Ga) proved more cost-effective than acid leaching ($1,321/kg Ga) but remained more expensive than primary production ($368/kg Ga). The high costs of bioleaching were attributed to extensive reactor usage (57 reactors required) due to long processing times.

Balchandani et al. (2023) demonstrated that utilizing waste materials as feedstocks substantially improves economic viability. Biolixiviant production from nonrecyclable paper ($0.04/kg) proved more cost-effective than corn stover ($0.06-0.08/kg) or date palm clippings ($0.06-0.09/kg).

Cost Comparison with Conventional Technologies

Direct cost comparisons between biological and conventional technologies were limited. Atkinson et al. (1998) conducted a pilot-scale feasibility study comparing biosorption using waste activated sludge with chemical precipitation for metal plating effluent treatment, finding chemical precipitation more cost-effective at that time.

The carbon footprint assessment by Becci et al. (2020) demonstrated environmental advantages for bioleaching, with a 4-fold reduction in CO_2-eq./kg treated PCB emissions compared to best available bioleaching processes in the literature. Higher PCB concentration in solution (up to 5% w/v) reduced bioreactor size requirements and associated energy and raw material demands.

Organism-Specific Performance

Bacterial Systems

Bacterial systems dominated the literature for metal recovery from manufacturing waste, with acidophilic chemolithotrophs showing particular promise for bioleaching applications.

Acidithiobacillus species: *At. ferrooxidans* and *At. thiooxidans* achieved the highest reported bioleaching efficiencies. A consortium approach using *At. ferrooxidans* CF3 with ten acidophilic Fe^{2+}-oxidizers achieved >99.6% Cu extraction from electrodeposited components in only 12 days in a bioreactor system at 32℃. Pure *At. thiooxidans* (DSM 9463) mobilized >99% of Ce, Eu, and Nd from WEEE shredding dust within 8 days by lowering pH from 3.5 to 1.0.

Leptospirillum species: Often used in consortium with *Acidithiobacillus*, *L. ferrooxidans* contributed to achieving 94% Cu and 70% Zn yields from PCBs. In bioreactor experiments, *Leptospirillum* dominated the microbial community (83.2% to 59% of total reads) during initial leaching phases.

Methylobacterium extorquens AM1: This mesophilic methylotroph demonstrated unique capabilities for REE recovery from NdFeB magnet waste. The organism naturally acquires REEs to power methanol metabolism and can be grown using electronic waste as the sole REE source, scalable to 10 L with consistent metal yields. Genetic engineering to eliminate exopolyphosphatase activity increased metal accumulation by enhancing polyphosphate granule storage of REEs.

Pseudomonas putida: The cyanide-producing strain WSC361 mobilized 48% of gold within 3 hours from *At. thiooxidans*-leached WEEE dust, demonstrating potential for precious metal recovery in sequential processes.

Magnetospirillum magneticum AMB-1: This magnetotactic bacterium provided unique operational advantages due to magnetic separation capability. La^{3+} biosorption reached 37.2 mg/g dry biomass at pH 6.0 with rapid 8-minute equilibration and tolerance to elevated salinity and temperature.

Fungal Systems

Fungal systems showed lower absolute recovery efficiencies but offered advantages in organic acid production for indirect bioleaching.

Aspergillus niger: This commonly studied fungus achieved 60% Cu and 40% Zn extraction from PCBs using citric acid production as the leaching mechanism. The process was enhanced using milk whey as a substrate for fungal growth, improving sustainability. In mixed culture with *Bacillus megaterium*, *A. niger* demonstrated oxalic acid overproduction (1,350-7,460 mg/L) through bacterial metabolite-induced bio-stress, achieving 100% Mn and Pt extraction.

Yarrowia lipolytica: Identified as effective for REE leaching from fluorescent phosphor alongside bacterial strains *Komagataeibacter xylinus* and *Lactobacillus casei.*

Microalgal Systems

Microalgal applications for manufacturing waste treatment were notably underrepresented despite their documented metal binding capabilities.

Galdieria sulphuraria: This extremophilic red alga demonstrated exceptional tolerance to acidic conditions (pH 1.5-2.5) and high temperatures, making it suitable for harsh industrial waste streams. Freeze-dried biomass achieved 41.61 mg/g dm total REE adsorption within 5 minutes. The organism selectively accumulated lanthanoids over Cu(II) at pH 1.0-1.5, and acid pretreatment enhanced binding sites for improved metal recovery.

Chlamydomonas sp. (ChlSG): This acidophilic extremotolerant strain achieved biouptake of 7.3 µg/g Zn, 4.1 µg/g Al, 1.3 µg/g Cu, and 0.7 µg/g Mn from e-scrap leachate, with higher efficiency than *Euglena* sp. (EugVP) for most elements.

Coccomyxa genus: Microalgae isolated from extreme environments demonstrated non-selective metal uptake with accumulation up to 15 mg REE/g dry weight and >90% decontamination within a few hours.

Engineered Systems

Genetic engineering and surface modification strategies substantially improved recovery performance:

Lanthanide binding tag (LBT)-displaying *E. coli*: Encapsulated in PEGDA hydrogel microbeads, this system achieved 97% REE purity from NdFeB magnet leachate with 85% capacity retention after nine reuse cycles.

Silk-LBT membranes: Recombinant silk-elastin-like proteins containing LBTs recovered up to 85% of Tb^{3+} with ~50% recovery maintained in the presence of 100× concentrations of interfering metals (Ca^{2+}, Cu^{2+}, Fe^{3+}, Zn^{2+}).

Lanmodulin-SpyCatcher on magnetic nanoparticles: This biosorbent demonstrated fast adsorption kinetics, >90% desorption efficiency, and retained ~95% activity after eight cycles, achieving 967-fold REE purity increase from complex waste streams.

Operational Scale and Process Integration

Scale of Demonstration

The vast majority of studies operated at laboratory batch scale (<10L), with limited progression to larger scales:

Scale	Number of Studies	Representative Example
Laboratory batch (<10L)	~60	Most biosorption studies
Bench-scale continuous (10-100L)	~8	*M. extorquens* scaled to 10L
Pilot-scale (>100L)	~3	Chalcopyrite column leaching
Commercial deployment	0	None documented for manufacturing waste

The *Methylobacterium extorquens* AM1 system demonstrated successful scale-up from laboratory to 10 L with consistent metal yields, maintaining 25-hour hydraulic retention time at pH 6.9 and 29.5℃. Column bioleaching studies with moderately thermophilic bacteria showed that metal tolerance could be improved from 12 g/L to 20 g/L after nearly two years of adaptation.

Process Duration and Kinetics

Processing times varied dramatically based on mechanism and target metals:

Process Type	Duration	System	Reference
Biosorption	5-30 min	*G. sulphuraria*, *M. magneticum*	
Bioleaching (optimized)	12 days	Acidophilic consortium in bioreactor	
Bioleaching (batch)	59-78 days	Shake flask experiments	
Sequential bioleaching	8 days + 3 hours	*At. thiooxidans* + *P. putida*	
Fungal bioleaching	21 days	*A. niger*	
Biofilm production	10-14 days	*Serratia* sp. N14	

Biosorption processes offered rapid kinetics suitable for flow-through applications, while bioleaching required extended residence times. Bioreactor optimization significantly accelerated copper bioleaching from 59-78 days in shake flasks to only 12 days.

Integration Considerations

Few studies addressed integration with existing manufacturing waste treatment infrastructure. The continuous bioreactor system coupled with electrowinning demonstrated >99% Cu deposition with 98.5-99.9% purity in 3.33 hours at 80-92% current efficiency. The bio-hydrometallurgical route combining bioleaching with electrodialysis achieved 76% Zn purity from iron oxide residue.

Two-step bioleaching processes showed promise for sequential recovery of different metal classes: base metals first by acidophilic bacteria, followed by precious metals using cyanide-producing organisms.

Synthesis

Reconciling Heterogeneous Findings

The substantial heterogeneity in reported recovery efficiencies can be explained through several mechanistic and methodological factors:

Context and Population Distinctions: Studies achieving >95% recovery efficiencies typically operated with optimized conditions (controlled pH, temperature, pulp density) on pre-characterized synthetic solutions or processed waste streams. Lower efficiencies (40-70%) were observed with raw industrial waste containing complex matrices, interfering substances, and variable metal speciation.

For REE recovery specifically, bioleaching from oxidic matrices (WEEE phosphors) achieved >99% extraction, while recovery from metallic waste (magnet scrap) showed greater variability (24-100%) due to the need for initial acid dissolution.

Study Quality Hierarchy: The highest quality evidence comes from bench-scale continuous systems with statistical validation. The *M. extorquens* AM1 studies employed ANOVA and Tukey's HSD tests, while the *G. sulphuraria* studies used two-way ANOVA. Many laboratory batch studies lacked statistical rigor documentation, limiting confidence in reported efficiencies.

Studies with industry partnerships (LYRSA-Derichebourg, voestalpine) and governmental funding (DOE, Horizon 2020) generally provided more comprehensive characterization of waste streams and process parameters.

Mechanistic Explanations: The divergence between bacterial and fungal recovery efficiencies reflects fundamentally different mechanisms. Acidophilic bacteria employ direct enzymatic oxidation and ferric iron regeneration, achieving rapid and complete solubilization of sulfide and metallic phases. Fungi rely on organic acid secretion (citric, oxalic, gluconic acids) for indirect leaching, which is less efficient but operates under milder conditions.

Biosorption systems show rapid initial uptake followed by slower intracellular accumulation. The 8-minute equilibrium for *M. magneticum* represents surface binding, while higher capacities achieved over longer periods (e.g., by *M. extorquens*) involve metabolically active bioaccumulation into polyphosphate granules.

Dose-Response Relationships: Metal concentration significantly impacts organism performance. High concentrations inhibit microbial growth and activity, as demonstrated in MICP experiments where elevated metal levels suppressed *Sporosarcina pasteurii* urease activity. The optimal pulp density for PCB bioleaching ranged from 5-7.5 g/L, with higher concentrations causing toxicity.

Conversely, very low metal concentrations reduce economic viability. The break-even analysis indicated that REE bioleaching profitability requires either elevated REE prices (26-fold increase) or waste processing tipping fees ($155/ton).

Organism Selection Criteria

Based on the synthesized evidence, organism selection for specific manufacturing waste streams should consider:

For semiconductor CMP wastewater (Cu, trace REEs):

- Bacterial biosorption (*M. magneticum* AMB-1) offers rapid recovery with magnetic separation advantages
- Mixed bacterial biofilms (*L. casei/P. pastoris*) demonstrate regeneration capability but lower capacity than ion exchange resins
- *Galdieria sulphuraria* tolerates low pH and provides selectivity for lanthanoids

For NdFeB magnet production waste (Nd, Dy, Pr):

- *Methylobacterium extorquens* AM1 offers direct bioleaching capability without acid pretreatment, scalable to 10L
- Engineered LBT-displaying systems achieve highest selectivity (97% REE purity) but require genetic modification
- *At. thiooxidans* bioleaching achieves >99% extraction but requires acidic conditions (pH 1.0)

For mixed metal streams:

- Sequential bacterial processes (*At. thiooxidans* followed by *P. putida*) enable base metal then precious metal recovery
- Acidophilic bacterial consortia outperform pure cultures for complex waste matrices
- Mixed fungal-bacterial cultures show synergistic organic acid production

Critical Gaps and Limitations

The reviewed literature reveals significant gaps specific to the research question:

1. **Limited CMP wastewater studies**: Only three studies directly addressed semiconductor CMP waste, with none providing comprehensive quantitative data on the full spectrum of metals (Ce, Pr, La, Y, Ga, Ge, In) present in CMP slurries.

2. **Absence of microalgae comparative studies**: No study provided direct side-by-side comparisons of microalgae, bacteria, and fungi treating identical manufacturing waste samples. The specified microalgal genera (Nannochloropsis, Spirulina, Scenedesmus) were absent from manufacturing waste applications.

3. **Economic data scarcity**: Only 4-5 studies provided quantitative cost metrics, with no comprehensive CAPEX/OPEX breakdowns for manufacturing waste applications. Break-even analyses were limited to REE bioleaching from catalysts and Ga recovery from LEDs.

4. **Scale-up limitations**: No commercial deployments were documented for manufacturing waste streams. The longest-running studies (2 years) focused on culture adaptation rather than process validation.

5. **Regulatory compliance**: No studies documented compliance with specific EPA discharge limits, EU REACH requirements, or WEEE Directive standards for biological recovery systems.

6. **Interfering substance tolerance**: While some studies noted tolerance to salinity and temperature variations, systematic evaluation of tolerance to EDTA chelators, HF residues, and organic solvents characteristic of semiconductor waste was absent.

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